LABG SUITE · OPEN SCIENCE IN THE BROWSER

CorrelationPro

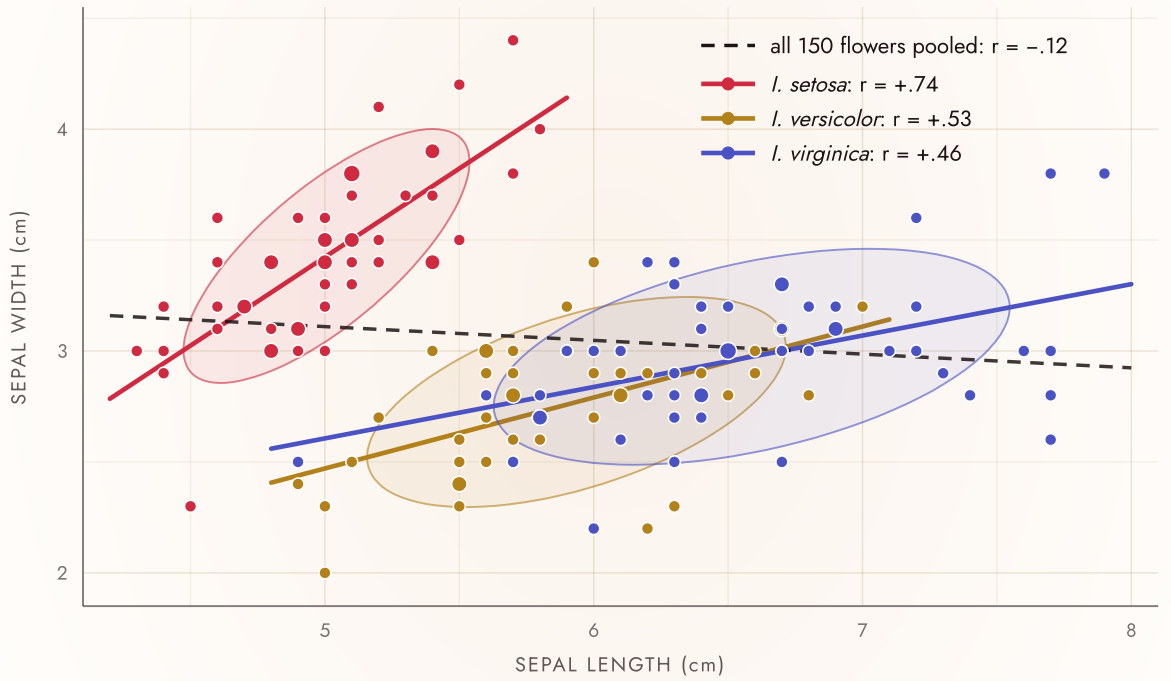
User's Manual

Manual de usuario

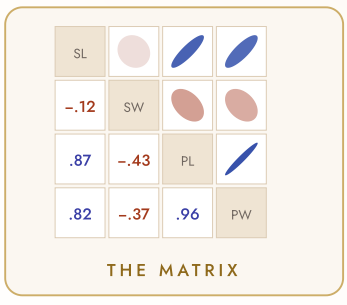
Correlation analysis of every kind, with its theory

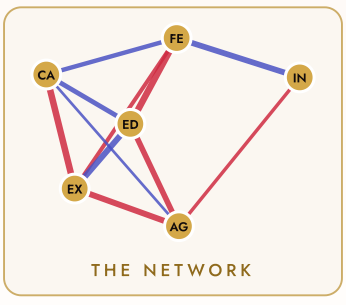
Análisis de correlación de todo tipo, con su teoría

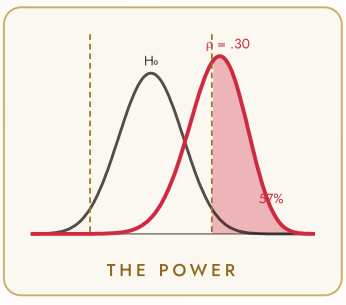
within each species they rise; pooled, they fall: Simpson's paradox











LUIS ÁNGEL BARRERA-GUZMÁN

ENGLISH EDITION · VERSION 1.0 · 2026

CorrelationPro

User's Manual · Correlation analysis of every kind, with its theory

AUTHOR	**Luis Ángel Barrera-Guzmán** ORCID 0000-0001-8057-2583
PROGRAM	CorrelationPro, version 1.0.0 (October 2026)
MANUAL	First English edition, 2026, translated from the first Spanish edition. Describes version 1.0 of the program.
PROGRAM LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL RIGHTS	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.
INTERFACE LANGUAGE	Spanish or English, with the ES · EN switch. In this manual the names of buttons, fields and menus are written as they appear on screen in English, for example Compute from the data.

HOW TO CITE THE PROGRAM

Barrera-Guzmán, L. Á. (2026). *CorrelationPro: a browser-based platform for correlation analysis of every kind, with its theory* (Version 1.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23867439

HOW TO CITE THIS MANUAL

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The program's citation also appears, ready to copy in two formats, in the How to cite section of the app's home page. The Zenodo DOI 10.5281/zenodo.23867439 always resolves to the most recent version; version 1.0.0 also has its own, 10.5281/zenodo.23867440. Code: https://github.com/luisangelbg/CorrelationPro.

> **PRIVACY**
>
> CorrelationPro runs entirely inside your browser. The files you open are read on your computer and **are never sent to any server**: you can analyze the data of a thesis in progress, a clinical trial or a confidential survey without any of it leaving your machine. The program uses no accounts, keeps your data nowhere outside your computer and needs no internet connection to compute.

The illustrations, diagrams and screenshots in this manual are original: they were generated with the program itself or with code written for the manual. The nine datasets used as examples are in the public domain; when an example is simulated, the text says so.

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How to read this manual

The manual follows the same order as the app: thirteen numbered blocks, walked from top to bottom in the left-hand bar. Each chapter explains one block and carries its number—Chapter 5 is Block 5, and its sections, figures and tables are numbered 5.1, 5.2…. The introduction has no number: its sections are I.1, I.2….

Each chapter opens with the idea the block puts to the test, goes on with what is on screen in the order it appears, and closes with the rules for reading the result and the most common mistakes. There is no need to read it from cover to cover: if you already know what you want to measure, the table in Section 1.4 tells you which measure to use and in which block it is, and Appendix B gathers the reading rules of every block.

A color for each block

The block's color appears in the opening of its chapter, in the number of each section, in the tab on the edge of the page and in the contents, and it is the same as in the band on the cover.

Block	Name in the app	What it does
1	Home	The idea of correlation in plain words, six labs that really compute, the catalog of 98 measures, the glyphs, the palettes and the reporting styles
2	Data	Opening spreadsheets or text, the kind and role of each column, missing data, transformations and the "which correlation should I use?" assistant
3	Exploration	Scatterplot matrix, normality, linearity, outliers, influence and the recommended coefficient for each pair
4	One pair in depth	Every coefficient for two variables, with its test, its intervals, the bootstrap, the permutation test and the Bayes factor
5	Matrix	The matrix of many variables with its multiple-testing corrections, its diagnosis, nine glyphs and the table to publish
6	Nonlinear	Measures that see any form of dependence, the scan of the whole matrix and the copulas
7	Partial and sets	Controlling for third variables, the multiple correlation and the correlations between two sets of variables
8	Categorical	Contingency tables, their measures, the latent correlations and the matrix with the right coefficient for each pair
9	Agreement	Concordance, Bland–Altman, intraclass coefficients and the κ family: when the question is whether two measurements say the same thing
10	Structure	Groups, repeated measures, time series, spatial data, phylogenies, angles and compositions
11	Networks	Marginal, partial and regularized correlation networks, with centralities and communities

Block	Name in the app	What it does
12	Compare and plan	Comparing correlations and matrices, correcting for attenuation and range, planning the sample and combining studies
13	Figures and report	Exporting figures, designing a palette, the table to publish, the written methods, the report and the project

Kinds of boxes

NOTE

Clarifications and practical tips that save time.

IMPORTANT

What can change your conclusions if it is overlooked.

THEORY IN BRIEF

The idea behind a computation, without unnecessary formulas.

WORKED EXAMPLE

A case with the data included in the program, with the numbers the app gives.

DECISION RULE

If the result falls in this case… **then** this is the recommended reading.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

COMMON MISTAKE

What is seen again and again in papers and theses, and how to avoid it.

98

KEY FACT

A figure or a fact worth remembering; for example, the number of measures described in the catalog of Block 1.

Writing conventions

Written like this	Means
Use across the app	A button, field or menu of the app, with its text as it appears on screen in English.
Block 12 › Meta-analysis	A path: open the first and look inside it for the second.
`sepal_length`, `.csv`	A column name, a file name or a value to be typed as is.

Written like this	Means
r, ρ, τ, *n*, *p*	The symbols of statistics, in italics when they are Latin letters; ρ is the population correlation and *r* the sample one.
.74 · 0.74	Numbers are written as the app writes them in the chosen style. This manual uses the social sciences style: no leading zero for what cannot exceed 1 (*r*, *p*) and a leading zero for everything else.
● ◐ —	Available · partly available or with conditions · does not apply.

I

INTRODUCTION

Introduction to CorrelationPro

A correlation looks like the simplest computation in statistics, and that is why it is the one most often misused: with data that cannot support it, with the wrong coefficient, without an interval, without looking at the cloud, ignoring that the cases come in groups or in time. CorrelationPro brings together in one place the measures of association that the literature has proposed over a hundred and forty years, with their tests and their conventions, and walks you through the analysis in the order in which it is done right.

I.1 What is CorrelationPro?

I.2 Questions it answers

I.3 Six ideas behind this manual

I.4 Data it accepts and what it delivers

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The nine example datasets

I.8 How an analysis is carried out

I.9 How it was checked

I.10 What the program does not do

In

I.1 What is CorrelationPro?

It is a program to measure, test, compare and communicate the association between variables. It runs in the browser, with nothing to install, and brings its own computing engine written from scratch: no part of it depends on other people's libraries.

It covers the classical correlations (Pearson, Spearman, Kendall) and the robust ones; those of nonlinear dependence, which see U-shaped or wave-shaped relations that Pearson does not; the partial ones and those that link two sets of variables; those for categorical, ordinal and latent variables; those of agreement between methods and raters; those for structured data—groups, repeated measures, time, space, phylogenies, angles and compositions—; correlation networks; and the tools that come after computing: comparing two correlations, correcting for measurement error, planning the sample size and combining studies.

Every result says which convention it used—what to do with ties, which divisor, which correction—because two programs that give different numbers for "the same" coefficient almost always differ in that. Every formula is explained, in the catalog and in the theory notes of each block. And the figures come out ready for a journal, with palettes that read the same with color blindness and in black and white.

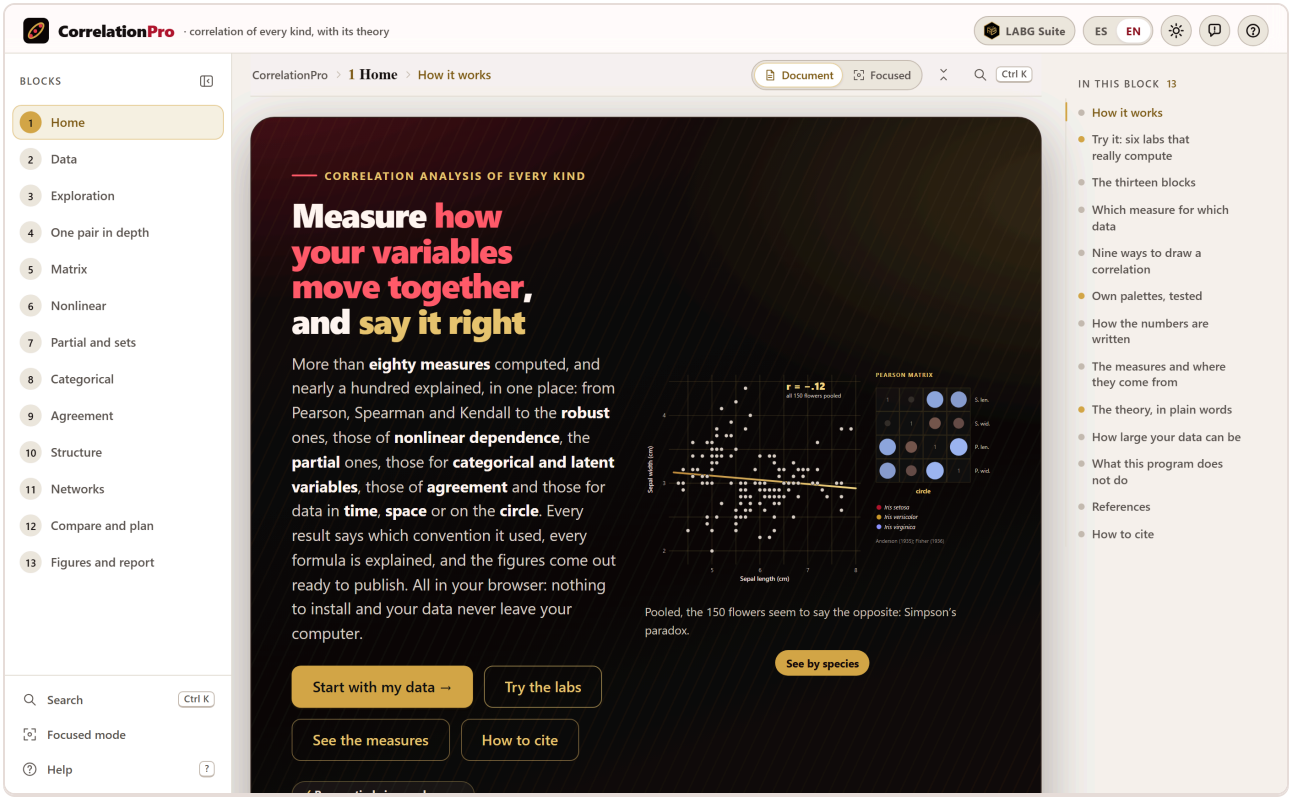


Figure I.1. The home page of CorrelationPro when it opens. On the left, the bar of the thirteen blocks; in the center, the four buttons to get started and the figure of Anderson's irises, which shows Simpson's paradox; on the right, the list of sections of the open block.

I.2 Questions it answers

A good way to know whether the program is useful to you is to see which questions it answers, and in which block.

Question	Block
Which coefficient should I use with these two variables?	2 3
Do my data meet Pearson's assumptions? Are there outliers that decide everything?	3
How large is the correlation, with what interval, and is it different from zero?	4
What is the evidence for and against a relation?	4 12
How are twenty variables related to one another, and which of those correlations survive the multiple-testing correction?	5
Is there a relation that is not a straight line, or that only shows up in the tails?	6
Are two variables still related when I control for a third?	7
Are two sets of variables alike (two ordinations, two distance matrices)?	7
How strongly associated are two categorical variables, or an ordinal and a continuous one?	8
Do two instruments, two methods or two raters say the same thing?	9
Are my cases independent? What if they come in groups, in time or in space?	10
Which variables are directly connected with which, and which one is at the center?	11
Is this correlation larger than that one? How many cases do I need? What do all the studies say together?	12
How do I publish the figure, the table and the methods?	13

I.3 Six ideas behind this manual

Everything that follows rests on six ideas. If you keep them in mind, half of the mistakes made with correlations are no longer possible.

1. **Look first, compute afterwards.** Four completely different clouds of points can have the same r (Anscombe, 1973). The app draws the cloud before the number, in every block.

2. **The data choose the coefficient, not habit.** Pearson measures the *linear* relation between two continuous variables; Spearman and Kendall, the *monotonic* one; the nonlinear dependence measures, any of them. With categories, ranks or latent variables, others are needed. Block 3 recommends one for each pair and explains why.

3. **A correlation without an interval does not say how much.** The p-value only answers whether r is different from zero, and with large samples anything is. The confidence interval says which values are compatible with the data (Cumming, 2008).

4 **Cases are almost never independent.** Measurements on the same subject, on neighboring sites, in consecutive years or on related species resemble one another, and that inflates the correlation and shrinks the *p*-value. Block 10 is made for that.

5 **Correlation is not agreement, nor cause.** Two methods can correlate at .99 and differ by ten units (Bland and Altman, 1986). And two variables can move together because a third one moves them, or by pure chance among many tests.

6 **What is not reported cannot be checked.** The coefficient, the convention, the handling of missing data, the multiple-testing correction and the program version belong in the methods. Block 13 writes them from what you ran.

1.4 Data it accepts and what it delivers

A table: one row per case and one column per variable. It is opened from a file, pasted from a spreadsheet or taken from the examples.

In	Format
Spreadsheets	`.xlsx`, `.xlsm` and `.ods` workbooks, with a choice of sheet
Text	`.csv`, `.tsv`, `.txt` and `.dat`, with comma, semicolon, tab or bar as delimiter, decimal point or comma, and UTF-8, UTF-16 or Latin-1 encoding, all detected automatically
Pasted	Whatever you copy from a spreadsheet, with its header
Projects	A `.json` package saved earlier by the app itself (Block 13)
Summaries	To compare correlations, correct them, plan or run a meta-analysis, typing *r* and *n* is enough (Block 12)

Out	Format
Figures	PNG, SVG, vector PDF and TIFF, at the journal widths of 85, 114 and 180 mm, through the figure studio
Tables	Copied to the word processor with their formatting, or as CSV and HTML
Text	The written methods with their references, and a printable report
Project	A `.json` package with the data, their types and the preferences, to resume the analysis

1.5 What you need and how to open it

Only a modern browser, in a version from the last few years. There is nothing to install and no internet connection is needed; the whole program fits in one folder.

1. **Double-click `index.html`.** It is the simplest way and everything works the same: all the computations and the examples are inside the program.
2. **Or double-click `Open CorrelationPro.bat`.** It starts a minimal local server on port 9950 and opens the app at `http://localhost:9950`. It is only needed to open it from another computer or a tablet on the same network; the file `server.ps1` explains how.

Without copying anything, the same app opens online at `https://luisangelbg.github.io/CorrelationPro/`, and the program folder can be downloaded from `https://github.com/luisangelbg/CorrelationPro`.

Preferences—language, theme, palette, number style—are saved in the browser itself and remembered the next time. Data are not: every time you open the app it starts empty, and to resume an analysis there is the project package of Block 13.

How large your data can be

The engine is written so that the browser does not freeze. These are times measured on a desktop computer with the same engine that runs in the app:

Computation	Time	In practice
Pearson matrix of 200 × 200 variables, 1,000 rows	34 ms	instant
Pearson matrix of 1,000 × 1,000 variables	83 ms	instant; drawn on a canvas
Kendall's τ-b with 100,000 pairs	64 ms	instant
Distance correlation with 2,000 pairs	33 ms	instant; up to 3,000 pairs
Mantel test, 9,999 permutations, 200 sites	0.34 s	a moment
BCa bootstrap of *r* with 1,999 resamples, 1,000 pairs	0.9 s	a moment
Graphical lasso with EBIC, 30 variables and 300 rows	1.0 s	a moment

When a computation is going to take more than a moment, the app says so with an animated wait and, when it can, adjusts the number of resamples to the time available on its own, and tells you.

I.6 A tour of the interface

All the applications of the suite share the same bar, the same navigation and the same figure tools, so what you learn here also serves you in the others.

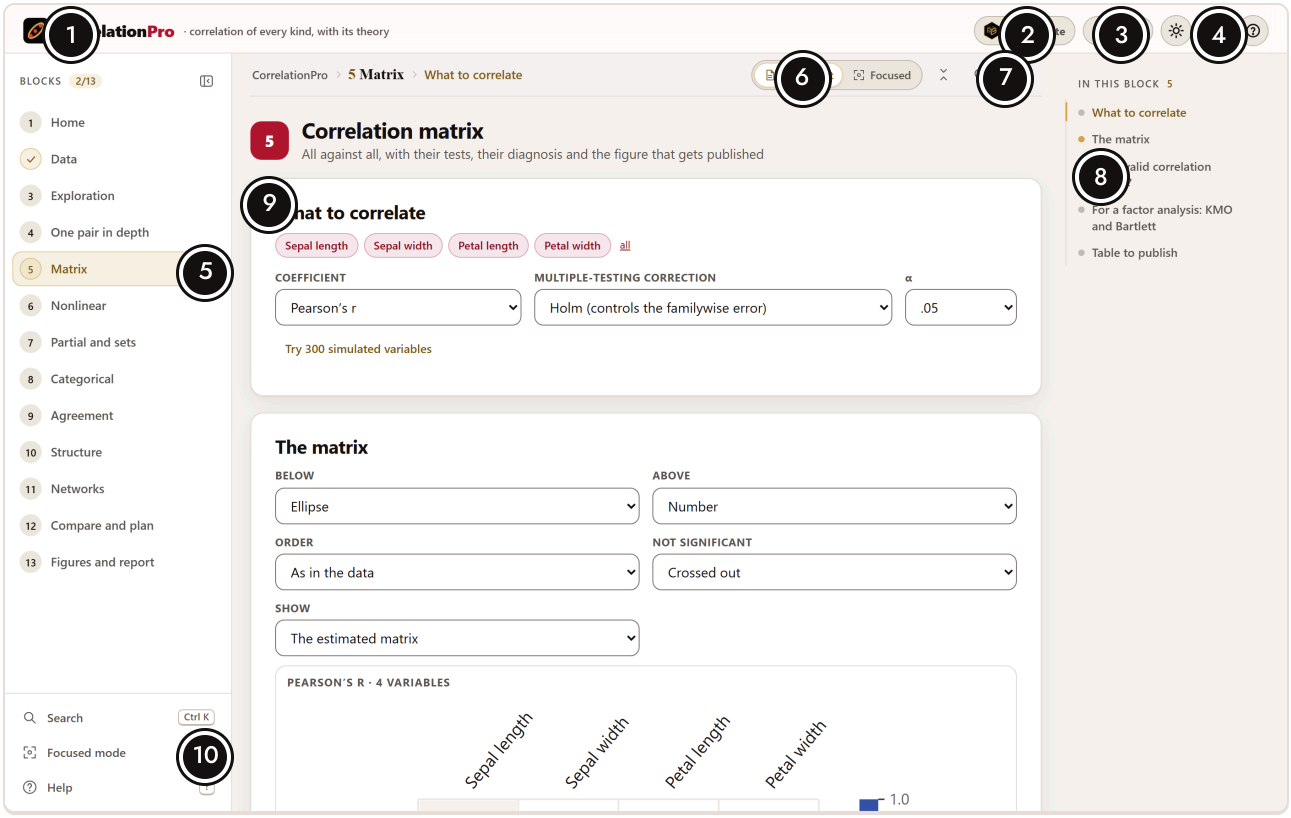


Figure I.2. The Block 5 screen with the irises open. The numbers correspond to the list below.

1. **App name.** Takes you to the home page (Block 1).
2. **LABG Suite.** Goes back to the portal with the other applications.
3. **ES · EN.** Changes the language of the whole interface, including that of the figures.
4. **Theme, report and shortcuts.** Light or dark; report a problem on the project's public page; the list of keyboard shortcuts (key `?`).
5. **The bar of the thirteen blocks.** The number turns into a check mark when the block already has a result; `[` collapses it.
6. **Document · Focused.** All the sections of the block one after another, or one at a time.
7. **Search** (`Ctrl+K`). A block, a section, a control or a measure, by its name.
8. **In this block.** The sections of the open block; the current one is highlighted.
9. **A card.** Each section is a card with its controls at the top and its results below.
10. **Search, Focused mode and Help,** also at the foot of the block bar.

Finding your way

At the top of each block, the breadcrumb says where you are—CorrelationPro › 5 Matrix › What to correlate—, and at the foot of each block two buttons lead to the previous and the next one. With `Alt+←` and `Alt+→` you move from one block to another from the keyboard; in focused mode, from one section to another. When you change a control and the results are recomputed further down, a floating button appears, Last change, that takes you back to the control you touched.

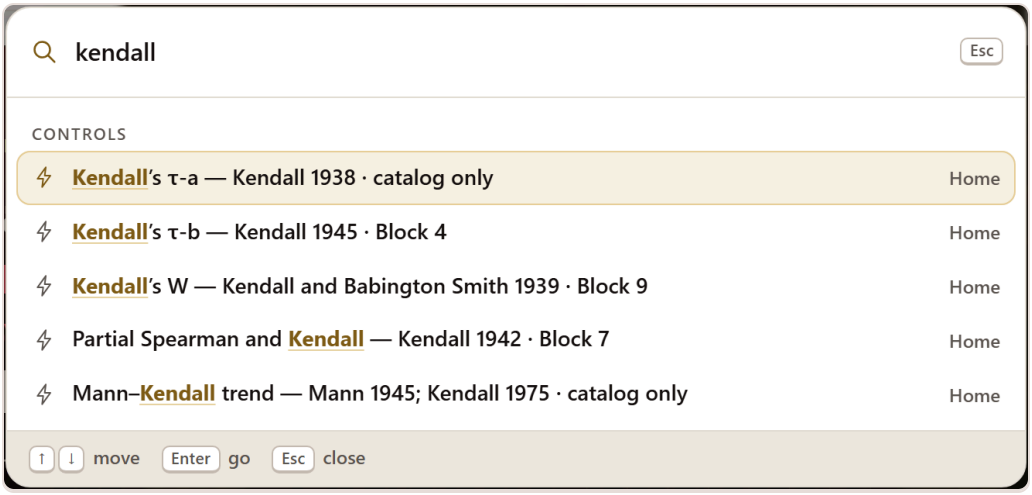


Figure I.3. The search (`Ctrl+K`) with "kendall" typed: it finds the five catalog measures that carry the name and says in which block each one is computed. `Enter` goes to the chosen result.

The figures

Every figure in the app can be opened in the **figure studio**, a window shared by the whole suite where the texts, the font size, the legend and the colors are changed without touching the data, and which exports at the widths and resolutions journals ask for. Chapter 13 explains it in depth. Next to each figure, the ✎ button opens the figure editor, to change its texts, its colors and its legend.

I.7 The nine example datasets

To learn without data of your own, the app brings nine public-domain datasets, all classics in the history of correlation. They are opened in Block 2 › An example dataset or with the buttons of the blocks that use them. When a block needs an example that does not exist among them—spatial data, a phylogeny, a panel of raters—it simulates one and says so in its title: "Simulated… (not real data)".

Dataset	*n*	What it holds and what it is good for
Anderson's irises (1935, 1936)	150	Four measurements of the flower of three *Iris* species. The correlation matrix, the correlations within and between species, and Simpson's paradox.
Galton's sweet peas (1877)	700	Diameter of the mother seed and of the daughter seed in the sweet pea (*Lathyrus odoratus*). Many ties: the conventions of Spearman, Kendall and ξ.
Galton's heights (1886)	928	Height of children and of their parents. The table with which regression to the mean was described.
Darwin's maize (1876)	15	Height of pairs of maize plants (*Zea mays*), cross-pollinated and self-pollinated, in the same pot. Paired data.
Switzerland, 1888	47	Six indicators of 47 French-speaking districts: relations of every sign, the partial network and the matrix that feeds a factor analysis.
US economy, 1947–1962	16	Seven nearly collinear annual series: very high correlations that come from time, not from a direct relation.

Dataset	*n*	What it holds and what it is good for
Titanic, 1912	2,201	Class, sex, age and survival: contingency tables, Cramér's *V*, the odds ratio and the tetrachoric correlation.
Car braking (Ezekiel, 1930)	50	Speed and stopping distance: a relation that grows faster than a straight line.
Sleep with two drugs (1905)	10	Extra hours of sleep of ten patients with two drugs: paired data, the example with which the *t* test was introduced.

I.8 How an analysis is carried out

The blocks are in the order in which a sound correlation analysis is done. There is no need to go through all of them: those that do not apply to your data are skipped, and each block tells you when it is worth going back to an earlier one.

1. **Load** (Block 2). Open your data and check that each column has the right kind and role: which coefficients are offered depends on that.
2. **Explore** (Block 3). Look at the clouds, check normality and outliers, and note the coefficient the app recommends for each pair.
3. **Measure** (Blocks 4 to 9). One pair in depth, the full matrix, nonlinear dependence, partial correlations, categorical data or agreement, depending on the question.
4. **Respect the structure** (Block 10). If the cases come in groups, in time or in space, measure accordingly before concluding.
5. **Connect and compare** (Blocks 11 and 12). The network of the direct associations; the comparison with another correlation, another group or other studies.
6. **Publish** (Block 13). The figure at the journal's width, the table, the written methods and the report.

> **THE DATA ARE SHARED BETWEEN BLOCKS**
>
> What you open in Block 2 is used by all the others. If you change the kind of a column or the handling of missing data, or open other data, each block recomputes its part the next time you visit it, and its check mark in the bar goes off until then.

I.9 How it was checked

A statistics program is worth what its numbers are worth. Each engine of CorrelationPro was compared, block by block, with independent implementations written by other people and with formulas recomputed separately, on the example datasets and on simulated data with hard cases: ties, outliers, samples of five cases, correlations close to 1.

19,334

COMPARISONS WITHOUT DISAGREEMENT

Values the app computes that were compared, one by one, with the independent references: estimates, statistics, p-values, intervals, matrices. In addition, 284 automatic tests run in the browser and watch the identities every result must satisfy.

The comparison also served to find errors, our own and others'. When a published reference does not match the formula of the original paper, the app follows the original formula and the corresponding chapter says so. The details—what was compared, with what tolerance and which errors were found—come with the program in its validation folder.

I.10 What the program does not do

Knowing where a tool ends is as useful as knowing what it does.

It does not do	Why, and what to do
Causal inference	A correlation does not say what causes what. The app reminds you where it matters, but it does not estimate causal effects.
Full regression models	Association is measured here. For linear and mixed models with their diagnostics there is another application of the suite, SigmaPro.
Factor and component analysis	The correlation matrix is their starting point, and Block 5 says whether it is fit for that (KMO and Bartlett); the analysis itself is in PCAPro, of the same suite.
Multilevel correlation with a mixed model	Block 10 separates the within- and between-group correlation descriptively; the bivariate mixed model is not included.
Data that do not fit in memory	Up to tens of thousands of rows and a thousand variables; beyond that, or with costly measures, the app warns before starting.

1

CHAPTER ONE · BLOCK 1

Home: the theory, the labs and the catalog

The first block asks for no data: it teaches. It explains in plain words what a correlation is and what it is not, lets you play with six labs that really compute with the same engine the app will use on your data, describes the 98 measures of the catalog with their formula and their original paper, and sets three things that hold for the whole app: the glyph of the matrices, the color palette and the way numbers are written.

1.1 The home page and its figure

1.2 The route of a sound analysis

1.3 Six labs that really compute

1.4 Which measure for which data

1.5 Nine ways to draw a correlation

1.6 Own palettes, tested

1.7 How the numbers are written

1.8 The catalog of measures

1.9 The theory, the references and the citation

1.1 The home page and its figure

The figure on the home page is not decoration: it is the most important lesson of the program, with real data. It shows the 150 flowers of Anderson's irises, sepal length against sepal width.

Taken together, the flowers say that the longer the sepal, the narrower it is: $r = -.12$. But the See by species button separates the three species, and within each one the relation is the opposite and much stronger: +.74 in *Iris setosa*, +.53 in *I. versicolor* and +.46 in *I. virginica*. The correlation of all of them together mixes two different things—how flowers vary within a species and how the species differ from one another—and describes none of them. This is **Simpson's paradox** (Simpson, 1951), and Block 10 is made to measure it.



Figure 1.1. The home page with See by species pressed: one line and one ellipse per species, each with its *r*. On the right, the Pearson matrix of the four flower measurements, drawn with the chosen glyph. The button changes to See them pooled to go back.

The four buttons on the left lead to what follows: Start with my data → opens Block 2; Try the labs, See the measures and How to cite scroll down to their sections on this same page.

1.2 The route of a sound analysis

Below the home figure, How it works sums up in seven steps the order in which the app walks through an analysis: **load** the data and their types, **explore** the cloud and the assumptions, **choose** the right coefficient, **measure** with a test and an interval, **control** for third variables and the structure, **compare** groups, matrices or studies, and **publish** the figure, the table and the report. Further down, The thirteen blocks shows one card per block with what it does; a click on the card opens it.

1.3 Six labs that really compute

None of them is a recorded animation: every move recomputes each coefficient and its p-value with the same engine the app uses on your data. They are chosen with the tabs at the top.

Drag the points

A cloud of 30 points you can touch: drag a point, click an empty spot to add one and double-click a point to remove it. On the right, twelve coefficients in four families—**linear**, **rank**, **robust** and **general dependence**—move at the same time. The buttons at the top set starting clouds: Linear, With an outlier, Parabola, Circle, Step, Random and Irises (petal).

> **WORKED EXAMPLE · A SINGLE DATA POINT**
>
> Press With an outlier: 28 points with no relation and one far away, at the top right. Pearson climbs to a high correlation because of that single point; the rank and robust coefficients barely move. Drag the outlier toward the center and watch Pearson collapse. This is why Block 3 checks the influence of each data point before recommending a coefficient.

The p-values are two-sided except for the general dependence measures, which can only be positive and are tested one-sided. Hoeffding's uses 199 permutations with a fixed seed, and the τ^* of Bergsma and Dassios gives only the estimator, because its exact test is too slow to recompute at every move.

Same r, different shape

Pick a figure—Axolotl, Prickly pear, Pyramid, Agave, Star, Circle or X—and press Morph. The points move little by little toward the figure, but a move is only accepted if the two means, the two standard deviations and Pearson's r stay equal to two decimals. In the end you have a completely different cloud with the same five numbers. The method is the simulated annealing of Matejka and Fitzmaurice (2017); the figures are this app's own.

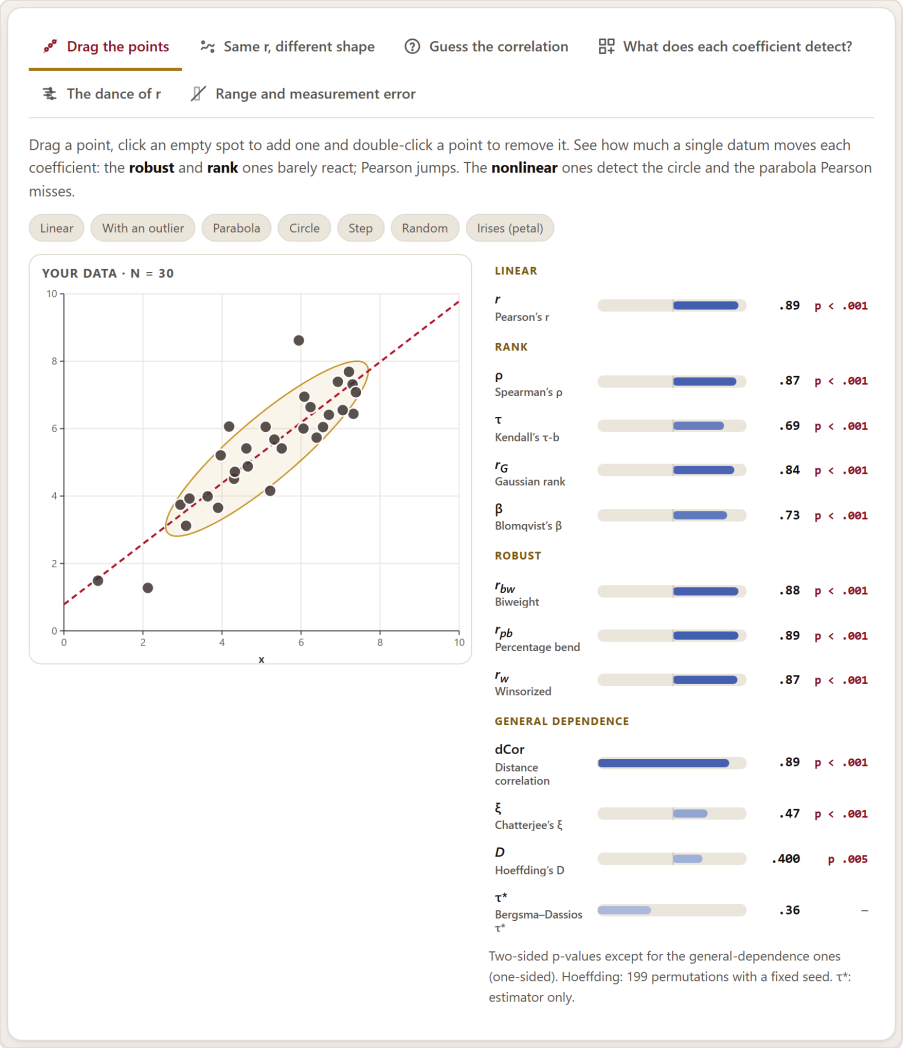


Figure 1.2. Drag the points with the linear cloud. The ellipse holds 68% of a bivariate normal with the means and covariances of the cloud; the dashed line is the least-squares line. Each bar runs from −1 to 1, and the figure on the right is the *p*-value.

Guess the correlation

How well do you read a cloud? Move the control to the *r* you believe and press Check; Another cloud moves on to the next of ten rounds. The score does not count the raw error: it divides it by the difference the eye can notice, which is large near zero and small near ±1 (Rensink and Baldridge, 2010). Missing by .10 around .70 is worth less than missing by .10 around zero.

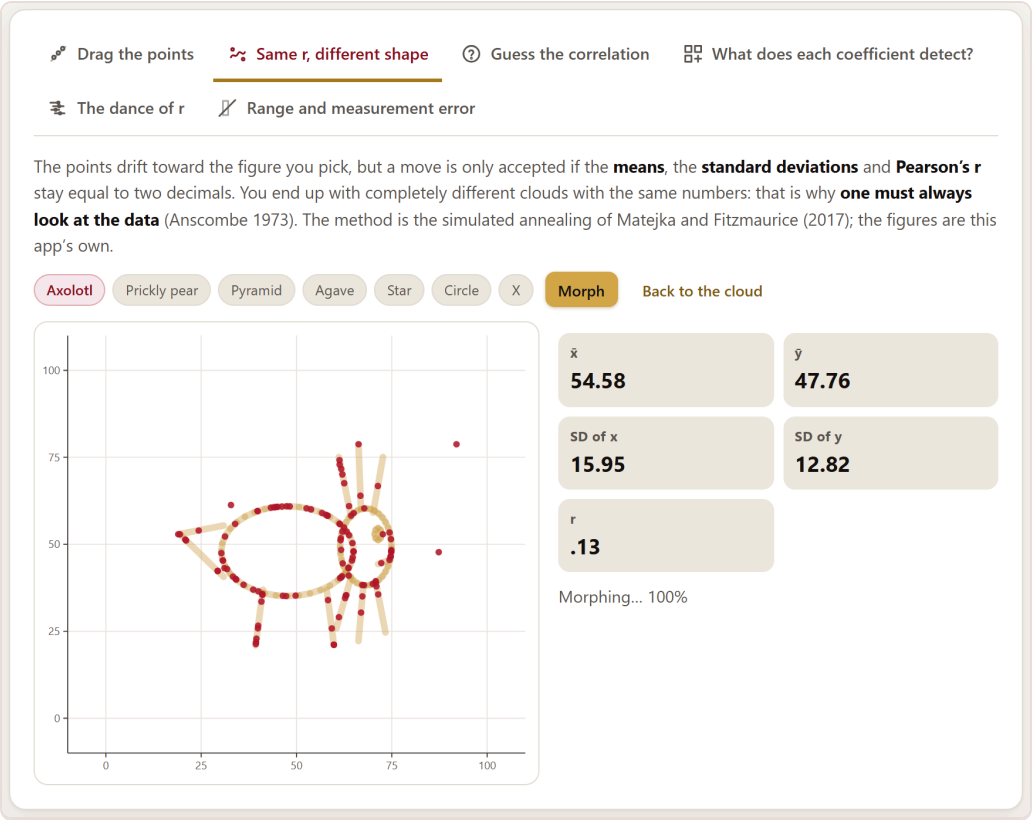


Figure 1.3. The cloud turned into an axolotl. The five numbers on the right—$\bar{x}$ = 54.58, $\bar{y}$ = 47.76, the two standard deviations and r = .13—are the same as those of the starting cloud. Back to the cloud returns it.

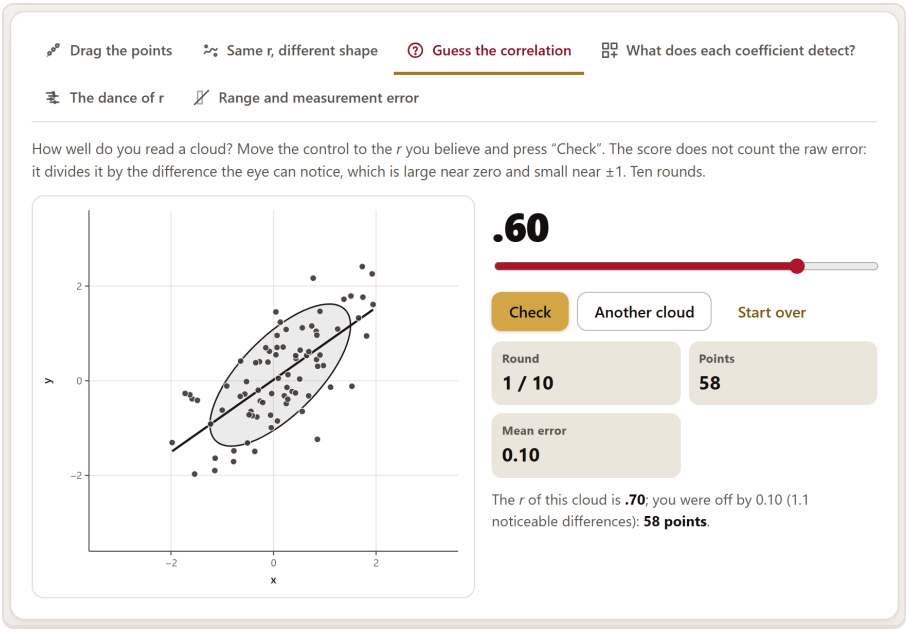


Figure 1.4. One round: the answer was .60 and the cloud had .70. The deviation of 0.10 equals 1.1 noticeable differences and gives 58 points.

What does each coefficient detect?

Nine shapes of relation against seven coefficients, in a table colored by the absolute value, with an asterisk where the test is significant at 5%. Pearson only sees the line; Spearman and Kendall, what is monotonic; distance correlation, Chatterjee's ξ and Hoeffding's *D* also see the parabola, the sine, the circle and the X. The *n* control changes the size of the samples and Another sample draws new ones.

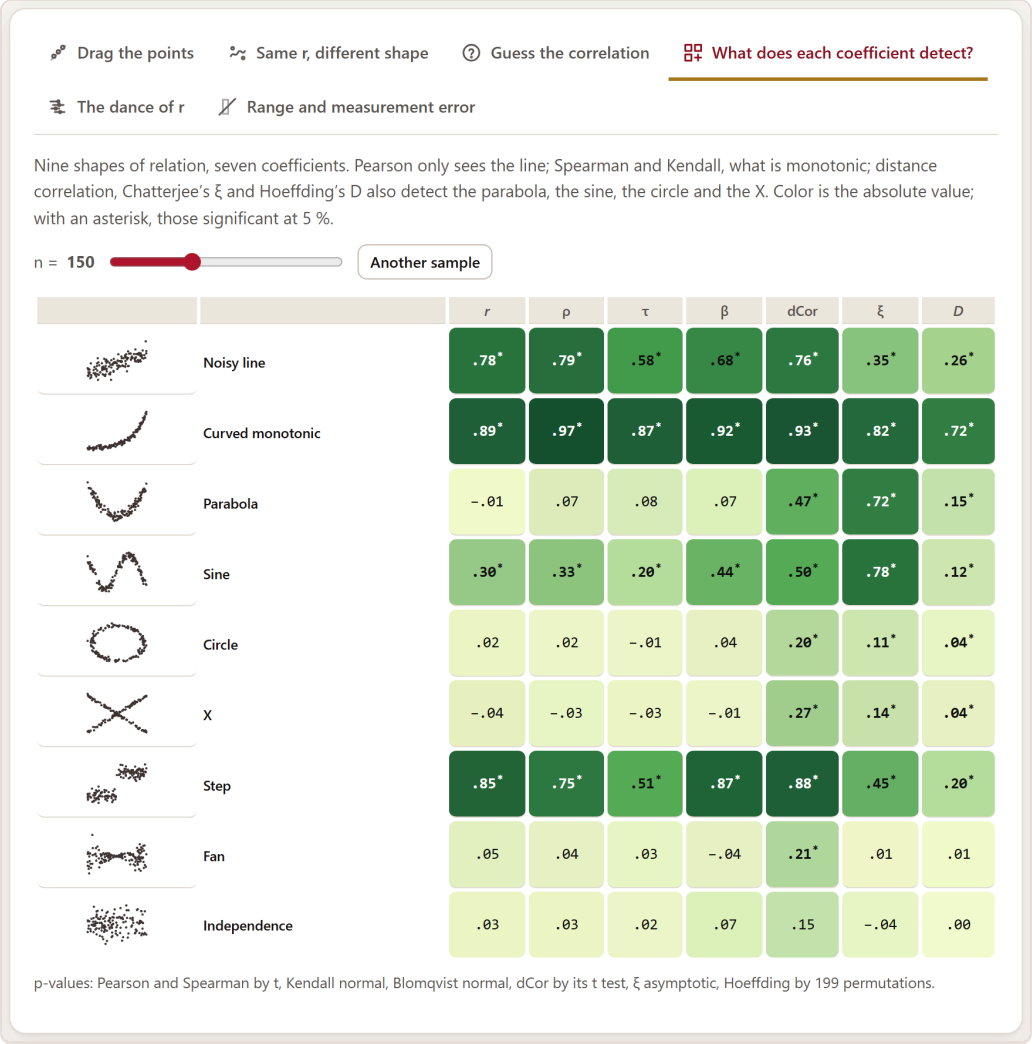


Figure 1.5. With *n* = 150, the parabola, the circle and the X have *r* close to zero, but dCor and ξ detect them. In the independence row nobody should light up, and nobody does.

> **AN R OF ZERO DOES NOT MEAN "NO RELATION"**
>
> It means "no *linear* relation". If the question is whether two variables have anything to do with each other in any way, a general dependence measure is needed. Block 6 computes them and scans the whole matrix looking for what Pearson does not see.

The dance of r

Set a population—its ρ and the sample size—and press **Dance**: each row is a new sample with its *r* and its 95% interval, and alongside, the distribution of the *r* values builds up, with the theoretical curve on top. With samples of 20 cases, *r* "dances" from .1 to .8 even though ρ is always .50, and in the long run 95% of the intervals contain ρ (Cumming, 2008). **Clear** starts over.

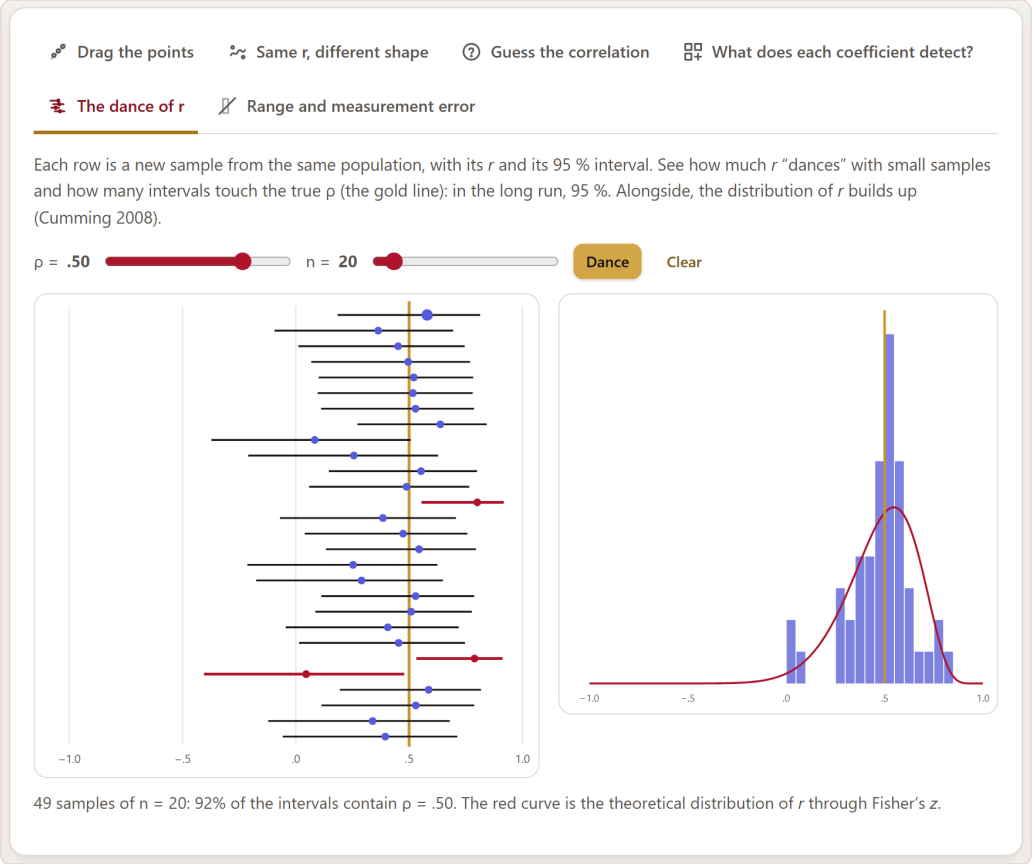


Figure 1.6. Forty-nine samples of $n = 20$ with ρ = .50. The intervals in red do not contain ρ (the gold line); in this run there were four, and 92% did contain it. The red curve on the right is the theoretical distribution of *r* through Fisher's *z*.

Range and measurement error

The two most common ways for a real correlation to come out smaller. **Range restriction**: if only part of the values of *x* is observed—only the plants that survived, only the students admitted—*r* drops; the percentile control chooses which part is seen. **Measurement error**: if *x* and *y* are measured with noise, *r* is "attenuated"; the reliability control sets it. The two cards on the right apply the corrections of Thorndike (case II) and of Spearman, which Block 12 offers for your own numbers.

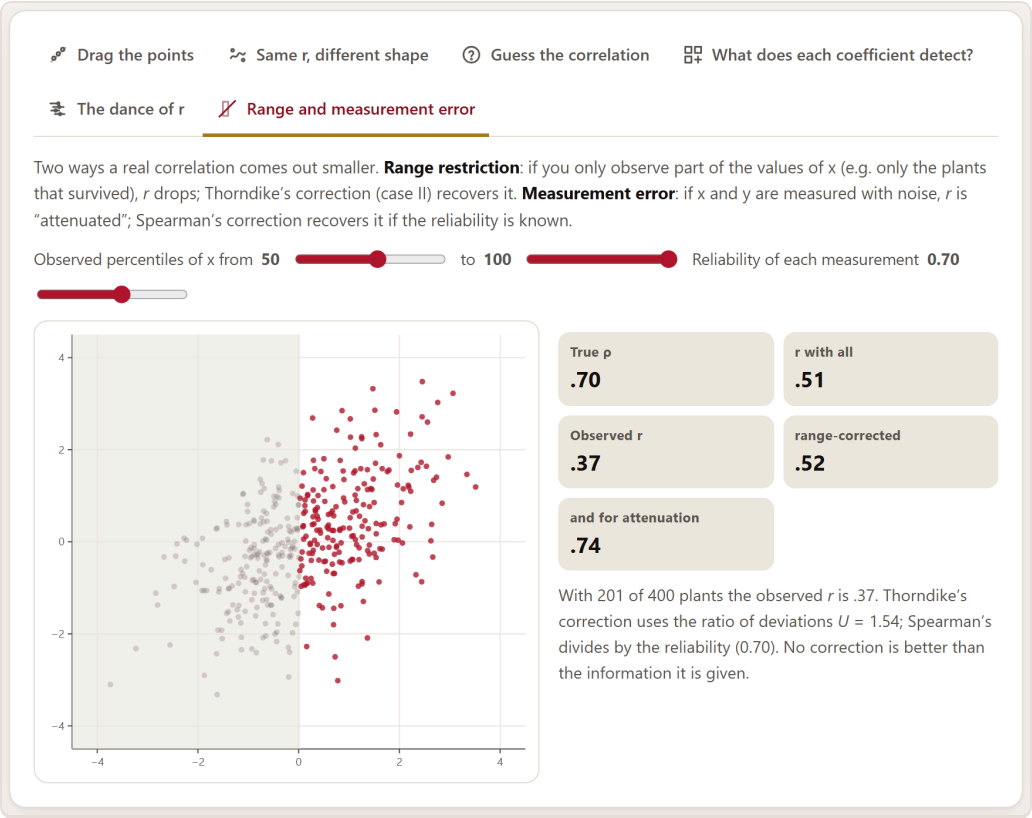


Figure 1.7. With ρ = .70, observing only the upper half of *x* and measuring with reliability .70 leaves the observed *r* at .37. The range correction raises it to .52 and the attenuation correction, after that, to .74: close to the true value, but not exact, because no correction is better than the information it is given.

1.4 Which measure for which data

The kind of the two variables decides which coefficients make sense. The section **Which measure for which data** lays out the full map in nine cards, each with the block that computes it:

Your two variables	Measures	Block
Continuous and continuous	*r*, ρ, τ, the robust ones, dCor, ξ, MIC	4 6
Ordinal and ordinal	ρ, τ-b, τ-c, γ, Somers' *d*, polychoric	8
Nominal and nominal	φ, *V*, *C*, λ, *U*, odds ratio	8
Binary or groups and continuous	point-biserial, biserial, rank-biserial, η	8
Raters or methods	CCC, ICC, κ, *W*, α, Bland–Altman	9
Angles and dates of the year	circular–circular, circular–linear	10
Proportions of a whole	ρ_p, the centered log-ratio	10
In time, in space or on a tree	repeated measures, prewhitening, Dutilleul, contrasts	10
Two tables or two matrices	canonical, RV, Procrustes, Mantel, co-inertia	7

There is no need to memorize it: the assistant of Block 2 applies it on its own to each pair of your columns.

1.5 Nine ways to draw a correlation

In a correlation matrix each cell is a **glyph**. The app offers nine, and all of them carry the sign in their shape too, not only in the color: with a divergent palette, $+r$ and $-r$ print as the same gray, and a glyph that depends only on color loses the sign in a photocopy.

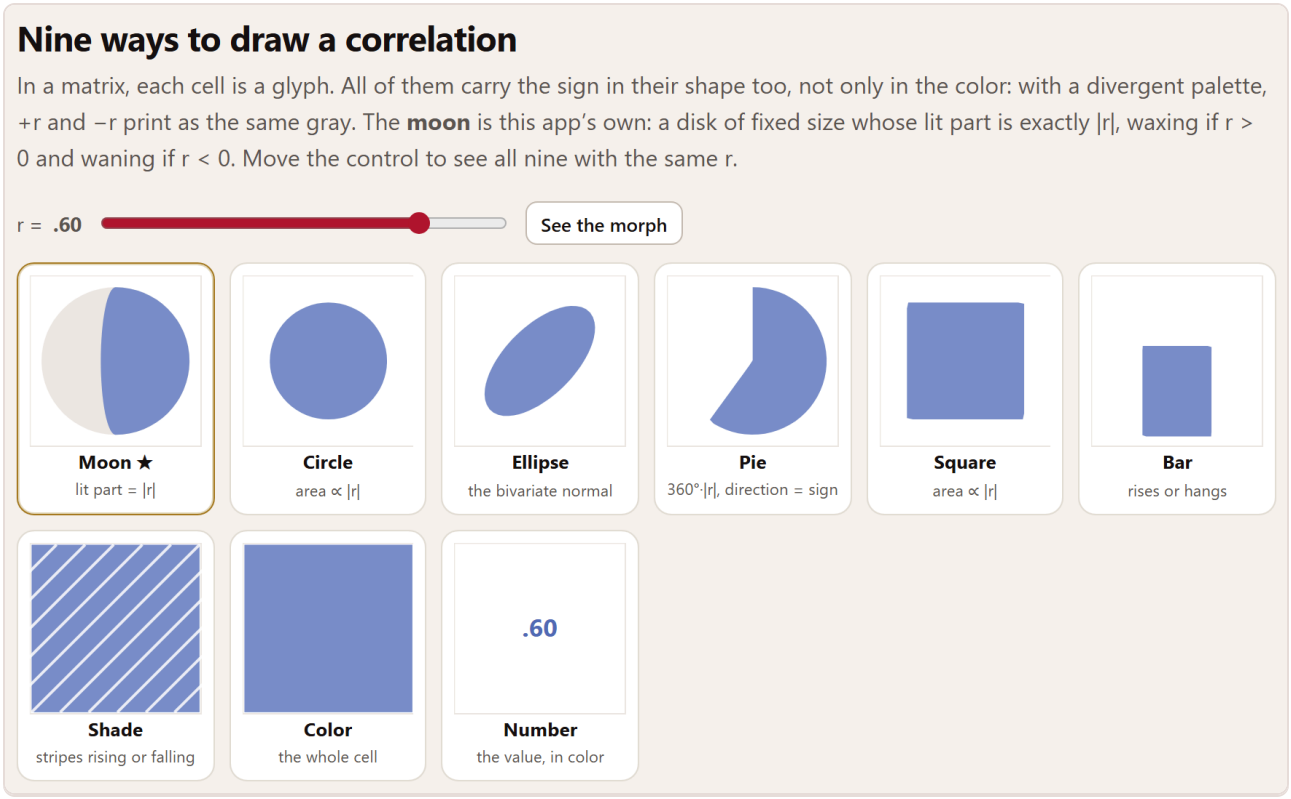


Figure 1.8. The nine glyphs with $r = .60$. The control moves r in all nine at once, and See the morph sweeps from −1 to 1 to show how each one changes.

Glyph	What it encodes
Moon ★	This app's own: a disk of fixed size whose lit part is exactly $\lvert r\rvert$, waxing if $r > 0$ and waning if $r < 0$. It reads the same in color, in gray and with color blindness.
Circle · Square	The area is proportional to $\lvert r\rvert$; the sign, by the color.
Ellipse	The shape of a bivariate normal with that r: tilted to the right if it is positive, thinner and thinner as it nears ±1 (Murdoch and Chow, 1996).
Pie	A sector of $360° \cdot \lvert r\rvert$, clockwise if it is positive.
Bar	Rises if it is positive and hangs if it is negative.
Shade	Stripes that rise or fall, with the intensity of $\lvert r\rvert$ (Friendly, 2002).
Color · Number	The cell filled with color, or the value written in the color of its sign.

1.6 Own palettes, tested

Nineteen palettes designed for this app, named after plants, places and crafts of Mexico: twelve divergent ones —ten for a light background and two for a dark one—and seven sequential ones, for what goes from 0 to 1 ($|r|$, r^2, densities). All of them were built in a perceptual color space, where equal steps of r look like equal steps of color, and each one passes five checks, which the card shows with a check mark:

- **fits:** its whole ramp can be shown on an ordinary screen, with no clipped colors;
- **uniform:** its steps have the same perceptual size;
- **color-blind:** the sign can be told apart with the three forms of color blindness, even with $|r| = .25$;
- **gray:** $+r$ and $-r$ have the same gray, so in black and white the glyph decides;
- **contrast:** its ends contrast at least 3 to 1 with the background.

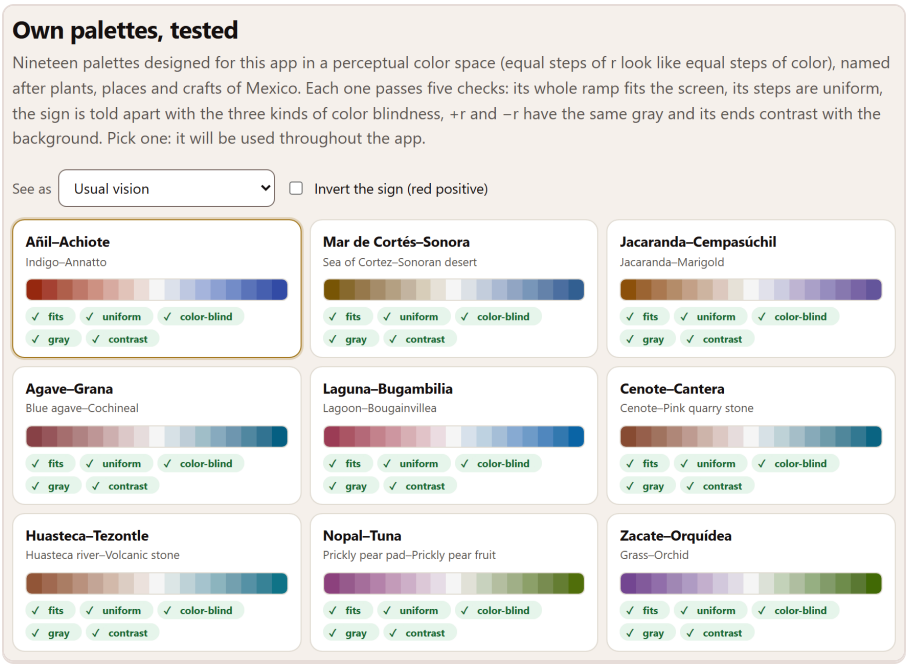


Figure 1.9. The first nine palettes, with their five checks. The chosen one—here Añil–Achiote (Indigo–Annatto), the default—is framed and is used throughout the app.

The See as menu draws all the palettes as a person with protanopia, deuteranopia or tritanopia would see them, or in grayscale, and the Invert the sign (red positive) box swaps the two hues for those used to the opposite convention. Below the gallery, the iris matrix is redrawn with the chosen palette. The palettes for figures with groups—one of our own, which keeps the distance between colors under the three forms of color blindness, and that of Okabe and Ito (2008)—are used automatically where they are needed. If none of them suits you, Block 13 has a designer to make your own.

1

1.7 How the numbers are written

Style manuals do not agree. The section **How the numbers are written** offers four styles, and the one you choose is used throughout the app: on screen, in the tables and in the report.

Style	Comes out as	Rule
Psychology and social sciences (APA 7)	`r(48) = .45, 95% CI [.20, .65], p < .001`	No leading zero for what cannot exceed 1
Biomedical (SAMPL guidelines)	`r(48) = 0.45, 95% CI, 0.20 to 0.65, P < 0.001`	Leading zero and a capital *P*
Mexican standard (NOM-008)	`r(48) = 0.45, 95% CI [0.20, 0.65], p < 0.001`	Leading zero required, decimal point
Standard with decimal comma	`r(48) = 0,45, 95% CI [0,20; 0,65], p < 0,001`	Decimal comma; the limits separated by a semicolon

The same style can also be chosen from the table to publish in Block 13.

1.8 The catalog of measures

The section **The measures and where they come from** describes 98 measures of association and inference tools, each on a card. They are searched by name in the field at the top, or filtered by family: classical, robust, nonlinear, partial and sets, categorical and latent, agreement, structured, special data and inference. Under the name of each card come the author and year that define it, and the block where it is computed.

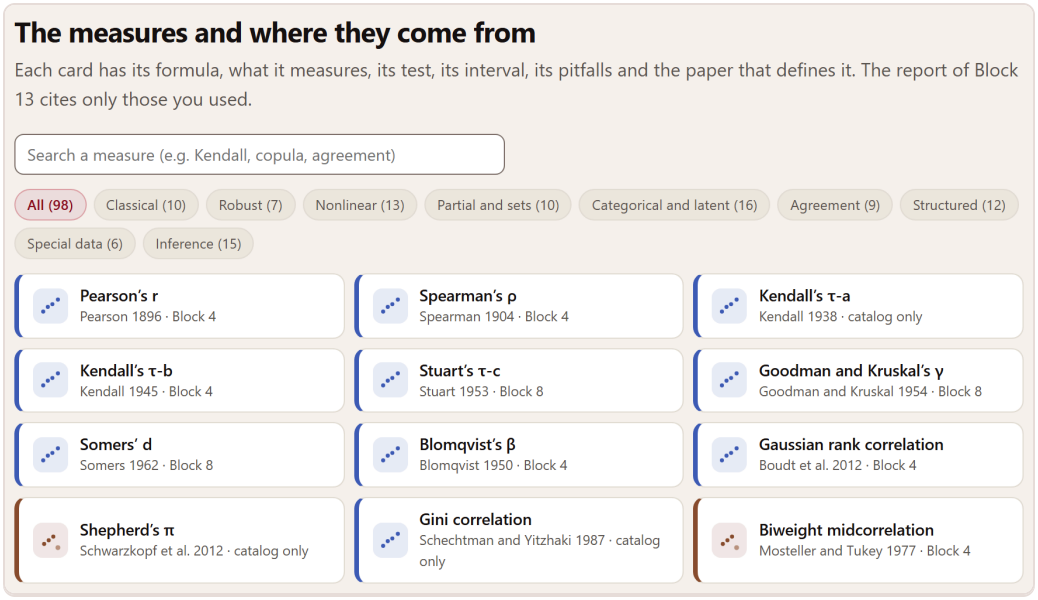


Figure 1.10. The catalog: the search field, the filters with the number of cards in each family and the first cards. The colored edge gives the family.

83

COMPUTED OUT OF THE 98

The other fifteen are described so that the catalog is complete—Shepherd's π, the median correlation, the tests of Blum, Kiefer and Rosenblatt or of Heller, Heller and Gorfine, the partial Mantel test, the moving correlation, among others—but the app does not compute them, and their card says `catalog only` instead of the block.

A click on a card opens it. Each one has the same parts: what it measures, its **formula** written as in a textbook, its hypothesis **test**, its confidence **interval**, its **pitfalls and conventions**—what it does with ties, which divisor it uses, where programs differ—and its full **references**.

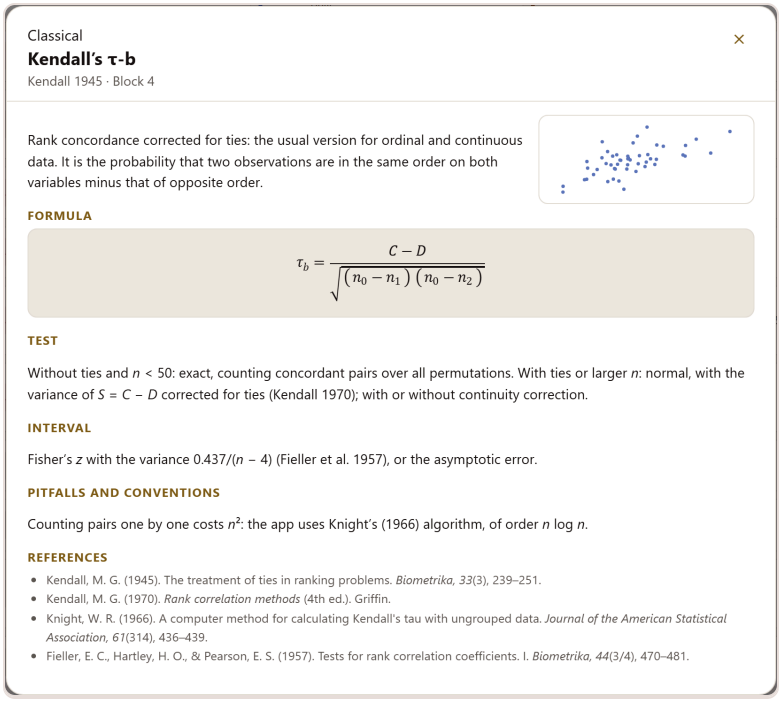


Figure 1.11. The card of Kendall's τ-b: the formula corrected for ties, the exact or normal test depending on the case, the interval with the variance of Fieller, Hartley and Pearson, the convention of Knight's (1966) algorithm and the four references.

I **THE CATALOG CAN ALSO BE SEARCHED FROM ANY BLOCK**

The search in the bar (`Ctrl+K`) finds the cards by their name and says in which block each measure is computed, as in Figure I.3.

1.9 The theory, the references and the citation

`The theory, in plain words` gathers in twelve collapsible topics the ideas that hold up everything else. Each topic says in which block it is used, and several bring their own figure.

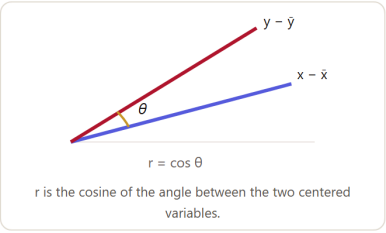
1

The theory, in plain words

The ideas that hold up everything else. Each one says in which block it is used.

What a correlation is — EVERY BLOCK

Galton (1888) called **co-relation** the tendency of two measurements to move together, and Pearson (1896) gave it its formula: the covariance divided by the product of the standard deviations. That frees it from the units: it goes from −1 (one rises as the other falls, on a straight line) to +1 (they rise together, on a line), with 0 when there is no linear trend.



r is the cosine of the angle between the two centered variables.

There are many equivalent ways to see it (Rodgers and Nicewander 1988): the slope of the line when both variables are in standard-deviation units; the geometric mean of the two regression slopes; the cosine of the angle between the two centered data vectors; the square root of the fraction of variance explained.

Everything else in the program answers what *r* does not do: measure curved relations, resist outliers, handle ranks and categories, measure agreement, respect data that are not independent.

- **Linear, monotonic or any dependence** — BLOCKS 4 AND 6
- **Correlation is not causation** — BLOCKS 7 AND 10
- **Is it significant? The p-value** — BLOCKS 4 AND 5
- **How precise? The intervals** — BLOCK 4
- **Is it large? The size** — BLOCKS 4 AND 12
- **Many tests at once** — BLOCK 5
- **Assumptions and outliers** — BLOCK 3
- **Groups: Simpson's paradox** — BLOCK 10
- **Correlation is not agreement** — BLOCK 9
- **How to read a correlation figure without fooling yourself** — BLOCKS 5 AND 13
- **How to report a correlation** — BLOCK 13

Figure 1.12. The theory with the first topic open: *r* as the cosine of the angle between the two centered variables, one of thirteen equivalent ways of seeing it (Rodgers and Nicewander, 1988).

Topic	Blocks
What a correlation is	all
Linear, monotonic or any dependence	4 and 6
Correlation is not causation	7 and 10
Is it significant? The *p*-value	4 and 5
How precise? The intervals	4
Is it large? The size	4 and 12

Topic	Blocks
Many tests at once	5
Assumptions and outliers	3
Groups: Simpson's paradox	10
Correlation is not agreement	9
How to read a correlation figure without fooling yourself	5 and 13
How to report a correlation	13

At the end of the page, References lists the 229 works the app cites, which can be filtered by topic with the buttons at the top (See every reference unfolds the full list), and How to cite gives the program's citation in APA format and in BibTeX, each with its button to copy it.

BEFORE YOU OPEN YOUR DATA

If you are going to publish the figures…	choose the palette and the glyph here, once: the whole app will use them.
If the journal asks for a number style…	choose it here; the screen, the tables and the report will follow it.
If you do not know which coefficient to use…	do not decide yet: Blocks 2 and 3 recommend one from your data.

2

CHAPTER TWO · BLOCK 2

The data and their kinds

Everything starts here, and so do almost all mistakes. This block opens your data, deduces the kind of each column and what it is for, counts the missing values and decides how to handle them, creates transformed columns and, for each pair of variables, proposes which correlation to use and why. What is decided here is used by the twelve blocks that follow.

2.1 Opening the data

2.2 The public-domain datasets

2.3 Your data at a glance

2.4 The kind and role of each column

2.5 Missing data

2.6 Transforming a column

2.7 Which correlation should I use?

2.8 Saving

2.1 Opening the data

One row per observation and one column per variable, with the names in the first row. It is the “long” layout of any well-made spreadsheet, and the only one the block needs.

There are three ways in. A file accepts delimited text (.csv, .tsv, .txt, .dat) and spreadsheet workbooks (.xlsx, .xlsm, .ods); drag it onto the box or pick it with a click. Paste opens a box where you paste the cells copied from a spreadsheet; Read what was pasted turns them into a table. An example dataset leads to the nine datasets of the next section.

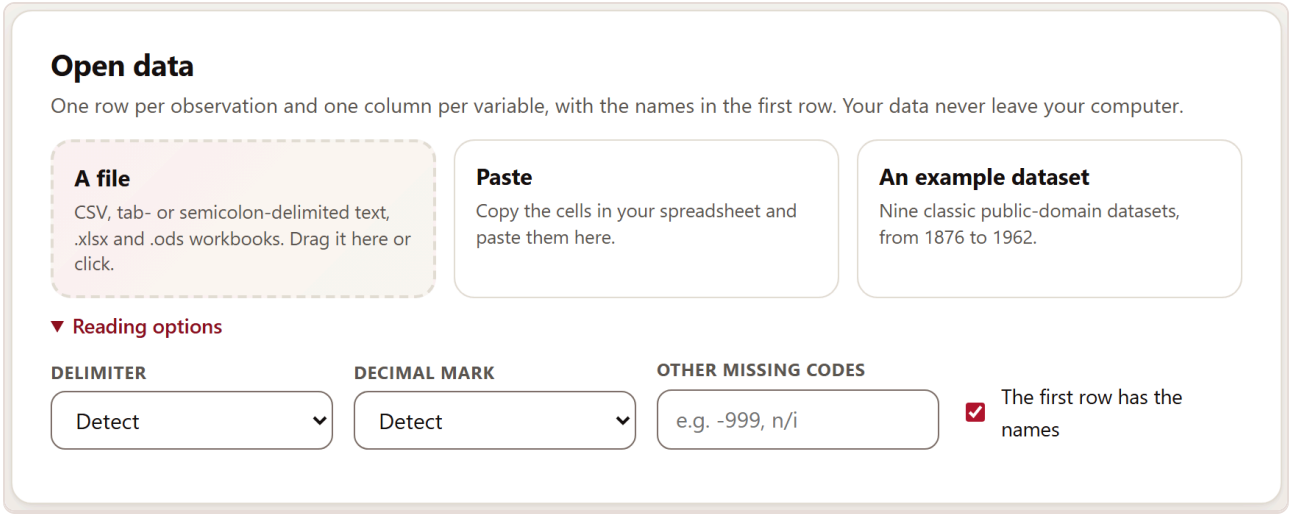


Figure 2.1. The three ways in and, expanded, the Reading options. Normally you leave everything on Detect: the app guesses the delimiter, the decimal mark and the encoding, and says what it chose.

The reading options are only needed when detection gets it wrong. Delimiter forces comma, semicolon, tab or bar; Decimal mark, point (1.5) or comma (1,5); Other missing codes adds your own, separated by commas (for example `-999, n/i`); and the box The first row has the names is unchecked if your data have no header. If a workbook has several sheets with data, the app asks which one to open.

> **WHAT THE APP RECOGNIZES ON ITS OWN AS "NO DATA"**
>
> Besides the empty cell, 28 ways of writing it that turn up in real spreadsheets, regardless of case: `NA`, `N/A`, `NaN`, `null`, `.`, `-`, `?`, `nd`, `n/d`, `s/d`, `sin dato`, `no disponible`, `faltante`, `missing` and spreadsheet errors such as `#N/A`, `#VALUE!` or `#DIV/0!`, among others.

2.2 The public-domain datasets

Nine real datasets, published long ago or by governments, each with its source, its license and what it is good for (the table in Section I.7 summarizes them). Use this dataset opens it as if it were your data, with the kind and role of each column already declared.

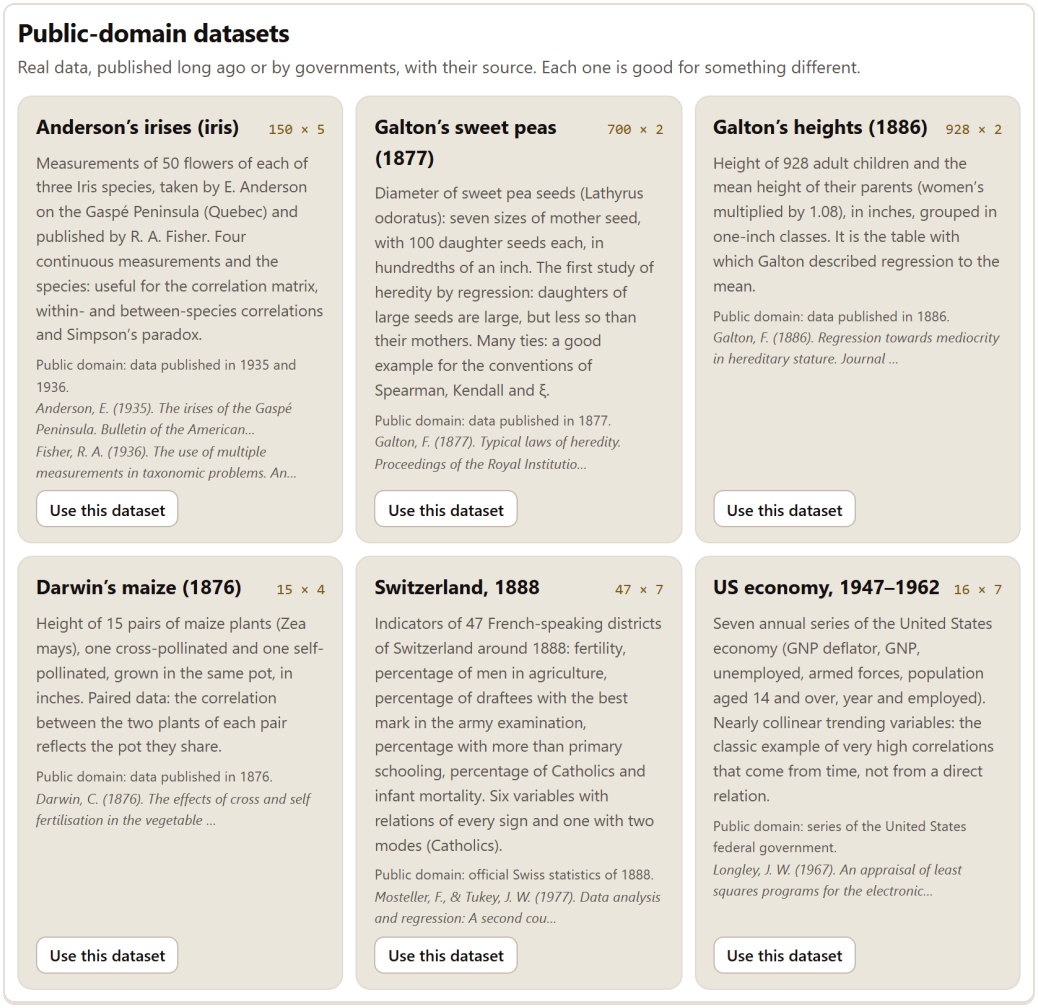
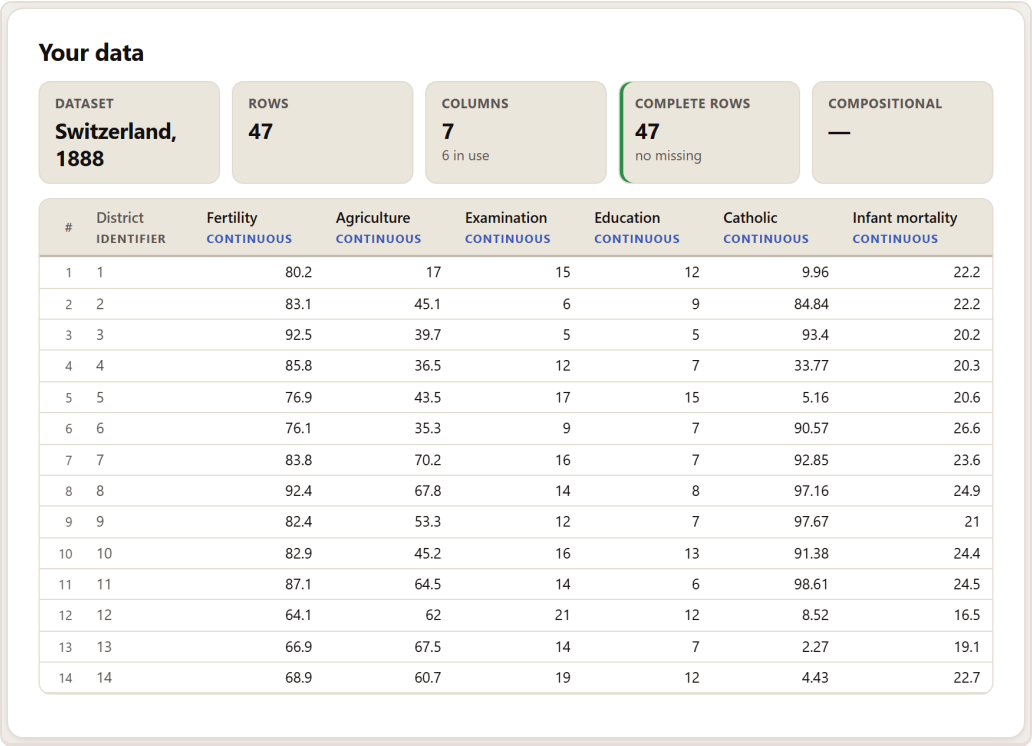


Figure 2.2. Six of the nine datasets. Each card states how many rows and columns it has, why it is in the public domain and which article it comes from.

2.3 Your data at a glance

When you open some data, a message says what was read—*Read "pasted data": 12 rows and 4 columns, separated by tab, with a decimal point.*—and the Your data card shows five figures and the table.

Compositional warns when the app finds three or more consecutive columns that add up to the same total in every row—percentages of a diet, fractions of sand, silt and clay—: they are parts of a whole, and their Pearson correlation is misleading by construction (Chapter 10). The table shows the first rows; if there are more, a note says so.



Your data

DATASET	ROWS	COLUMNS	COMPLETE ROWS	COMPOSITIONAL
Switzerland, 1888	47	7 6 in use	47 no missing	—

#	District IDENTIFIER	Fertility CONTINUOUS	Agriculture CONTINUOUS	Examination CONTINUOUS	Education CONTINUOUS	Catholic CONTINUOUS	Infant mortality CONTINUOUS
1	1	80.2	17	15	12	9.96	22.2
2	2	83.1	45.1	6	9	84.84	22.2
3	3	92.5	39.7	5	5	93.4	20.2
4	4	85.8	36.5	12	7	33.77	20.3
5	5	76.9	43.5	17	15	5.16	20.6
6	6	76.1	35.3	9	7	90.57	26.6
7	7	83.8	70.2	16	7	92.85	23.6
8	8	92.4	67.8	14	8	97.16	24.9
9	9	82.4	53.3	12	7	97.67	21
10	10	82.9	45.2	16	13	91.38	24.4
11	11	87.1	64.5	14	6	98.61	24.5
12	12	64.1	62	21	12	8.52	16.5
13	13	66.9	67.5	14	7	2.27	19.1
14	14	68.9	60.7	19	12	4.43	22.7

Figure 2.3. Switzerland, 1888: 47 rows, 7 columns of which 6 are in use (the district is an identifier), all complete. Under the name of each column goes its kind.

2.4 The kind and role of each column

This is the most important card of the block. The **kind** decides which coefficients make sense; the **role**, how the column is used. The app deduces both from the values and from the column name, and explains why in the last column of the table. If it got them wrong, change them with the menus; Back to what was deduced undoes the changes.

Table 2.1. The nine kinds and when the app deduces them

Kind	Deduced when…
continuous	the values are numbers with many distinct values. If many are repeated it says so, because ties matter for Spearman and Kendall.
ordinal	the values are integers with seven values or fewer and a short range (a 1-to-5 scale), or texts that form a known scale (low < medium < high, never < sometimes < always…), which are put in order automatically.
binary	there are only two values, numbers or texts.
nominal	the values are repeated texts with no order. With twelve categories or fewer it gets the group role.
circular	the name says angle, direction, hour, month or day of the year and the values fit its range (0 to 360, 0 to 24, 1 to 12, 1 to 366): the end lies next to the beginning.
date	all the values are dates; they are stored as days since 1970 and the column gets the time role.

Table 2.1. The nine kinds and when the app deduces them *(continued)*

Kind	Deduced when…
identifier	the name looks like a key (id, folio, code, subject, patient…) or there is a different text in every row. If its values repeat, each value is a subject measured several times.
text	there are too many distinct texts to be categories.
empty	there are no data at all.

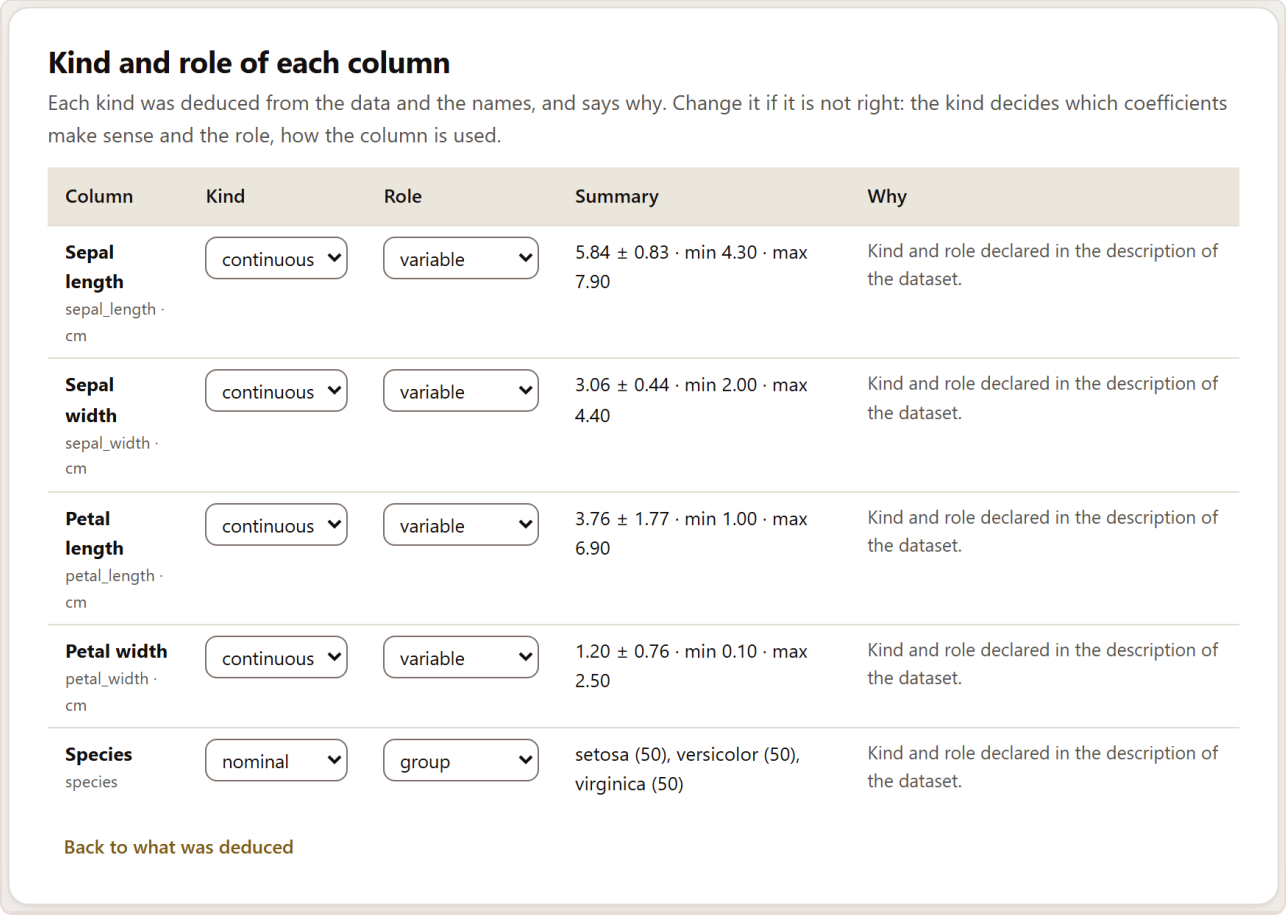


Figure 2.4. The irises: four continuous columns with the variable role and the species, nominal, with the group role. The summary gives mean ± standard deviation, minimum and maximum, or the categories with their frequencies.

Table 2.2. The eight roles

Role	What it is for
variable	it is analyzed: it enters the matrices, the pairs and the networks
group	it splits the cases: correlations within and between groups, matrices by group, the colors of the figures
subject	it identifies whoever was measured several times: the repeated-measures correlation
time	it orders the rows: time series and their cross-correlation

Table 2.2. The eight roles *(continued)*

Role	What it is for
X coordinate · Y coordinate	the position in space: Dutilleul's test and the correlogram (the app recognizes them by names such as longitude, latitude, east, north)
weight	a sampling weight or a frequency per row. In this version no block weights the cases: the column is kept but not analyzed
not used	it is kept but no block touches it

> **A 1-TO-5 SCALE IS NOT CONTINUOUS**
>
> If your satisfaction or damage column ended up as continuous because it has many values, but it is really a scale of ordered categories, change it to ordinal: Block 8 will compute the polychoric correlation, which is the right one for a latent variable measured in categories, and Block 3 will stop recommending Pearson.

Circular columns also have a **unit**: degrees (0 to 360), radians, hours (0 to 24), day of year or month. The app deduces it from the name and uses it to convert to angles in Block 10.

2.5 Missing data

How many cells are missing, how many rows are complete and how many distinct **patterns** there are: which combinations of columns are missing together. One dominant pattern—the same column is always missing—is less serious than many scattered patterns.

The menu `For each pair of variables, use` decides what to do with what is missing, and it applies to every block:

Option	Advantage	Cost
Pairwise (the default): the complete rows of that pair	Uses every datum	Each correlation of a matrix may come from different rows, and the matrix may not be valid; Block 5 checks and corrects it
Listwise: only the rows complete in every column	All the correlations come from the same cases; the matrix is always valid	Rows with a single missing value are lost

> **COMMON MISTAKE · A MISSING-VALUE CODE TAKEN AS DATA**
>
> A `−999` or a `99` left in as numbers changes any correlation: a single `−999` in a column of diameters makes Pearson say whatever that point wants. When the app sees a value like that, far away from the rest, write it in `Other missing codes` and open the file again.

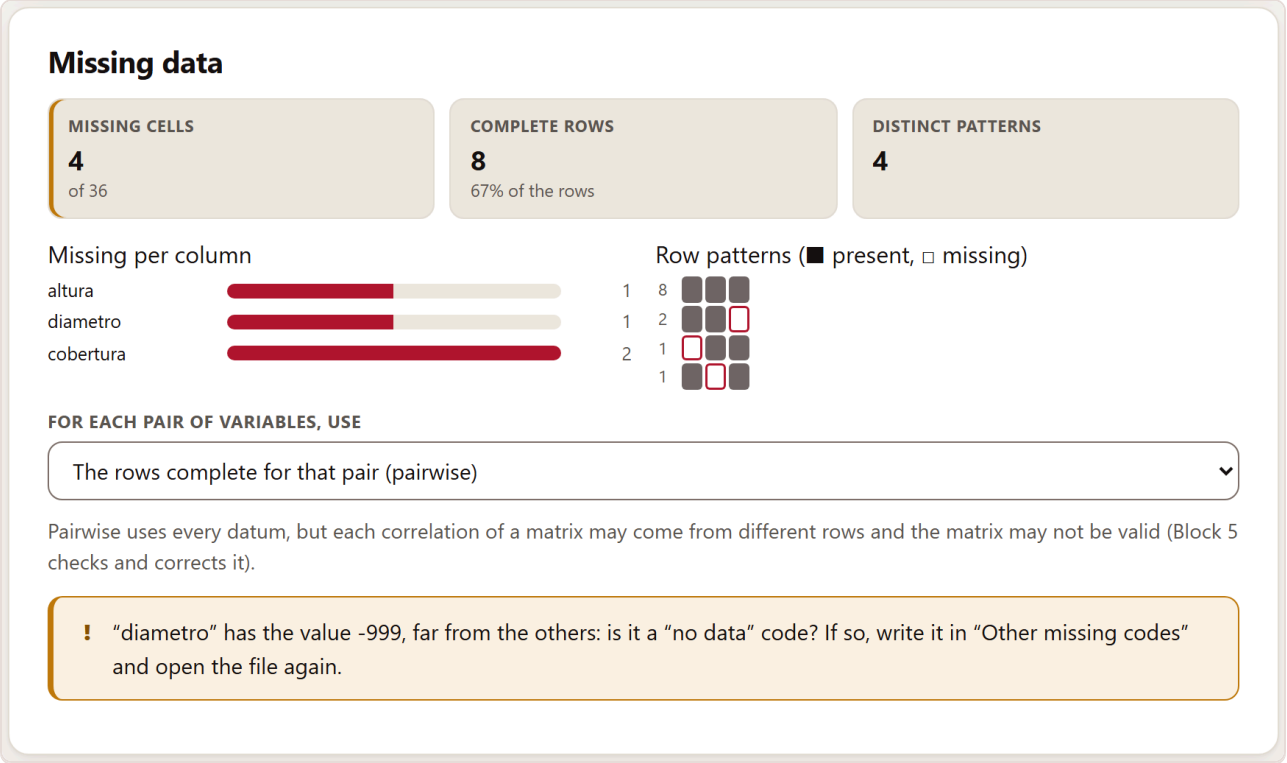


Figure 2.5. Some pasted data where "no data" was written in three ways (`n/d`, `NA` and the empty cell): the app recognized them all. The `-999`, on the other hand, is not taken as missing without asking: the app warns about it, because it could be a real value.

2.6 Transforming a column

Each transformation creates a new column and leaves the original as it was; the list below says where each one came from, and Remove deletes it.

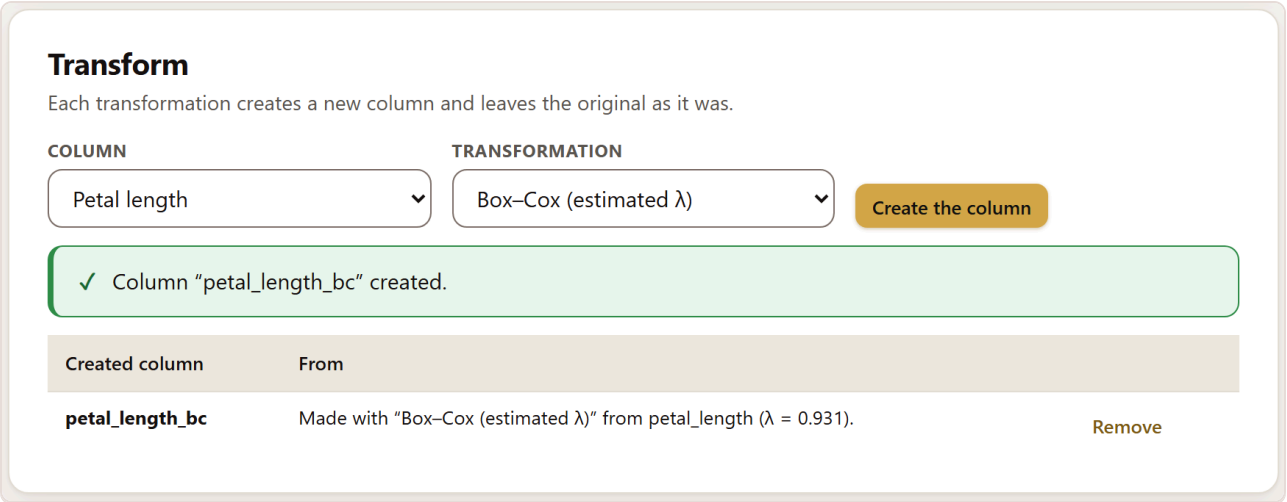


Figure 2.6. The Box–Cox transformation of petal length, with λ estimated by maximum likelihood (λ = 0.931).

Transformation	What for
Ranks (average for ties)	Working with the order; Pearson on the ranks is Spearman
Standardize (z) · Center on the mean	Putting the variables on the same scale; they do not change any correlation
Natural logarithm · Base-10 logarithm · $\log(x + 1)$	Multiplicative relations and long right tails; the last one allows zeros
Square root · Inverse $1/x$	Counts and rates
Box–Cox (estimated λ)	The power that brings the variable closest to the normal (Box and Cox, 1964)
Normal scores	The ranks turned into normal quantiles: the basis of the Gaussian rank correlation
Reverse the scale	Making "more" mean the same thing on two scales
Centered log-ratio	The parts of a whole (compositional data), before correlating them

The menu only offers the transformations that make sense for the chosen column: the logarithm, for example, does not appear if there are zeros or negative values.

2.7 Which correlation should I use?

Pick two columns in X and Y. The assistant looks at their kind and, if both are continuous, at the shape of their cloud: the outliers (beyond 3.5 on a robust scale), the curvature (the test of a quadratic term), the proportion of ties, r, ρ and the distance correlation. With that it proposes the measures in order, the first as Recommended and the others as Also, each with its reason, the block where it is computed and a What is it? link to its card in the catalog.

With variables that are not continuous the proposal changes completely. For two binary variables, such as sex and survival on the Titanic, it recommends φ, and as alternatives the odds ratio with Yule's Q, which do not depend on the margins, and the tetrachoric correlation, if each binary variable is the cut of a latent continuous one. For a continuous and a categorical variable, the point-biserial, the rank-biserial or η; for two ordinal variables, Spearman, τ-b or the polychoric; for angles, the circular correlations; if one column is a subject, the repeated-measures correlation.

WORKED EXAMPLE · WHEN THE RECOMMENDATION COMES "WITH CARE"

If the cloud has outliers, the assistant puts the percentage bend or the biweight first and leaves Pearson as "Pearson's r (with care)". If there is curvature, it proposes Spearman and the distance correlation. It does not choose for you: it tells you what to look at, and Block 3 checks it for all the pairs at once.

When the recommendation is computed in Block 4, the button Study this pair in Block 4 → takes the pair there.

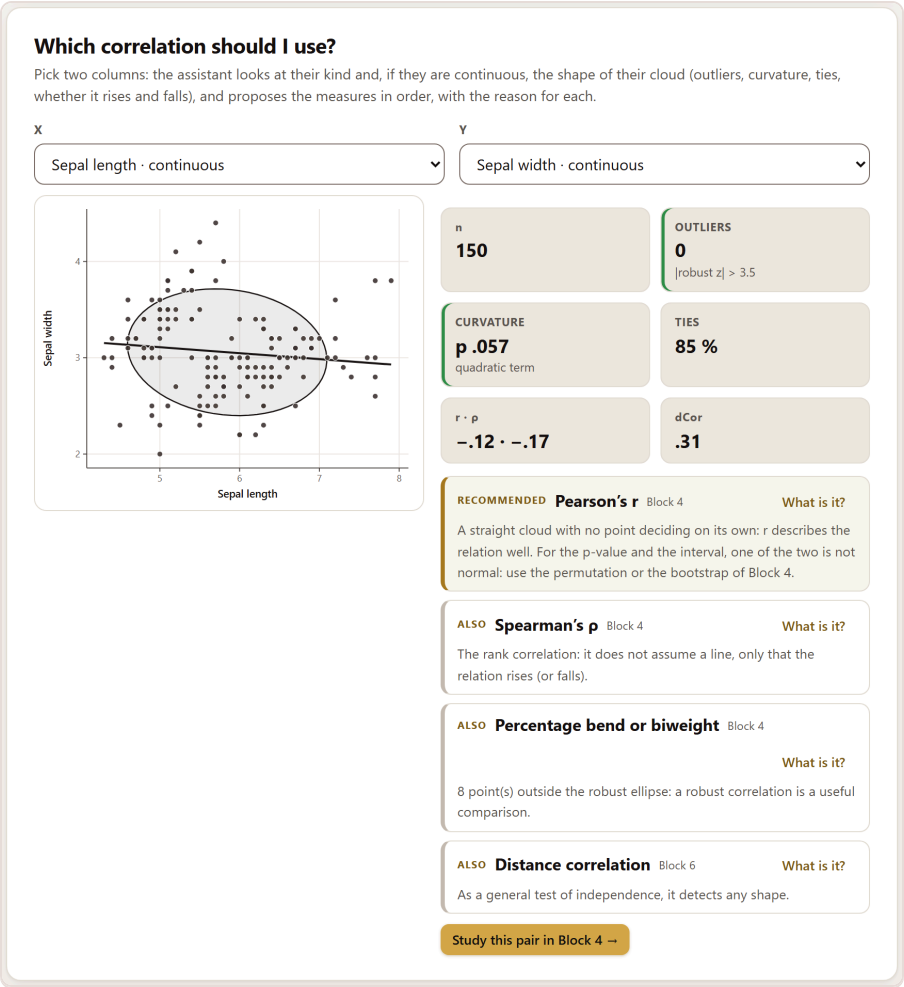


Figure 2.7. Sepal length and sepal width of the irises: a straight cloud, no outliers, non-significant curvature (p = .057) and 85% ties. It recommends Pearson, and suggests checking the p-value by permutation because one of the two is not normal.

2.8 Saving

Download CSV saves the table as it is now, with the transformed columns, as comma-delimited text that any spreadsheet opens. To save the kinds, the roles, the handling of missing data and the preferences as well, and pick the analysis up another day, there is the project package of Block 13.

✓ BEFORE LEAVING BLOCK 2	
If a column ended up with the wrong kind…	change it: the kind decides which coefficients the other blocks offer.
If the app warns about a value very far from the rest…	find out whether it is a missing-value code before going on.
If you are going to compare correlations from the same matrix or publish the matrix…	prefer listwise handling of missing data, or check the matrix in Block 5.
If the cases repeat (subjects, plots, years)…	mark the column as subject or time: Block 10 needs it.

3

CHAPTER THREE · BLOCK 3

Exploration and assumptions

A coefficient sums up a cloud of points in a single number, and that number is only faithful if the cloud is what the coefficient assumes. This block looks before measuring: the shape of each variable, of the whole set and of each pair; it searches for outliers, curvature, fanning, points that decide on their own and groups that reverse a relation, and for each pair it says which coefficient describes it best and why.

3.1 What to explore

3.2 The scatterplot matrix

3.3 The shape of each variable

3.4 Joint normality and multivariate outliers

3.5 The assumptions of each pair

3.6 The chosen pair, up close

3.7 From Block 3 to the others

3.1 What to explore

The block works with the numeric columns—continuous and ordinal—that have the variable role in Block 2. There is nothing to press to compute: every change redoes all the calculations.

The variables appear as buttons. When you open some data the first eight are selected; a click removes or adds one, at least one always stays selected and up to ten fit at once (for more, use the matrix of Block 5). Ordinal variables carry the mark ord., because they change the recommendation for their pairs.

Color by group offers the nominal or binary columns with two to eight categories; if one of them has the group role, it is selected automatically. The group colors the points of every cloud, adds r within each group and turns on the search for Simpson's paradox. Line in each cloud decides what is drawn in the clouds of the matrix: the LOWESS smoother (the default), the Least-squares line or None.

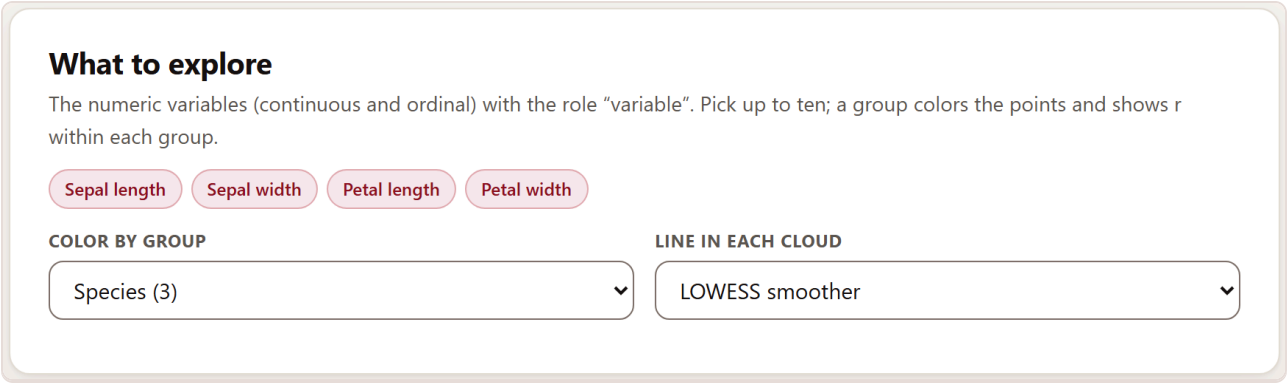


Figure 3.1. The irises: the four measurements selected, the species as group (with its three categories in parentheses) and the LOWESS smoother in the clouds.

WITH A LOT OF DATA

The calculations always use all the rows. To keep the page responsive, the clouds of the matrix draw at most 1,500 randomly chosen points, and a note below says so. With more than 400 rows or more than six variables, a wait indicator appears while the computation runs.

3.2 The scatterplot matrix

Each pair of variables has two cells. **Below** the diagonal goes its cloud, with the row variable on the vertical axis and the column variable on the horizontal one, and the chosen line. On the **diagonal**, the histogram of each variable. **Above**, Pearson's r with its stars (* $p < .05$, ** $p < .01$, *** $p < .001$) and, underneath, Spearman's ρ; the color of r follows the palette chosen in Block 1 and its size grows with $|r|$, so that strong correlations stand out. With a group, the within-group r values go at the foot of each upper cell, in the color of each group.

A click on an upper or lower cell takes that pair to the section The chosen pair (Section 3.6) and frames it in red in both of its cells; a click on the diagonal takes that variable to Each variable.

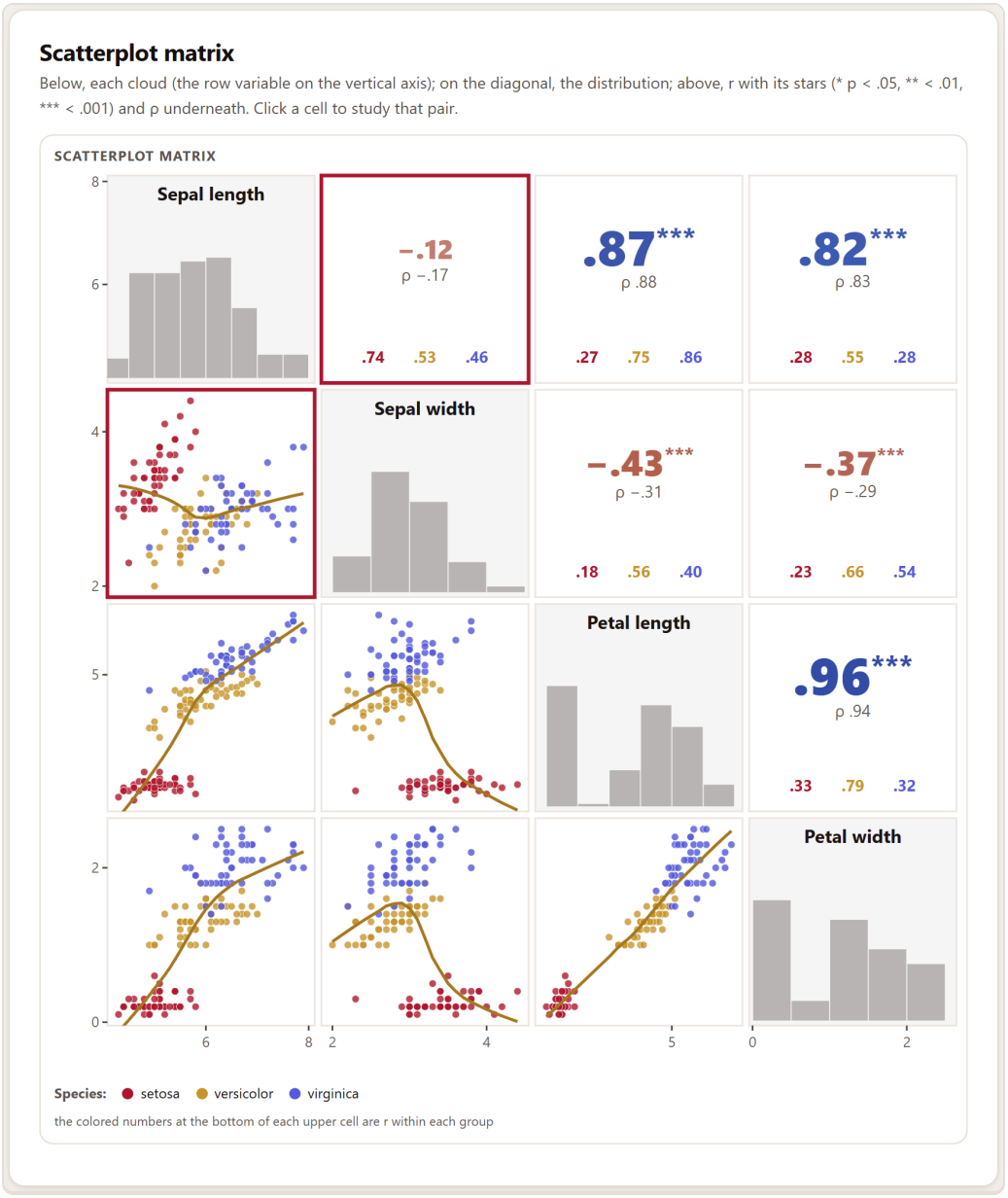


Figure 3.2. The matrix of the irises by species. Between sepal length and sepal width (framed cells), $r = -.12$, with no stars; within each species, .74, .53 and .46. Pooling the three species reverses the sign: Simpson's paradox, at a glance.

WORKED EXAMPLE · WHAT THE MATRIX SHOWS ABOUT THE IRISES

The histogram of petal length has a gap: the setosas, with short petals, sit apart from the other two species. Pooling everything, petal length and petal width give $r = .96$; within each species, .33, .79 and .32. Almost all of that very high correlation comes from the species differing in size, not from flowers with long petals also having wide ones within a species.

3.3 The shape of each variable

The table **Each variable** gives, for each one, the number of values, the mean, the standard deviation (SD), the skewness g_1 and the excess kurtosis g_2, the Shapiro–Wilk test (W and its p-value), the Anderson–Darling test (A^2 and its p-value), a miniature Q–Q plot and a **Reading** in words. The p-values are red below .01, amber between .01 and .05 and green from .05 up; g_1 is amber if it exceeds 1 in absolute value, and g_2 if it exceeds 2.

A click on a row (or `Enter` on it) draws below it the histogram of that variable with the normal curve of the same mean and SD, and its normal Q–Q plot: if the data were normal, the points would follow the dotted diagonal.

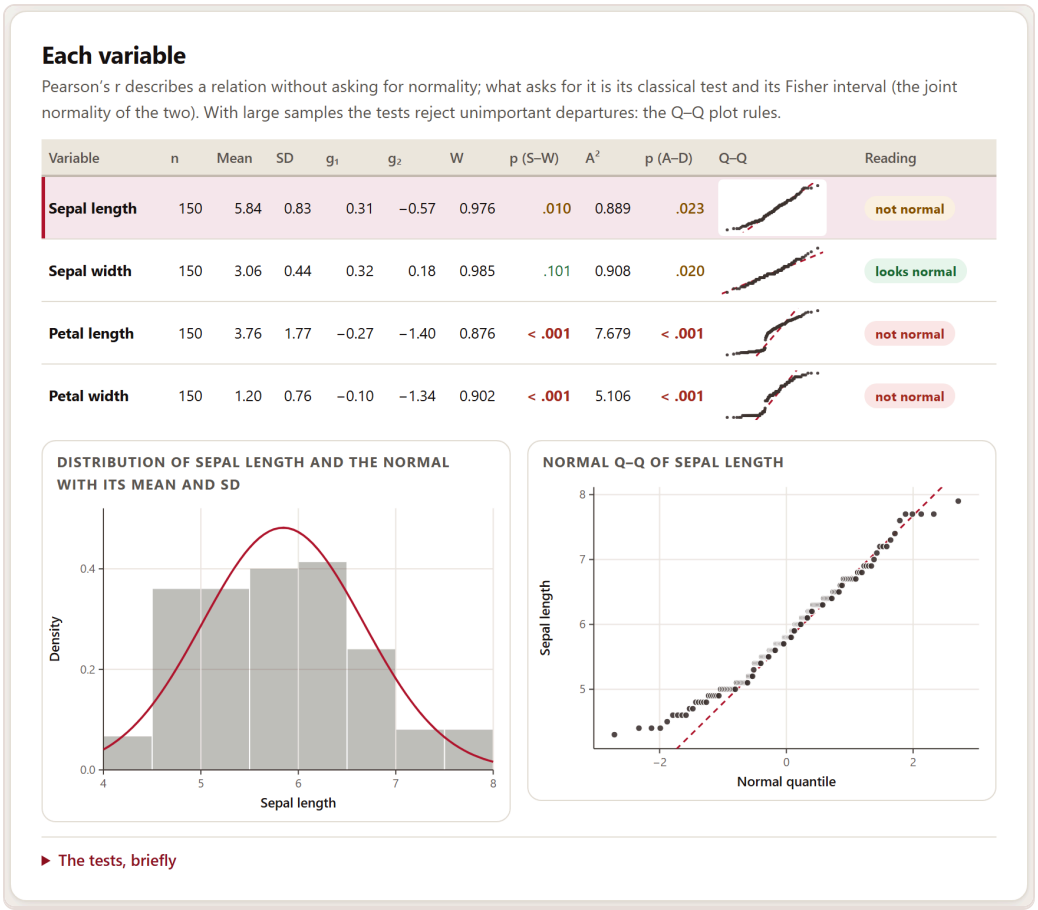


Figure 3.3. The four measurements of the irises. Sepal length gives $W = 0.976$ and $p = .010$, in amber: "not normal", although its Q–Q plot departs only at the ends. Sepal width "looks normal" ($p = .101$) even though Anderson–Darling gives .020: the reading follows Shapiro–Wilk. The two petal measurements, with $p < .001$, have a stepped Q–Q plot: the setosas, apart.

Table 3.1 says when each reading appears. The non-normal ones are red if $p < .01$ and amber if p is between .01 and .05.

Table 3.1. The reading of each variable

Reading	When it appears
looks normal	the Shapiro–Wilk p-value is .05 or more
departs little (large n)	$p < .05$, but with more than 300 values, $\lvert g_1 \rvert \le 1$ and $\lvert g_2 \rvert \le 2$: the test detects a departure that does not matter
right-skewed · left-skewed	$p < .05$ and $g_1 > 1 \cdot g_1 < -1$
heavy tails · light tails	$p < .05$, $\lvert g_1 \rvert \le 1$ and $g_2 > 2 \cdot g_2 < -2$
not normal	$p < .05$ without a large skewness or kurtosis to explain it (for example, a mixture of groups)

! PEARSON'S R DOES NOT NEED NORMALITY TO DESCRIBE

As a summary of how well a cloud lines up, r assumes nothing. What needs normality—that of the two variables together—is its classical test and its Fisher interval. And with large samples, normality tests reject departures of no practical importance: the Q–Q plot rules. If in doubt, Block 4 gives the permutation p-value and the bootstrap interval, which do not depend on normality.

THEORY IN BRIEF · THE NORMALITY TESTS

Shapiro–Wilk (1965) compares the ordered data with what would be expected from a normal; W close to 1 is normal. Its p-value follows Royston's approximation (1995), valid from 3 to 5,000 values; outside that range the table shows "—". **Anderson–Darling** measures the distance between the cumulative distribution of the data and the fitted normal, with more weight on the tails; its p-value is Stephens's (D'Agostino and Stephens 1986) and needs at least 8 values. The **skewness** g_1 and the excess **kurtosis** g_2 say how it departs: $\lvert g_1 \rvert > 1$ or $\lvert g_2 \rvert > 2$ already matter for the test of r.

3.4 Joint normality and multivariate outliers

Each variable being normal is not enough: the multivariate normal also requires all their combinations to be normal. This card uses the rows complete in all the chosen variables (it needs at least two variables and four more complete rows than variables) and shows five figures:

- **Complete rows**, with the number of variables.
- **Mardia skewness**, with its χ^2 and its $p(p+1)(p+2)/6$ degrees of freedom (corrected with fewer than 20 rows).
- **Mardia kurtosis**, with its z.
- **Henze–Zirkler**, an omnibus test that detects any departure from normality.
- **Robust outliers**: the rows beyond the cut with the robust distance, and how many with the classical one.

Mardia and Henze–Zirkler are computed with up to 2,500 rows. The colored edge of each figure follows the scale of the p-values of the previous table: red, amber or green.

The plot compares the squared Mahalanobis distances of each row—the classical one in gray, the robust one in red—with the quantiles of χ^2 with as many degrees of freedom as variables; under a normal, the points follow the diagonal. The horizontal dotted line is the $\chi^2(0.975)$ cut. Beside it, the table Farthest rows (robust distance) gives the eight rows that most exceed the robust cut, with their two distances, and how many more there are. The rows in gray italics are the ones the classical distance does not see: **masking**.

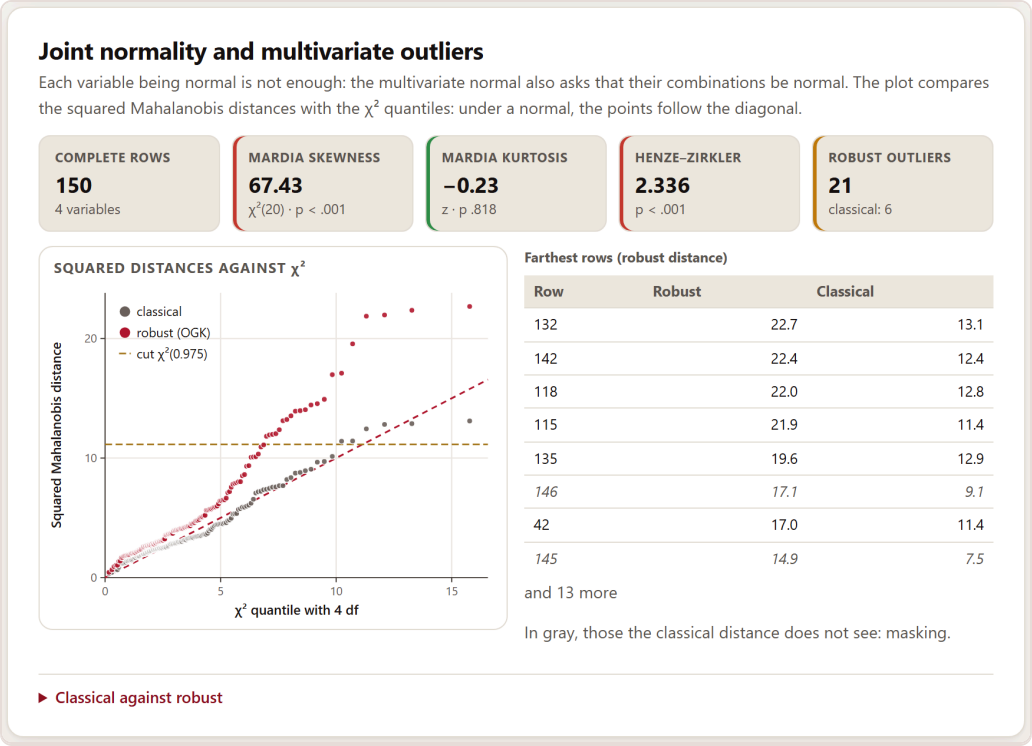


Figure 3.4. The four measurements of the irises together. Mardia skewness (67.43 with 20 degrees of freedom) and Henze–Zirkler (2.336) give $p < .001$: the set is not normal, even though the kurtosis looks fine ($p = .818$). 21 rows pass the cut (11.14) with the robust distance and only 6 with the classical one; rows 146 and 145 are seen only by the robust one.

THEORY IN BRIEF · CLASSICAL AGAINST ROBUST

The classical distance uses the mean and covariance of all the data, and outliers inflate them until they hide: this is the masking effect. The robust one uses the orthogonalized Gnanadesikan–Kettenring estimator (Maronna and Zamar 2002) with τ-scales and a reweighting that leaves out whatever exceeds the cut, so that outliers stay far away. **Mardia** (1970) tests multivariate skewness and kurtosis; **Henze and Zirkler** (1990) compare the characteristic function of the data with that of the normal, and their test is consistent against any alternative.

> **AN OUTLIER IS NOT AN ERROR**
>
> Before removing a row, look at it: it may be a data-entry error, a missing-value code or a real and valuable observation. And when different groups are mixed, like the three species of irises, the "outliers" may simply be a group far from the others; in that case, the right thing is to analyze within the groups.

3.5 The assumptions of each pair

The table Each pair checks all the pairs at once, each with its rows complete in both variables (that is why its n may vary). For each pair it gives n, r and ρ, and four diagnostics:

- Curve: the p-value of adding a quadratic term to the line.
- Fan: that of the Breusch–Pagan test in its studentized form (Koenker 1981), which checks whether the spread changes along the line.
- Out: how many points lie outside the 97.5% robust ellipse.
- Largest |Δr|: the most r changes when a single point is removed.

From them, the table gives the Signals of Table 3.2 and a Recommended coefficient: the one that best describes the pair. A click on a row takes the pair to the next section. A pair with fewer than five complete rows says "too few complete pairs".

Each pair: assumptions and recommended coefficient

Curvature: the quadratic term over the line. Spread: studentized Breusch–Pagan. Influence: the most r changes when a single point is removed. Click a row to see the pair.

Pair	n	r	ρ	Curve	Fan	Out	Largest |Δr|	Signals	Recommended
Sepal length Sepal width	150	−.12	−.17	.057	.098	8	0.033	not normal, Simpson	By group
Sepal length Petal length	150	.87	.88	< .001	.077	4	0.008	curved, not normal	Spearman's ρ
Sepal length Petal width	150	.82	.83	< .001	.419	12	0.010	curved, not normal, groups	By group
Sepal width Petal length	150	−.43	−.31	.013	.037	8	0.028	not normal, Simpson	By group
Sepal width Petal width	150	−.37	−.29	.007	.018	8	0.021	curved, not normal, Simpson	By group
Petal length Petal width	150	.96	.94	.732	< .001	14	0.002	fan, not normal, groups	By group

Figure 3.5. The six pairs of the irises with the species as group. Five call for analyzing By group: in three the relation changes sign within the species (Simpson) and in two pooling the species inflates it. Only sepal length with petal length, which curves, gets Spearman's ρ.

Table 3.2. The signals of each pair

Signal	Appears when…
curved	the quadratic term gives $p < .01$
fan	Breusch–Pagan gives $p < .01$
not normal	Shapiro–Wilk gives $p < .05$ in at least one of the two variables
influential point	the largest $\lvert\Delta r\rvert$ exceeds the limit of Section 3.6 and at least one point lies outside the robust ellipse
few values	one of the two takes ten distinct values or fewer
not monotonic	the distance correlation exceeds $\lvert\rho\rvert$ by more than 0.25: the cloud rises and falls (computed with up to 2,000 pairs)
Simpson	with a group: in at least half of the groups with five cases or more, r has the opposite sign and $\lvert r\rvert \geq .3$
groups	with a group and no Simpson: the mean r within groups differs by 0.3 or more from r pooling everything

The recommendation follows a fixed order: the first rule that holds wins. The reasons for the order are common sense: if the groups change the relation, no coefficient computed by pooling everything describes it; if the cloud rises and falls, no monotonic coefficient sees it; and only when the cloud is straight does it make sense to ask about points that decide on their own.

Table 3.3. How the recommended coefficient is chosen

#	If…	it recommends
1	there is Simpson or a group effect	**By group**: analyze within the groups (Block 10)
2	the relation is not monotonic	**dCor or ξ**: the distance correlation or Chatterjee's ξ (Block 6)
3	at least one of the two is ordinal	**Spearman's ρ**, or **Kendall's τ-b** if there are ten distinct values or fewer
4	there is curvature, or r and ρ differ by 0.1 or more	**Spearman's ρ** (or τ-b, with few values)
5	there is an influential point	**Percentage bend**, a robust correlation (Block 4)
6	none of the above	**Pearson's r**; if one of the two is not normal or there is fanning, with the note to use the permutation or the bootstrap of Block 4

The recommendation label is green when it is Pearson's r with no notes, amber when it is Pearson with a note and gold when the pair calls for another coefficient.

▸ WORKED EXAMPLE · FIFTEEN PAIRS FROM AN ECONOMY

The six annual series of the United States economy from 1947 to 1962 (Figure 3.6) give the three kinds of answer. Eight pairs get Pearson's r. Three get ρ: two because they curve, and unemployed with armed forces because $r = -.18$ and $\rho = -.34$ differ by more than 0.1. And four of the five pairs with the armed forces, which grow suddenly in 1951 with the Korean War and then fall slowly, are "not monotonic": dCor or ξ.

Each pair: assumptions and recommended coefficient

Curvature: the quadratic term over the line. Spread: studentized Breusch–Pagan. Influence: the most r changes when a single point is removed. Click a row to see the pair.

Pair	n	r	ρ	Curve	Fan	Out	Largest |Δr|	Signals	Recommended
GNP deflator GNP	16	.99	1.00	.110	.287	3	0.003		Pearson's r
GNP deflator Unemployed	16	.62	.66	.056	.400	0	0.094		Pearson's r
GNP deflator Armed forces	16	.46	.22	< .001	.118	4	0.116	curved, not monotonic	dCor or ξ
GNP deflator Population (≥ 14 years)	16	.98	1.00	.001	.442	2	0.004	curved	Spearman's ρ
GNP deflator Employed	16	.97	.98	.894	.736	1	0.009		Pearson's r
GNP Unemployed	16	.60	.64	.179	.428	0	0.099		Pearson's r
GNP Armed forces	16	.45	.22	.002	.074	4	0.106	curved, not monotonic	dCor or ξ
GNP Population (≥ 14 years)	16	.99	.99	.003	.961	3	0.001	curved	Spearman's ρ
GNP Employed	16	.98	.99	.297	.538	0	0.006		Pearson's r
Unemployed Armed forces	16	−.18	−.34	.283	.066	2	0.152		Spearman's ρ
Unemployed Population (≥ 14 years)	16	.69	.69	.555	.443	0	0.082		Pearson's r
Unemployed Employed	16	.50	.56	.756	.559	0	0.109		Pearson's r
Armed forces Population (≥ 14 years)	16	.36	.23	< .001	.555	4	0.109	curved, not monotonic	dCor or ξ
Armed forces Employed	16	.46	.23	< .001	.333	4	0.094	curved, not monotonic	dCor or ξ
Population (≥ 14 years) Employed	16	.96	.98	.043	.626	0	0.010		Pearson's r

Figure 3.6. The United States economy from 1947 to 1962: sixteen years, six variables and fifteen pairs, with the three kinds of recommendation.

3.6 The chosen pair, up close

The last card studies one pair: the one chosen in the matrix or in the table, or the one set in X and Y. The orientation matters for the line, the residuals and the Breusch–Pagan test, which are of Y on X; r, ρ and the recommendation are symmetric.

Seven figures sum up the pair: n (and how many rows do not have the pair complete), r with its p-value, ρ, the distance correlation dCor ("any shape": in the population it is zero only if the two are independent), the Curvature with its F, the Spread (BP) with its χ^2 and the Largest |Δr| with the row that causes it and the limit. Below, For this pair gives the recommendation with its reason and its notes; if the recommendation is a measure from the catalog, What is it? opens its card.

Three plots show why:

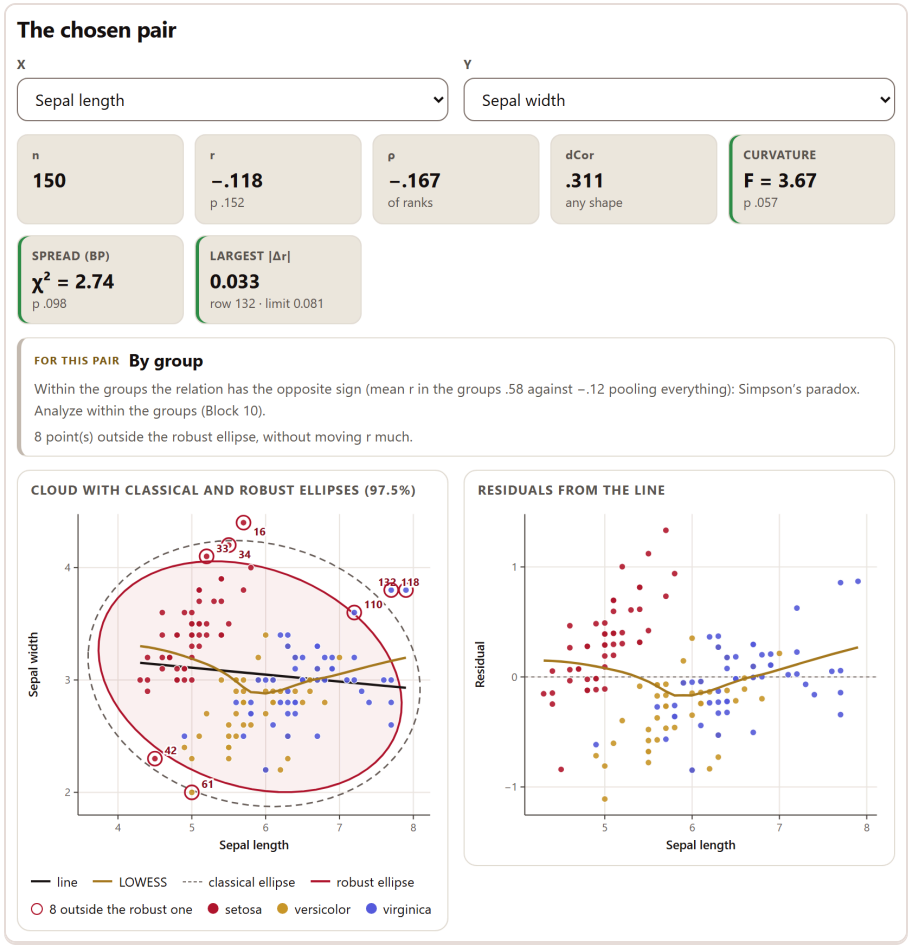


Figure 3.7. Sepal length and sepal width. Neither curvature (p = .057) nor fanning (p = .098), and no point decides (0.033, below the limit of 0.081); but within the species the mean r is .58 against −.12 pooling everything, and the recommendation is By group. The eight points outside the robust ellipse carry their row number.

- **The cloud with its ellipses.** The line in black, the LOWESS in gold, the classical 97.5% ellipse dashed and the robust one in red. The points outside the robust ellipse are circled, and the eight farthest carry their row number.
- **The residuals from the line**, with their own LOWESS. If the gold line curves, the relation curves; if the points open out toward one side, there is fanning.
- **The influence.** For each point, in row order, how much r changes if it is removed. The dashed lines mark the limit, and the four points that move r the most carry their row.

The influence limit is, roughly, one standard error of r: a point that by itself moves r more than that decides too much.

$$\text{limit} = \max\left(0.02\,,\ (1 - r^2) \,/\, \sqrt{(n-1)}\right)$$

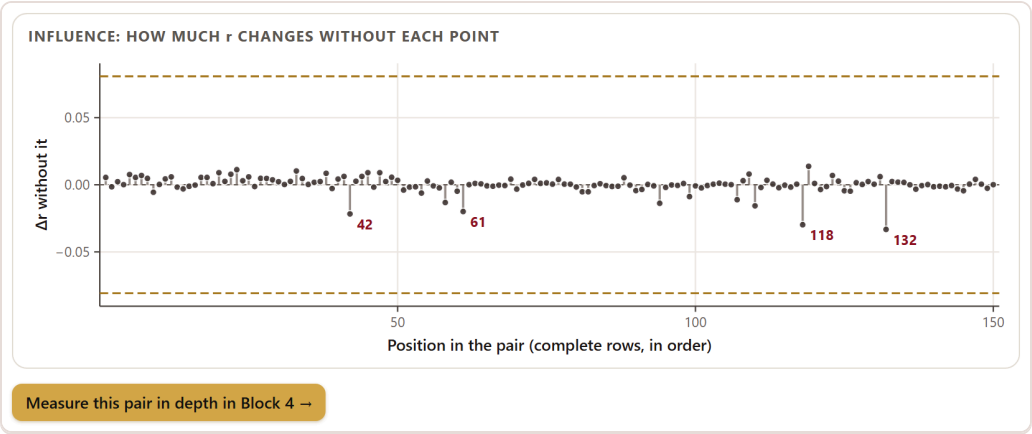


Figure 3.8. The influence in the same pair. The point that moves *r* the most, row 132, changes it by 0.033, less than half the limit: no flower decides on its own.

Measure this pair in depth in Block 4 → takes the pair to the block of single-pair measures, with all its coefficients, intervals, permutations and bootstrap. There, the coefficient studied in depth is, until you choose another, the one recommended here; with By group it is Pearson's *r*, because the within-group analysis belongs to Block 10.

WORKED EXAMPLE · A PAIR THAT CALLS FOR PEARSON, WITH A NOTE

The speed and braking distance of 50 cars (Ezekiel 1930): $r = .807$ and $\rho = .830$; neither curvature ($p = .136$) nor fanning ($p = .073$), and the most influential point, row 23, moves *r* by 0.027, below the limit of 0.050. The recommendation is Pearson's *r*, but braking distance is not normal, so the note asks for the permutation *p*-value or the bootstrap interval.

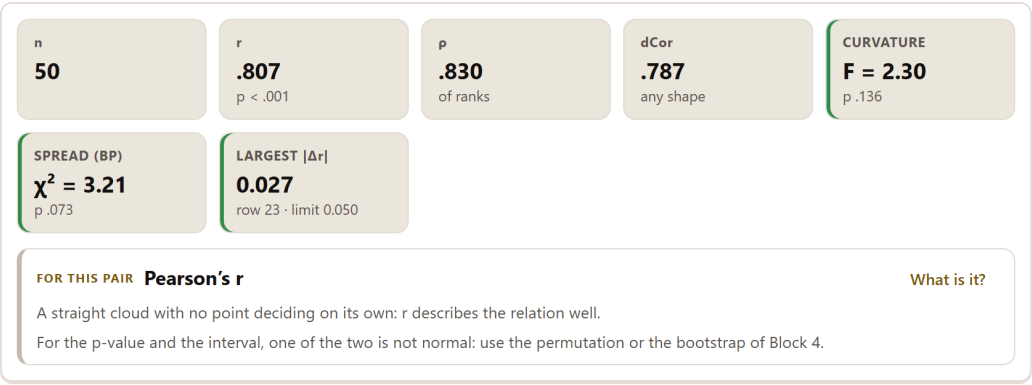


Figure 3.9. Ezekiel's cars: the three tests in green and the recommendation Pearson's r, with its note.

THEORY IN BRIEF · WHY EACH ASSUMPTION

Linearity. *r* measures only the straight part of the relation; if the cloud curves but keeps rising, ρ or τ measure all of it; if it rises and falls, neither *r* nor ρ sees it (Anscombe 1973). **Constant spread.** When the cloud opens like a fan, the standard error of *r* given by the classical formula is wrong; the permutation *p*-value and the bootstrap interval do not depend on it. **Outliers and influence.** A single point can create or erase a correlation; if *r* changes by more than one standard error when it is removed, that point decides too much. **Groups.** Pooling groups with different means can inflate, hide or reverse the relation (Simpson 1951). The app carries this same explanation in the collapsible note Why each assumption, at the foot of the card.

3.7 From Block 3 to the others

The block computes nothing for publication: it decides what to compute. What it finds is used by Block 4, which opens with the recommended coefficient, and by the drafted methods of Block 13, which describe the tests used to check the assumptions and cite their references.

BEFORE LEAVING BLOCK 3

If the recommendation is By group…	analyze within the groups in Block 10, and do not publish the pooled *r* without saying so.
If it is dCor or ξ…	look at the cloud: the relation rises and falls. Measure it in Block 6; *r* and ρ near zero do not mean independence.
If it is Spearman's ρ or Kendall's τ-b…	use it in Block 4; if you report *r* anyway, say that the relation curves.
If it is the Percentage bend…	find the point in the cloud and work out what it is before deciding; compare with the robust coefficients of Block 4.
If it is Pearson's *r* with a note…	give the permutation *p*-value or the bootstrap interval from Block 4.
If the normality tests reject with a lot of data…	look at the Q–Q plot before switching coefficients.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

4

CHAPTER FOUR · BLOCK 4

One pair in depth

Two numeric variables, and everything that can be said about their relation: twelve coefficients with their tests and intervals, the bootstrap and the permutation of the main coefficient, the Bayes factor of Pearson's *r* and a reading in words that ends in a sentence ready for the report.

4.1 The pair and the question

4.2 The result, in figures and in words

4.3 The cloud and its ranks

4.4 Every coefficient, side by side

4.5 Bootstrap and permutation

4.6 The Bayes factor

4.7 How it was computed

4.1 The pair and the question

The pair arrives in three ways: from the button Measure this pair in depth in Block 4 → of Block 3, from the assistant of Block 2, or by choosing it here in X and Y. The menus offer the continuous and ordinal columns that are not marked as "not used".

The first card gathers the eight decisions of the analysis. Every change recomputes everything; if the calculation takes a while (many rows or many resamples), a wait indicator appears.

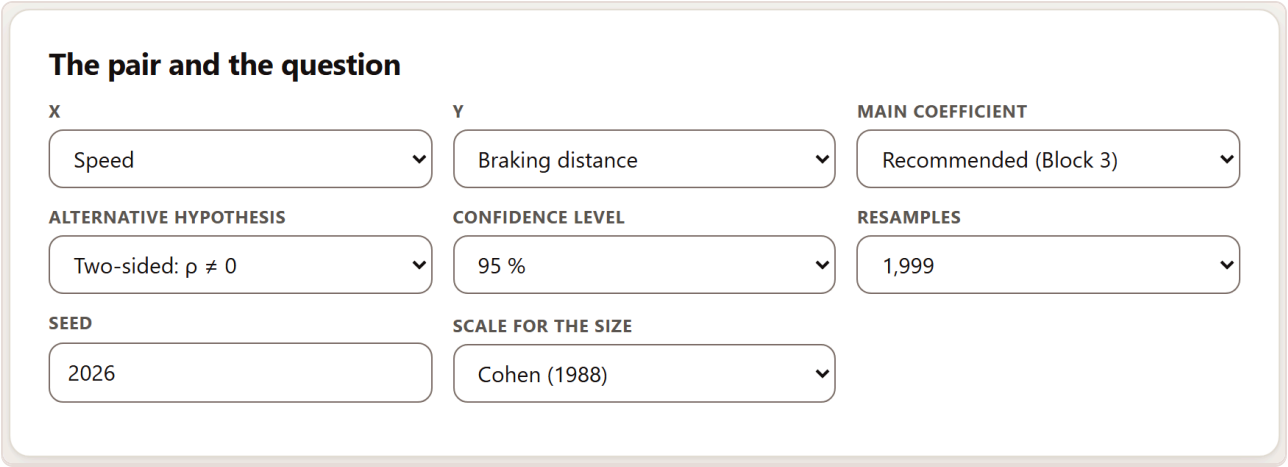


Figure 4.1. Ezekiel's cars, the example the block opens with, with the default options.

- Main coefficient: the one studied in depth. Recommended (Block 3) lets the assumption check of Section 3.5 choose it; you can also fix any of the twelve of Section 4.4.
- Alternative hypothesis: Two-sided: ρ ≠ 0, Positive: ρ > 0 or Negative: ρ < 0. Choose a one-sided one only if the direction was decided before seeing the data. The dependence measures have no sign, and their test is always for large values.
- Confidence level: 90, 95 or 99% for all the intervals; its complement (10, 5 or 1%) is the significance level of the reading.
- Resamples: 999, 1,999, 4,999 or 9,999 for the bootstrap and for the permutation. With a costly coefficient and many data, the app uses fewer and says so.
- Seed: the starting point of the random number generator. The same seed gives exactly the same numbers; write it down if you are going to publish resampling results.
- Scale for the size: the words the reading uses to rate the strength of the relation (Table 4.1).

Table 4.1. The three scales for the size of a correlation (by $|r|$)

Scale	Cuts and words
Cohen (1988)	negligible < .10 ≤ small < .30 ≤ medium < .50 ≤ large
Evans (1996)	very weak < .20 ≤ weak < .40 ≤ moderate < .60 ≤ strong < .80 ≤ very strong
Gignac and Szodorai (2016)	very small < .10 ≤ small < .20 ≤ typical < .30 ≤ large

WHICH SCALE TO CHOOSE

Cohen's and Evans's are general conventions. Gignac and Szodorai's comes from the correlations actually published in psychology: there, an r of .30 is already large. The best thing is to compare with what is known about the phenomenon in your discipline, and to say in the report which scale you used.

4.2 The result, in figures and in words

The Result card sums up the analysis in six figures: the main coefficient with the word for its size (and the mark chosen by Block 3 if the assumption check chose it), its confidence interval with the method that produced it, its own test with the statistic and the degrees of freedom, the permutation *p*-value, the bootstrap interval and the Bayes factor BF_{10}. In the tests, a green edge means significant at the chosen level and amber, not; in the Bayes factor, green is evidence for the correlation and amber, for $\rho = 0$.

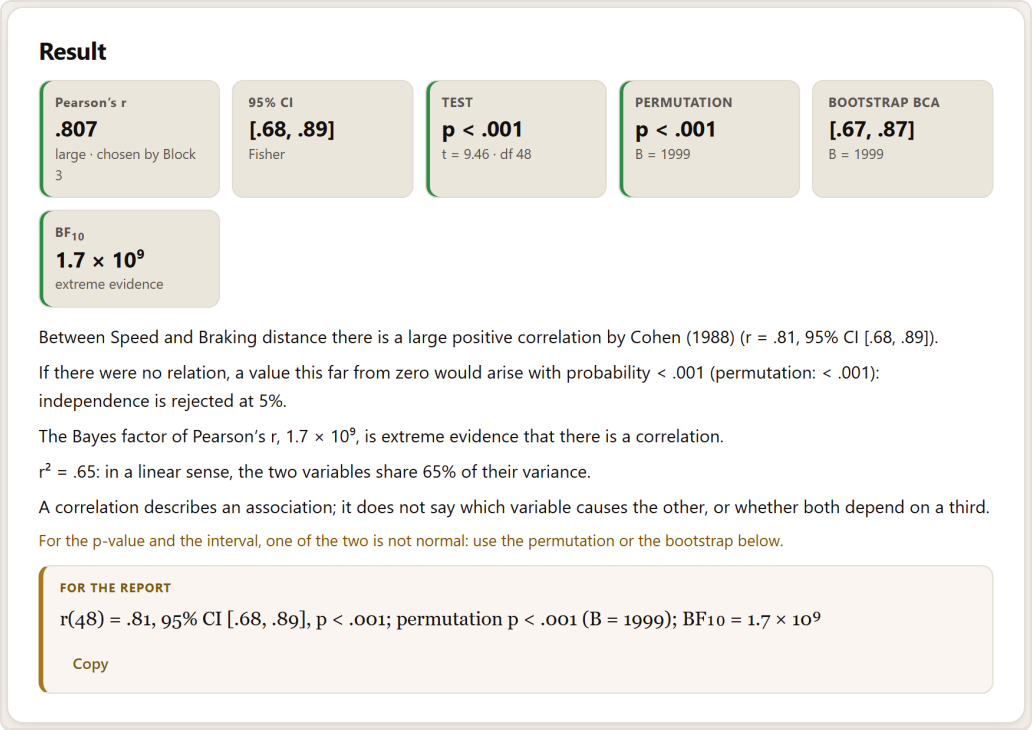


Figure 4.2. Ezekiel's cars: $r = .807$, "large" by Cohen, with Fisher interval [.68, .89], $t = 9.46$ with 48 degrees of freedom, permutation and BCa interval in agreement and a Bayes factor of 1.7×10^9. The line in amber is the note from Block 3.

Below, the reading says the same thing in words, in five short paragraphs:

1. **The direction and the size**, with the interval: "there is a large positive correlation by Cohen (1988)". For a dependence measure, which has no sign, it gives its value on the scale from 0 (independence) to 1.
2. **The test**: how likely a value like this would be if there were no relation, with the permutation one in parentheses, and whether independence is rejected. If it is not rejected, it says so carefully: the data are not enough to rule it out, which does not prove that there is no relation.
3. **The Bayesian evidence** of Pearson's *r*, in the words of Section 4.6.

4 **The shared variance**, only with Pearson: $r^2 = .65$ means that, in a linear sense, the two variables share 65% of their variance.

5 **The reminder** that a correlation does not say which variable causes the other, or whether both depend on a third.

The notes go in amber: those from Block 3 for this pair and, if you chose a coefficient other than the recommended one, which one Block 3 recommends and why.

For the report gives the sentence in the number style chosen in Block 1—"$r(48) = .81$, 95% CI [.68, .89], $p < .001$; permutation $p < .001$ (B = 1999); $BF_{10} = 1.7 \times 10^9$"—and Copy puts it on the clipboard.

WORKED EXAMPLE · WHEN THE MAIN COEFFICIENT IS A DEPENDENCE MEASURE

In the 1947–1962 economy, Block 3 recommends the distance correlation for armed forces with employed, because their cloud rises and then falls (Section 3.5). Here, dCor = .76 with BCa interval [.68, .86] and $p < .001$ both ways. The Bayes factor, which is always that of Pearson's r, gives only 1.70: anecdotal evidence. The two figures do not contradict each other: the relation is strong, but it is not a straight line.

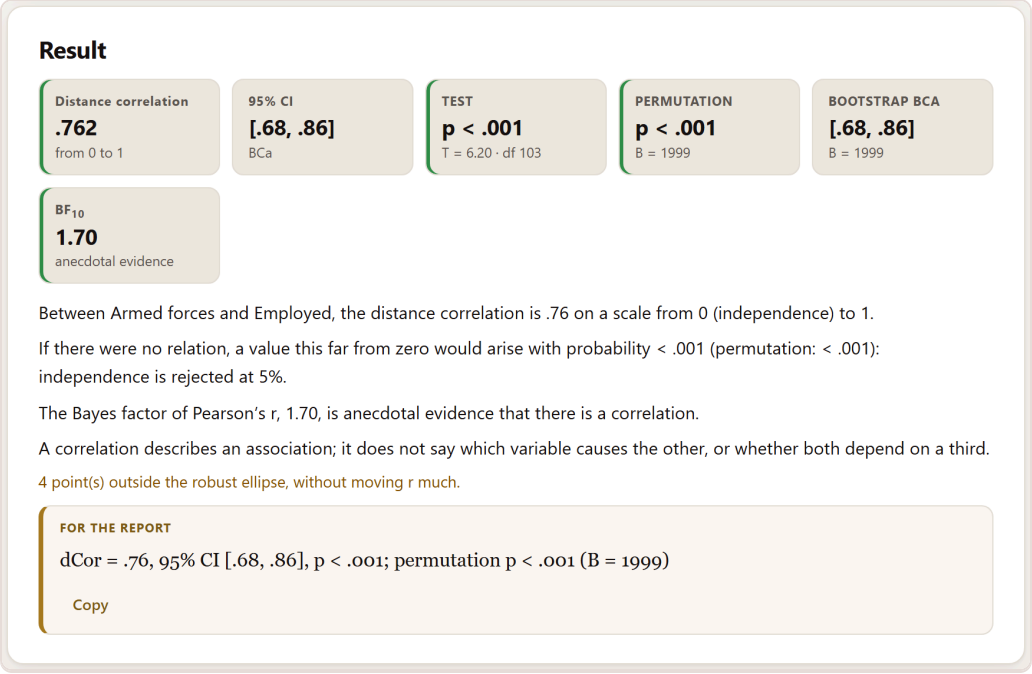


Figure 4.3. The distance correlation as the main coefficient. For a measure without sign, the reading gives its value on the scale from 0 to 1 instead of a direction and a size.

4.3 The cloud and its ranks

Two plots of the same pair. On the left, the data with the 95% ellipse, the line in black, the LOWESS in gold and, along the axes, a tick for each value. On the right, the same points turned into ranks (divided by $n + 1$, so that they run from 0 to 1): this is what Spearman's ρ and Kendall's τ see. A monotonic but curved relation becomes straight in the ranks; a point very far away in the data becomes just one more in the ranks.

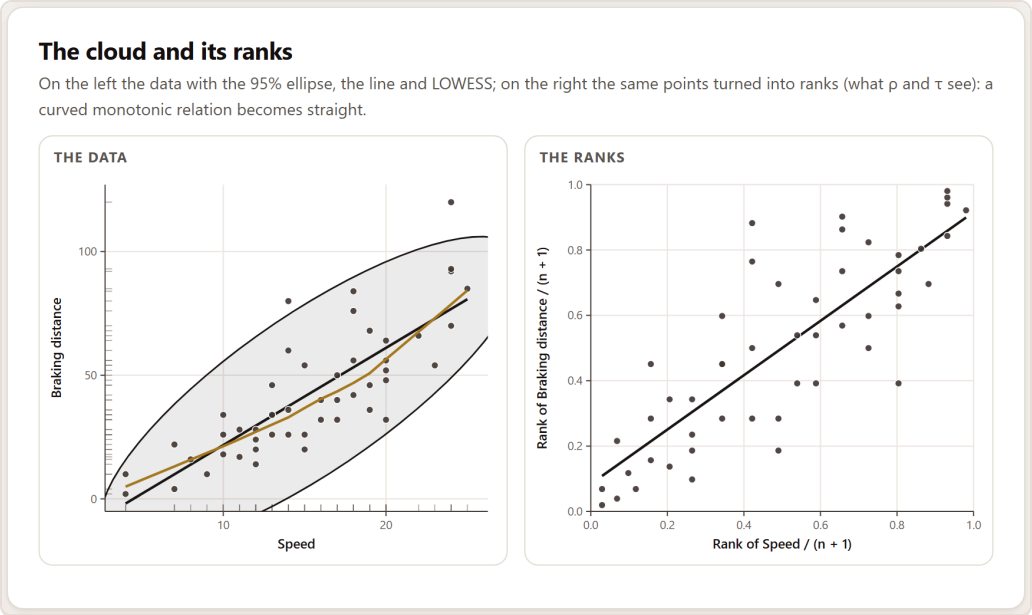
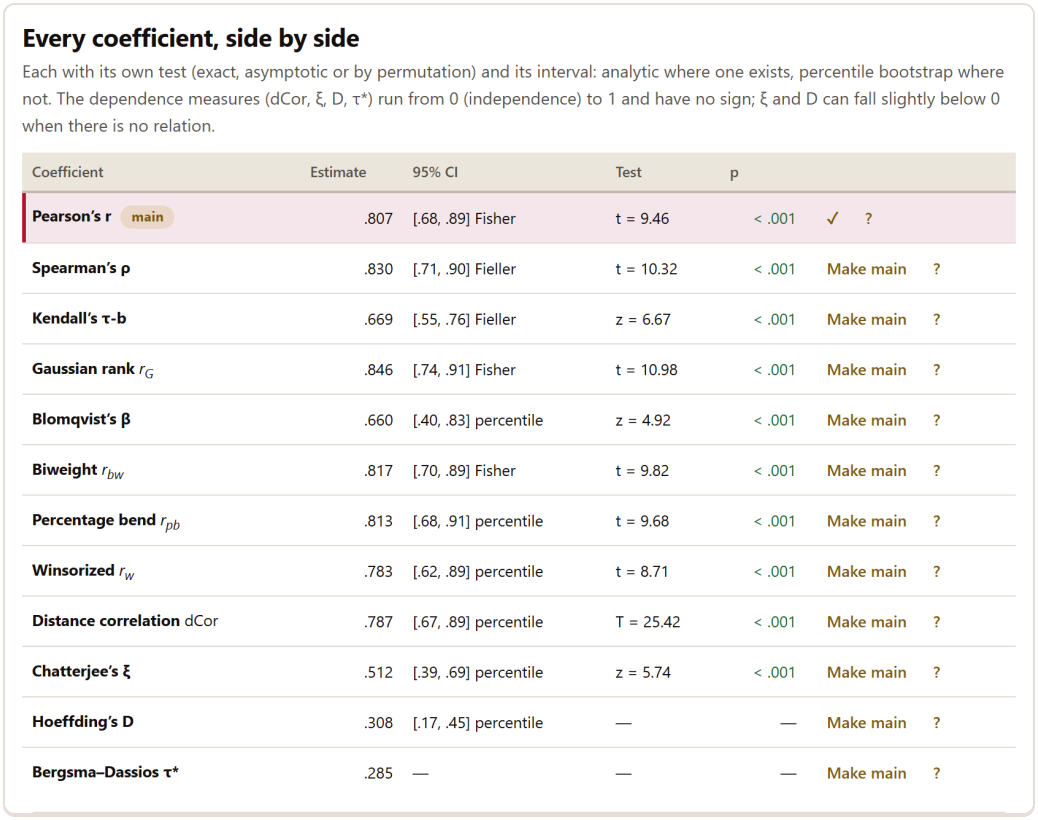


Figure 4.4. Ezekiel's cars. In the data, distance grows somewhat faster than the line at high speeds (the LOWESS curves upward) and spreads out more; in the ranks the curve disappears and the cloud is straight.

4.4 Every coefficient, side by side

The table computes the twelve coefficients that apply to two numeric variables, each with its own test—exact, asymptotic or by permutation—and its interval: the analytic one where it exists and, where not, a percentile bootstrap with 499 resamples. The main one carries the label `main`; `Make main` changes the main coefficient without going back to the menu, and `?` opens its card in the catalog. Significant p-values are in green.

> **DO NOT CHOOSE THE COEFFICIENT AFTER SEEING THE TABLE**
>
> With twelve coefficients in view, it is tempting to report the one that gave the highest value or the smallest p. That inflates the results. Choose the coefficient by the shape of the data—which is what Block 3 does—or before seeing them, and report that one; the table serves to see whether the conclusion depends on the choice.



Every coefficient, side by side

Each with its own test (exact, asymptotic or by permutation) and its interval: analytic where one exists, percentile bootstrap where not. The dependence measures (dCor, ξ, D, τ*) run from 0 (independence) to 1 and have no sign; ξ and D can fall slightly below 0 when there is no relation.

Coefficient	Estimate	95% CI	Test	p	
Pearson's r main	.807	[.68, .89] Fisher	t = 9.46	< .001	✓ ?
Spearman's ρ	.830	[.71, .90] Fieller	t = 10.32	< .001	Make main ?
Kendall's τ-b	.669	[.55, .76] Fieller	z = 6.67	< .001	Make main ?
Gaussian rank r_G	.846	[.74, .91] Fisher	t = 10.98	< .001	Make main ?
Blomqvist's β	.660	[.40, .83] percentile	z = 4.92	< .001	Make main ?
Biweight r_{bw}	.817	[.70, .89] Fisher	t = 9.82	< .001	Make main ?
Percentage bend r_{pb}	.813	[.68, .91] percentile	t = 9.68	< .001	Make main ?
Winsorized r_w	.783	[.62, .89] percentile	t = 8.71	< .001	Make main ?
Distance correlation dCor	.787	[.67, .89] percentile	T = 25.42	< .001	Make main ?
Chatterjee's ξ	.512	[.39, .69] percentile	z = 5.74	< .001	Make main ?
Hoeffding's D	.308	[.17, .45] percentile	—	—	Make main ?
Bergsma–Dassios τ*	.285	—	—	—	Make main ?

Figure 4.5. The twelve coefficients for Ezekiel's cars. The rank and robust ones lie between .66 and .85; the dependence measures, on their own scale. Hoeffding's D and τ* have no test in the table.

Table 4.2. The twelve coefficients of Block 4

Coefficient	What it measures	Test · interval
Pearson's *r*	the linear relation	*t* with $n - 2$ df · Fisher
Spearman's ρ	the monotonic relation: Pearson on the ranks	exact without ties, otherwise *t* · Fieller
Kendall's τ-b	the concordance of pairs of observations, corrected for ties	exact without ties ($n < 50$), otherwise *z* · Fieller
Gaussian rank r_G	Pearson on the normal scores of the ranks	*t* · Fisher
Blomqvist's β	the concordance with respect to the two medians	*z* · percentile
Biweight r_{bw}	robust: gives less weight to data far from the median	*t* · Fisher
Percentage bend r_{pb}	robust: bounds the farthest 20% of each variable	*t* · percentile
Winsorized r_w	robust: Pearson after pulling 20% of each tail in to its limit	*t* with $n - 2g - 2$ df · percentile
Distance correlation dCor	any dependence; 0 only under independence	*T* with $n(n-3)/2 - 1$ df · percentile
Chatterjee's ξ	how much *Y* is a function of *X* (it is not symmetric)	*z* · percentile

Table 4.2. The twelve coefficients of Block 4 *(continued)*

Coefficient	What it measures	Test · interval
Hoeffding's D	the distance between the joint distribution and that of independence (scale 30·D)	— · percentile
Bergsma–Dassios τ*	any dependence, with a statistic over all groups of four points; only up to $n = 80$	— · —

The dependence measures (dCor, ξ, D, τ*) run from 0 to 1 and have no sign; ξ and D can fall slightly below zero when there is no relation. D and τ* have no test of their own in the table: when made main, the permutation gives their p-value. The distance correlation is computed with up to 3,000 pairs.

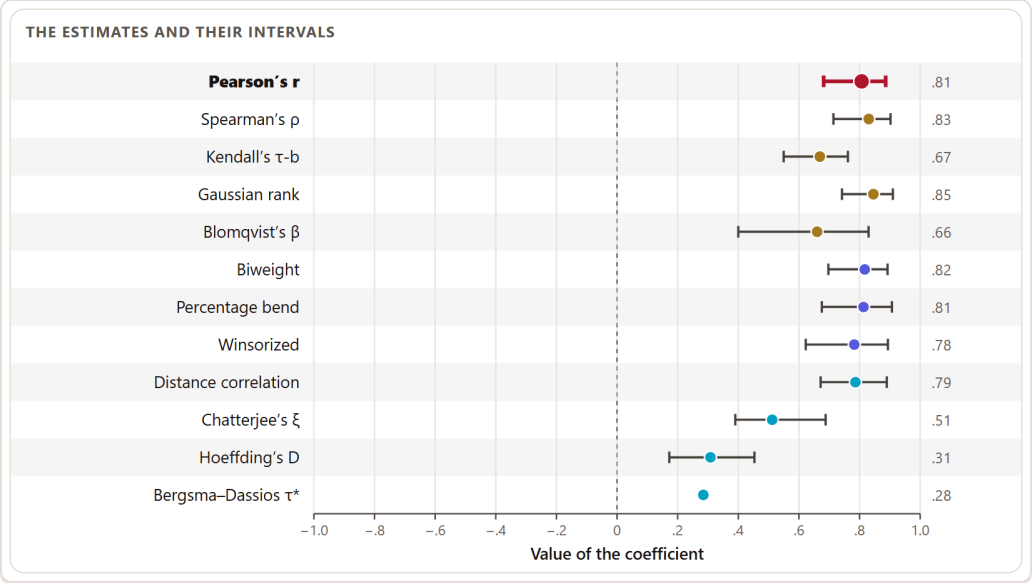


Figure 4.6. The twelve estimates with their intervals on a single scale from –1 to 1. The main one is in red and larger; the others, in the color of their family: rank, robust and dependence.

4.5 Bootstrap and permutation

Two ways of doing inference without assuming normality, both on the main coefficient. The **bootstrap** resamples the pairs with replacement and recomputes the coefficient in each resample; the spread of those replicates imitates that of the coefficient from one sample to another. The **permutation** shuffles Y against X, which breaks any relation and preserves everything else; its p-value is the proportion of shuffles that give a value as far from zero as the observed one.

Six figures sum up the bootstrap: its standard error, its bias (with the correction z_0 and the acceleration a of the BCa) and four intervals: percentile, basic, normal and BCa (the last one when the coefficient is not too costly to recompute). The two plots show the replicates: in the bootstrap one, the band is the BCa interval, the solid line the estimate and the dashed ones the percentile limits; in the permutation one, the solid line is the observed value and the dashed one its mirror image, which also counts in the two-sided test.

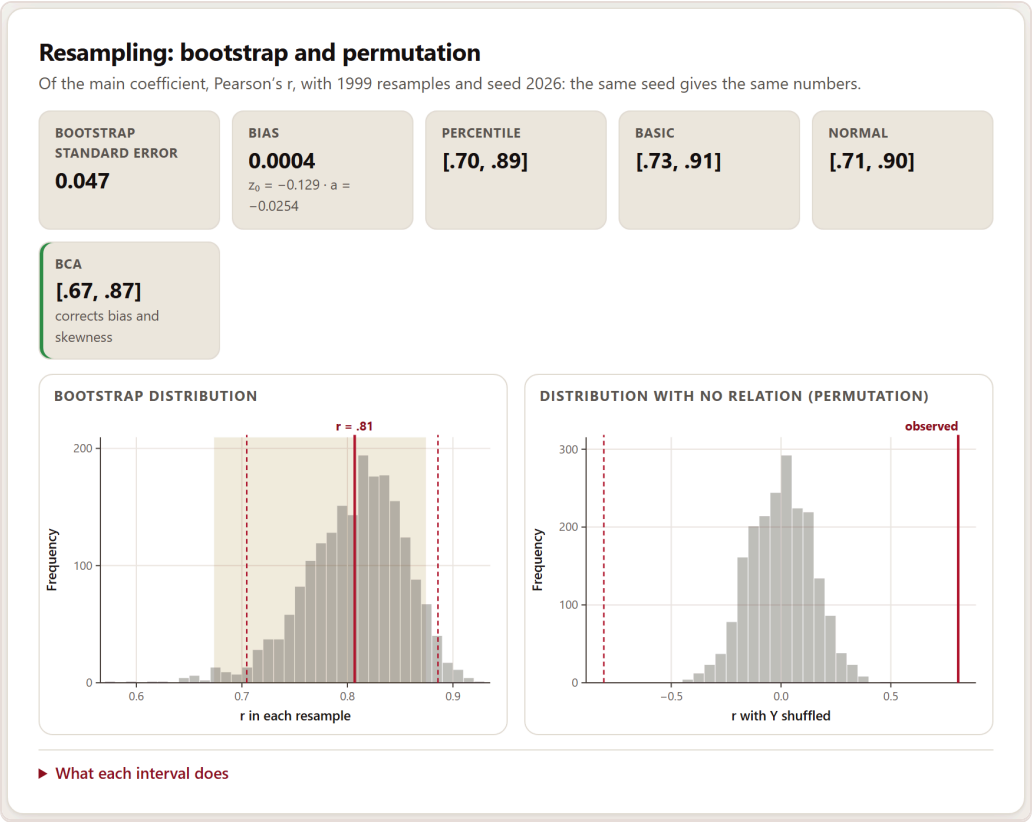


Figure 4.7. Ezekiel's cars with 1,999 resamples and seed 2026. The bootstrap distribution has a long left tail, and the BCa [.67, .87] shifts in that direction relative to the percentile [.70, .89]. None of the 1,999 shuffles comes close to .81.

THEORY IN BRIEF · WHAT EACH INTERVAL DOES

The **percentile** interval takes the quantiles of the replicates; the **basic** one reflects them around the estimate; the **normal** one uses their standard error; the **BCa** (Efron 1987) also corrects for bias (z_0) and for the spread changing with the value (the acceleration a, estimated with the jackknife). With normal data the Fisher interval covers what it promises; with very skewed data it falls short, and the bootstrap ones come closer to the requested level, without quite reaching it with small samples (Bishara and Hittner 2017). The permutation (Pitman 1937) gives $p = (k + 1)/(B + 1)$, where k is the number of shuffles as extreme as the observed value: it never gives zero.

HOW MANY RESAMPLES

With 1,999 the smallest possible p-value is 1/2000 = .0005, and the interval limits settle at the second digit. For a 99% interval or to publish a p-value close to .05, use 4,999 or 9,999.

4.6 The Bayes factor

The Bayes factor BF_{10} says how many times more probable the data are if there is a correlation than if $\rho = 0$; its reciprocal BF_{01}, the opposite. Unlike the *p*-value, it can give evidence that there is no relation. It is always computed for Pearson's *r*, whatever the main coefficient.

The prior distribution of ρ is a beta stretched to the interval from −1 to 1, and Prior (κ) controls its width: κ = 1/3, the common default, κ = 1/√3, wider or κ = 1, uniform (Jeffreys). The figures are BF_{10}, BF_{01}, the two one-sided ones (BF_{+0} for $\rho > 0$ and BF_{-0} for $\rho < 0$; with a one-sided alternative, only its own), the median and mean of the posterior of ρ and its central credible interval at the chosen level.

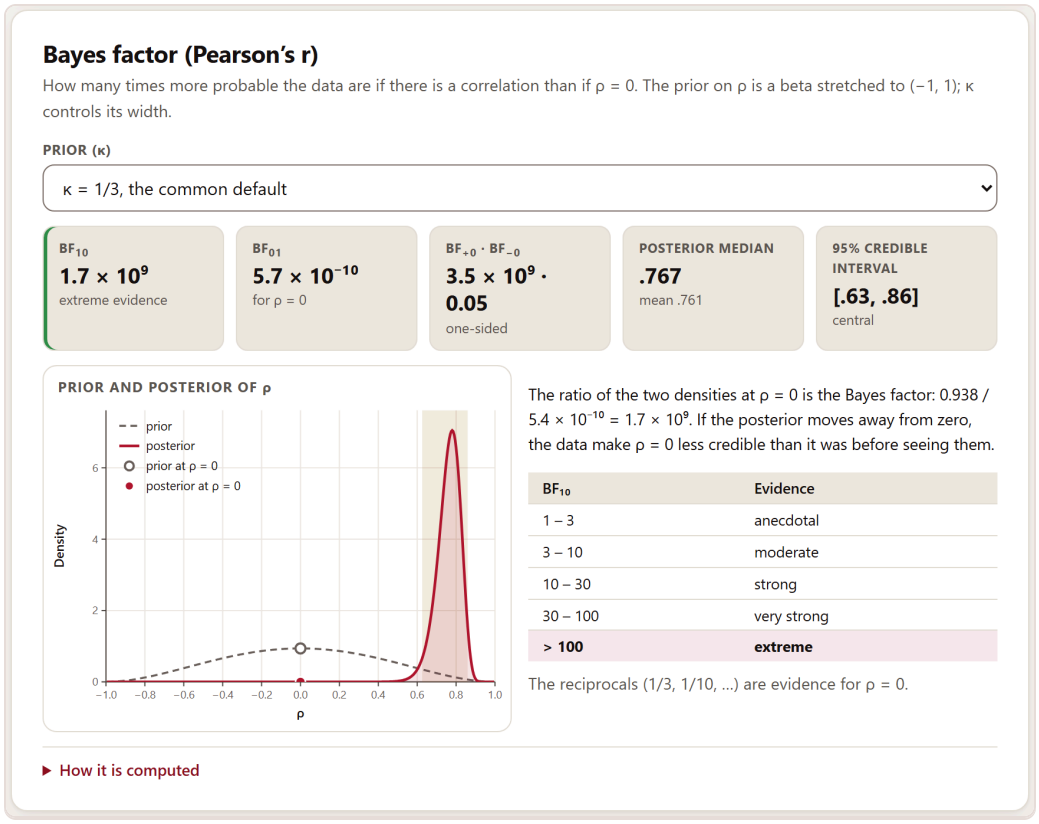


Figure 4.8. Ezekiel's cars. The posterior of ρ (in red) is concentrated around .77, with credible interval [.63, .86]; at ρ = 0 its density is practically nil against 0.938 for the prior, and the ratio of the two is the Bayes factor: extreme evidence.

Table 4.3. The words for the evidence (Jeffreys 1961, as given by Lee and Wagenmakers 2013)

BF_{10}	Evidence that there is a correlation
1 to 3	anecdotal
3 to 10	moderate
10 to 30	strong
30 to 100	very strong
more than 100	extreme

The reciprocals (from 1/3 to 1, from 1/10 to 1/3 and so on) are the same scale of evidence for ρ = 0.

> **THEORY IN BRIEF · HOW IT IS COMPUTED**
>
> With the means and variances integrated out under Jeffreys' priors, the likelihood of ρ depends only on *r* and *n* (Ly, Verhagen and Wagenmakers 2016). The Bayes factor is its average under the prior divided by its value at ρ = 0, and it equals the ratio of the prior and posterior densities at ρ = 0 (the Savage–Dickey ratio): these are the two dots on the plot. The app uses that exact form and integrates it numerically, with an adaptive method.

4.7 How it was computed

The last card lists the conventions of this calculation: the test and interval of the main coefficient, the bootstrap with its number of resamples and its seed, the permutation with its own (the seed plus one), the prior of the Bayes factor and how many rows were left out because of missing values. It is what the methods section needs, and what Block 13 drafts for you.

How it was computed

- t test with n − 2 degrees of freedom; interval by Fisher's z transformation with standard error 1/√(n − 3).
- Nonparametric bootstrap of the pairs: 1999 resamples with seed 2026; normal, basic, percentile and BCa intervals with the quantiles interpolated on the normal scale (Davison and Hinkley 1997) and the jackknife acceleration.
- Permutation: 1999 shuffles of Y with seed 2027; p = (k + 1)/(B + 1), which counts the observed value as one of the shuffles.
- Jeffreys' Bayes factor for Pearson's r in its exact form (Ly, Verhagen and Wagenmakers 2016), with a beta(1/κ, 1/κ) stretched to (−1, 1), κ = 0.333; adaptive Gauss–Kronrod numerical integration.

Figure 4.9. The conventions for Ezekiel's cars with Pearson's *r* as the main coefficient.

READING AND WRITING UP A PAIR

If the classical test and the permutation agree…	report the classical one; the permutation confirms that it does not depend on normality.
If they do not agree…	report the permutation and the bootstrap interval, and say why.
If the interval includes both small and large values…	the sample is not enough to know how strong the relation is; say so, even if *p* is small.
If *p* is not significant…	do not write "there is no relation": look at the interval and the Bayes factor, which can support ρ = 0.
If the main coefficient is not Pearson's *r*…	remember that the Bayes factor and r^2 belong to Pearson, not to the main coefficient.
If you used resampling…	report the number of resamples and the seed.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

5

CHAPTER FIVE · BLOCK 5

The correlation matrix

All the variables against all: the matrix with its tests corrected for multiple testing, the figure for publication with the glyph and the order that read it best, the diagnosis that says whether the matrix is fit for what follows—regression, factor analysis, simulation—, the KMO and Bartlett's test, and the table of means, standard deviations and correlations ready for the article.

5.1 What to correlate

5.2 The matrix

5.3 Multiple testing

5.4 Four matrices to show

5.5 Is it a valid matrix?

5.6 KMO and Bartlett

5.7 The table to publish

5.8 Large matrices

5.1 What to correlate

The numeric columns—continuous and ordinal—with the variable role come in. When you open some data all of them are selected; a click removes or adds one (at least two always remain) and `all` puts them all back.

Coefficient offers eleven of the twelve coefficients of Block 4 (all except τ*, which is too slow for a matrix). Multiple-testing correction and α decide which cells count as significant (Section 5.3). Each cell uses the rows complete in its two variables, unless listwise handling of missing data was chosen in Block 2.

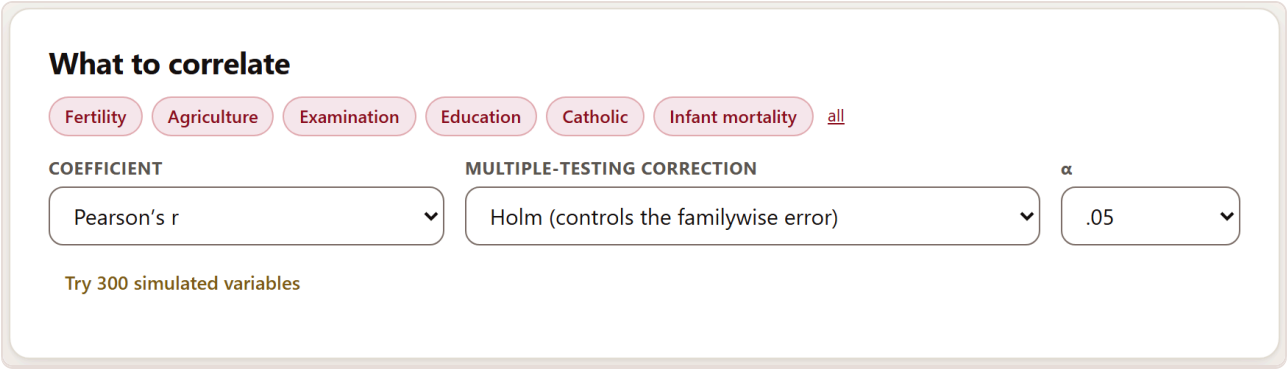


Figure 5.1. Switzerland, 1888, the example the block opens with: six indicators for 47 French-speaking districts, Pearson's *r* and Holm's correction at 5%.

5

COSTLY MEASURES AND MANY VARIABLES

The distance correlation stores $n \times n$ matrices and is computed with up to 3,000 rows. The distance correlation, Hoeffding's D and Chatterjee's ξ allow up to 60 variables when there are more than 300 rows. With more than 30 complete variables, Pearson and Spearman are computed as products of standardized columns, with the *p*-values of the *t* approximation, and a message says so.

5.2 The matrix

The figure has two halves. Below and Above choose the glyph for each one among the nine of Section 1.5—ellipse, circle, moon, pie, square, bar, shade, color and number—or Nothing; when you change it, the shape glyphs morph into the new one. The color follows the palette chosen in Block 1, and the bar on the right gives its scale from −1 to 1.

- Order: As in the data; three of hierarchical Clustering (average, complete or Ward), with the distance 1 − *r*; Angle of the eigenvectors (Friendly 2002), which orders by the angle of each variable in the first two; First principal component, by the loading on the first; or Alphabetical.
- Not significant: the cells whose corrected *p*-value does not reach α are Crossed out, Faded or drawn Like the others.
- Show: which of the four matrices of Section 5.4 is drawn.

Hovering the pointer over a cell shows its two variables, its value, its corrected *p* and its *n*. A click selects it: it is framed in both halves, a line below gives its uncorrected and corrected *p*, and Study the chosen pair in Block 4 → takes that pair to Block 4.

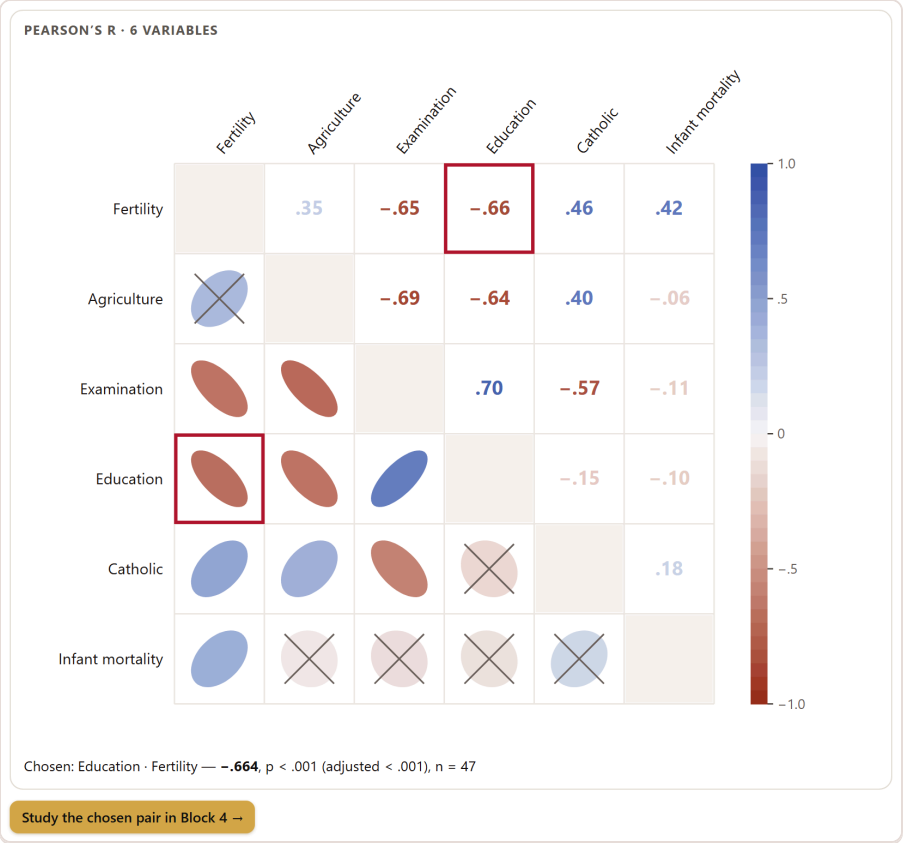


Figure 5.2. Switzerland, 1888, with ellipses below and numbers above. Education and fertility (framed) give −.66. Agriculture and fertility, with .35, are crossed out: uncorrected they would give $p < .05$, but they do not survive Holm's correction.

THEORY IN BRIEF · GLYPHS AND ORDER

Each glyph encodes r in a different way: the ellipse is the shape of a cloud with that correlation (Murdoch and Chow 1996), the moon and the pie show $|r|$ as a fraction, color and shading (Friendly 2002) read well from a distance and the number is exact. A good arrangement combines two: a glyph that reads at a glance in one half and the number in the other. The order matters as much as the glyph: grouping similar variables reveals blocks that the order of the data hides.

5.3 Multiple testing

With p variables there are $p(p-1)/2$ tests: with six variables, fifteen; with thirty, 435. Without correction, at 5% false positives are expected even if no variable is related to any other: one in every twenty pairs, on average. The corrections adjust the p-values of the whole family of tests.

Table 5.1. The multiple-testing corrections

Correction	What it controls
Holm (the default)	the probability of having at least one false positive in the whole matrix (the familywise error), with more power than Bonferroni and no assumptions
Benjamini–Hochberg	the expected proportion of false positives among the significant cells (the false discovery rate); more powerful when there are many tests
Benjamini–Yekutieli	the same rate, valid under any dependence between the tests; more conservative
Hochberg · **Hommel**	the familywise error, with somewhat more power than Holm; they assume that the tests are independent or positively associated
Bonferroni	the familywise error, by multiplying each p by the number of tests: the simplest and the most conservative
None	nothing: each cell with its own p

DECIDE THE CORRECTION BEFORE SEEING THE MATRIX

If the question is confirmatory ("are these correlations real?"), use Holm. If it is exploratory ("which pairs are worth following up?"), Benjamini–Hochberg. What is not acceptable is to try several and keep the one that gives the most stars. And always state in the report which one you used: the table to publish puts it in its note.

5.4 Four matrices to show

- The estimated matrix: the one from the data, cell by cell.
- The nearest positive-definite one: the valid correlation matrix most similar to the estimated one (Section 5.5).
- The shrunk one (Schäfer–Strimmer): each correlation shrunk toward zero in the proportion λ estimated from the data; useful with few rows for many variables.
- The partial correlations: each pair after removing the linear effect of all the other variables, with its t test on $n - p$ degrees of freedom and the same correction. It requires a positive-definite matrix.

THE PARTIAL CORRELATION OF A SINGLE PAIR

This view removes all the other variables at once. To control for only some of them, with their interval and their robust version, there is Block 7.

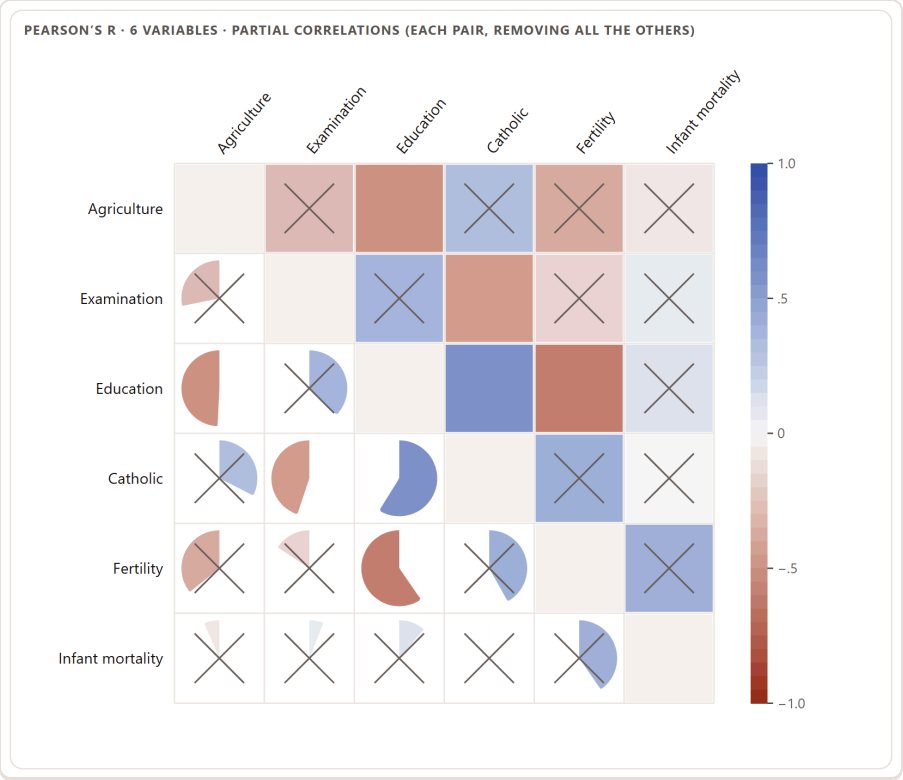


Figure 5.3. The partial correlations of Switzerland, 1888, ordered by clustering, with pies below and color above. With the other variables removed, education remains closely tied to Catholics and to fertility; examination and education, which gave .70, are crossed out: almost everything they share goes through the other variables.

5.5 Is it a valid matrix?

A correlation matrix must be **positive definite**: all its eigenvalues greater than zero. Computed cell by cell it may not be—with pairwise missing data, with robust coefficients or with more variables than rows—, and then it cannot be used for regression, factor analysis, structural equations or simulation. The card gives five figures: whether it is positive definite (and, if not, how many eigenvalues are negative or zero), the smallest and largest eigenvalue (with the share of the variance taken by the largest), the condition number—above 1,000 the matrix is nearly collinear—and the logarithm of the determinant.

The eigenvalue plot helps to see how many dimensions the data have. On the right, the Corrections: Compute the nearest positive-definite one applies Higham's algorithm (2002), with up to 150 variables, and states how many iterations it used, the Frobenius distance to the estimated matrix and the largest change in a single cell; the Schäfer and Strimmer shrinkage gives its λ and the factor by which each correlation is multiplied.

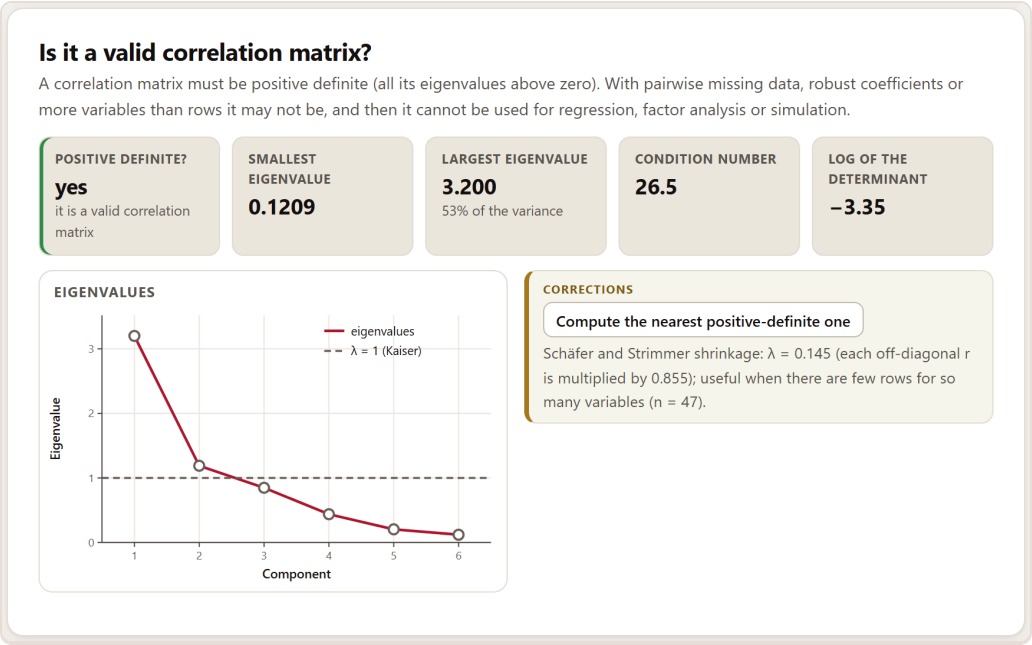


Figure 5.4. Switzerland, 1888: positive definite, with the smallest eigenvalue at 0.121 and a condition number of 26.5. The first component takes 53% of the variance; the dashed line is the Kaiser criterion (λ = 1).

IF THE MATRIX IS NOT POSITIVE DEFINITE	
If missing data are handled pairwise…	try listwise handling in Block 2; if things change a lot, the missing values are not at random.
If the coefficient is robust…	use the nearest one, and say in the report that you used it and how much it changed.
If there are more variables than rows…	the matrix is necessarily singular: use the shrunk one, or reduce the variables.
If the largest change of the nearest one is large (more than .05)…	the estimated matrix has a deeper problem: check the data before going on.

5.6 KMO and Bartlett

Before a factor analysis it is worth knowing whether the variables share anything. The **Kaiser–Meyer–Olkin** measure (KMO) compares the correlations with the partial ones: if the partials are small compared with the correlations, the variables share factors. It is given for the whole matrix and, as MSA, for each variable (up to 60 variables). **Bartlett's test** asks whether the matrix is the identity, that is, whether nothing correlates; it uses the smallest *n* in the matrix. Both require a positive-definite matrix.

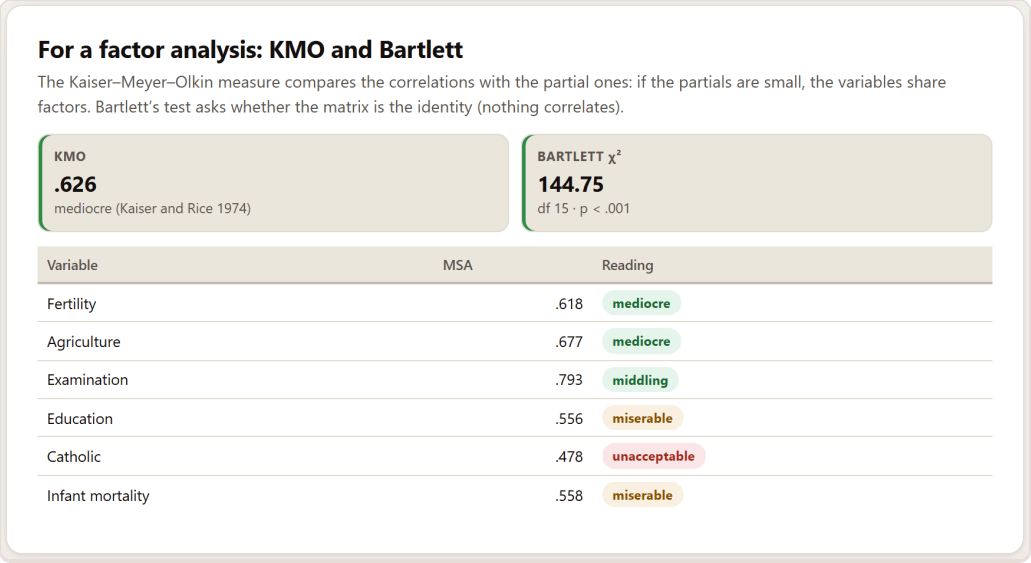


Figure 5.5. Switzerland, 1888: KMO = .626, "mediocre"; Bartlett rejects the identity (χ^2 = 144.75 with 15 degrees of freedom). Catholic, with MSA = .478, is the variable that shares least with the others.

Table 5.2. The words for the KMO (Kaiser and Rice 1974)

KMO or MSA	Reading
.90 or more	marvelous
.80 to .90	meritorious
.70 to .80	middling
.60 to .70	mediocre
.50 to .60	miserable
less than .50	unacceptable

In the table for each variable, the labels are green from .60 up, amber between .50 and .60 and red below. An unacceptable variable is the first candidate to be left out of the factor analysis.

5.7 The table to publish

The classic table of an article: each variable numbered with its mean (M) and its standard deviation (SD), and below the diagonal the correlations with stars according to the corrected *p*. The footnote gives the *n* (or its range, with pairwise missing data) and the correction used. The numbers follow the style of Block 1. It is built with up to 30 variables, and it shows whichever matrix is selected in Show.

Download r, p and n (CSV) saves one line per pair with its two variables, *r*, the uncorrected *p*, the corrected *p* and the *n*: what you need for supplementary material or for another program, with any number of variables.

Table to publish

Table 1. Means, standard deviations and correlations (Pearson's r)

Variable	M	SD	1	2	3	4	5
1. Fertility	70.14	12.49					
2. Agriculture	50.66	22.71	.35				
3. Examination	16.49	7.98	−.65***	−.69***			
4. Education	10.98	9.62	−.66***	−.64***	.70***		
5. Catholic	41.14	41.70	.46**	.40*	−.57***	−.15	
6. Infant mortality	19.94	2.91	.42*	−.06	−.11	−.10	.18

Note. n = 47. p-values corrected by Holm: * p < .05, ** p < .01, *** p < .001.

Copy the table Download r, p and n (CSV)

Figure 5.6. Switzerland, 1888, ready for the article. Copy the table puts it on the clipboard separated by tabs, so that when you paste it into a word processor or a spreadsheet each value falls into its own cell.

5

5.8 Large matrices

With more than 40 variables the figure drops the glyphs and becomes a canvas of colors, one cell per pair. When you hover the pointer, a **magnifier** shows the numbers of the 9 × 9 window of cells around it, and a line below gives the pair, its value, its corrected *p* and its *n*; a click selects the pair. Ordering by clustering is what makes a large matrix readable. Try 300 simulated variables opens a set of 300 variables and 120 rows in six modules—simulated, not real data—to try it out.

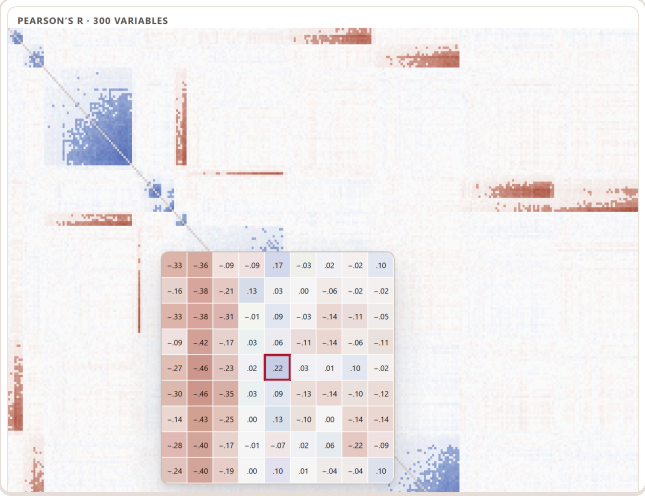


Figure 5.7. The 300 simulated variables, ordered by clustering: the six modules are the blocks along the diagonal. The magnifier gives the numbers around the pointer.

The diagnosis is computed with up to 400 variables. With 300 variables and only 120 rows, the matrix cannot be positive definite: it has 181 zero eigenvalues (300 minus 119), its condition number is infinite and the nearest one is not computed with so many variables. For a case like this there is shrinkage: λ = 0.199 multiplies each correlation by 0.801 and leaves a valid matrix.

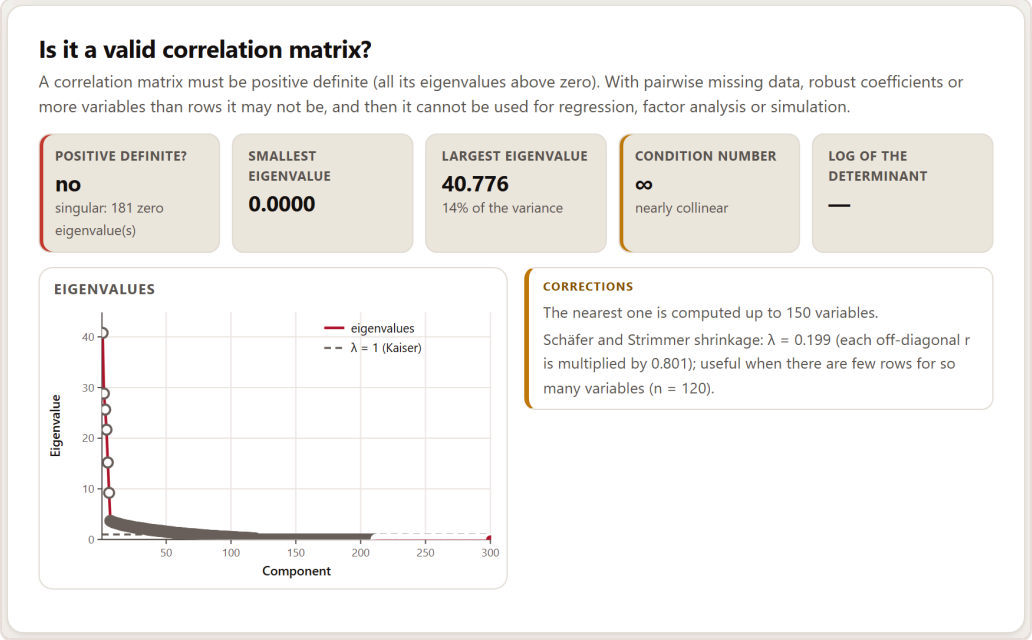


Figure 5.8. A matrix with more variables than rows: singular, with 181 zero eigenvalues. The first six eigenvalues, far above the rest, are the six modules.

6

CHAPTER SIX · BLOCK 6

Nonlinear dependence and copulas

Pearson's *r* measures straight lines, and Spearman's ρ measures curves that always go up or always go down. A U-shaped, bell-shaped or cyclic relation escapes both. This block gathers the measures that are zero only when there is no relation at all, sweeps every pair to find the ones Pearson misses, and fits five copulas that tell you whether extreme values happen together.

6.1 One pair: what each measure sees

6.2 The cloud and the common scale

6.3 MIC and its characteristic matrix

6.4 The sweep: what Pearson misses

6.5 Copulas: the shape of the dependence

6.1 One pair: what each measure sees

You choose two numeric variables in `X` and `Y`, and the block computes nine figures at once: Pearson's *r*, Spearman's ρ and the dependence measures that see any shape. Kendall's τ appears in the chart of Section 6.2.

At least 10 complete pairs are needed. Permutations for MIC computes the p-value of MIC by shuffling Y 199, 499 or 999 times; since MIC is slow, it is off by default (Do not compute its p-value). With more than 250 rows, or with permutations, a waiting screen appears.

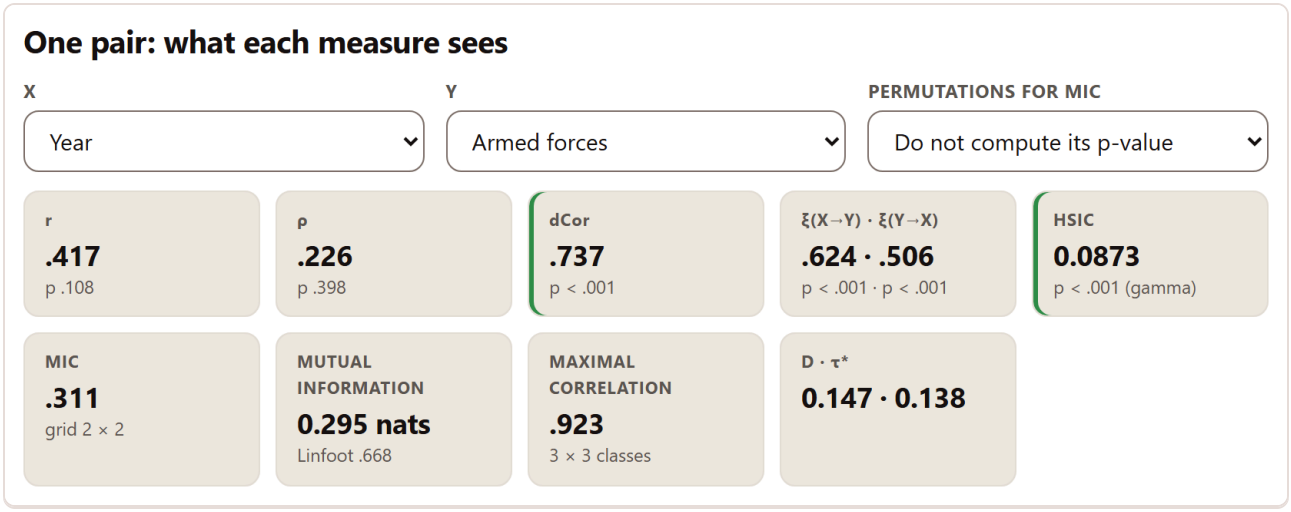


Figure 6.1. The year and the United States armed forces, 1947 to 1962, the example the block opens with. Neither r (.417, p = .108) nor ρ (.226) is significant; the distance correlation (.737), ξ and HSIC are, with p < .001.

Table 6.1. The measures of Block 6

Measure	What it measures	Test · limit
***r*, ρ, τ**	the linear part (r) and the monotonic part (ρ, τ) of the relation	those of Block 4
dCor	any dependence; in the population it is 0 only under independence (Székely, Rizzo and Bakirov 2007)	T · up to 3,000 rows
ξ(X→Y) · ξ(Y→X)	how much Y is a function of X, and the other way around: it is not symmetric (Chatterjee 2021)	z · no limit
HSIC	dependence measured in a space of functions, with Gaussian kernels (Gretton et al. 2005)	gamma approximation · up to 1,500
MIC	the largest mutual information a grid can capture, normalized so that a noiseless function of any shape scores 1 (Reshef et al. 2011)	permutation, if requested · up to 1,200
Mutual information	how much uncertainty about Y is removed by knowing X, in nats; Linfoot's coefficient puts it on the scale of a correlation (Kraskov, Stögbauer and Grassberger 2004)	— · up to 3,000
Maximal correlation	the largest r obtained by transforming X and Y with any function (Gebelein 1941; Rényi 1959), on quantile classes	— · no limit
D · τ*	Hoeffding's D and the τ* of Bergsma and Dassios, as in Block 4	— · τ* up to 80

MUTUAL INFORMATION, IN STANDARD UNITS

The nearest-neighbor estimator measures distances with the maximum norm, so the app standardizes both variables first: in their own units, the one with the larger spread would decide on its own which points are neighbors. Linfoot's coefficient, $\sqrt{(1 - e^{-2I})}$, equals $|\rho|$ when both variables are normal.

6.2 The cloud and the common scale

On the left, the cloud with the straight line and the LOWESS curve. On the right, the nine measures on a single scale from 0 (independence) to 1, in the color of their family: r in red, the rank-based ones in gold, the dependence measures in turquoise and the maximal correlation in violet. When the dependence measures lie far to the right of $|r|$, there is a relation that a straight line does not see.

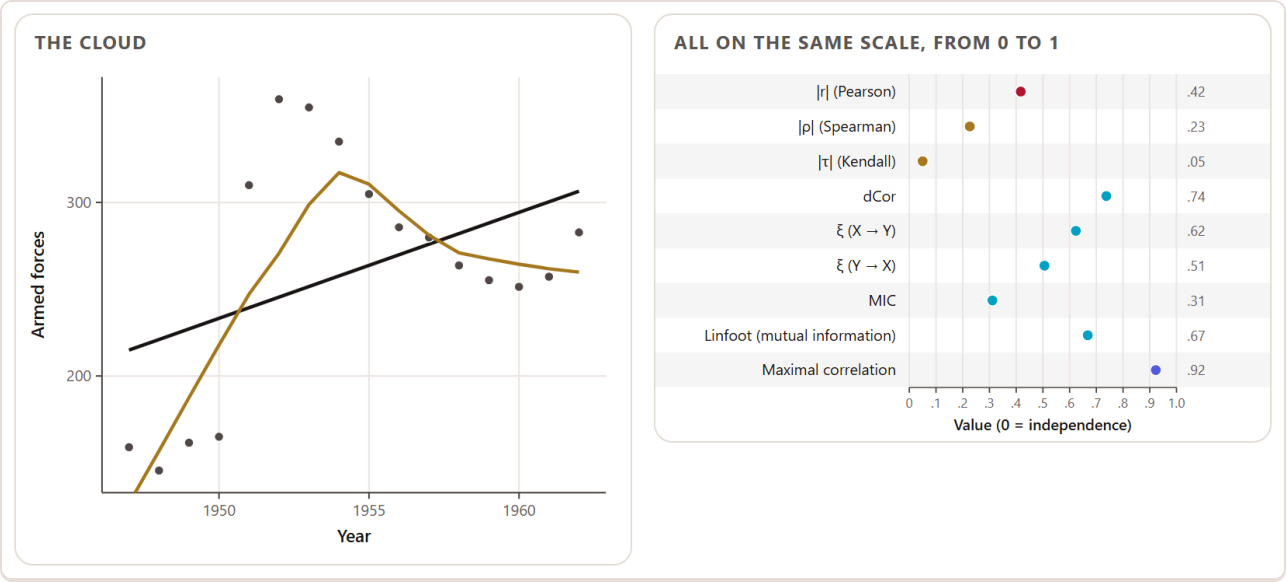


Figure 6.2. The armed forces grow suddenly in 1951, with the Korean War, and then decline slowly. The straight line barely captures it; the distance correlation (.74), ξ (.62 and .51) and the maximal correlation (.92) do.

Below, the reading in words compares the distance correlation with $|r|$: if it exceeds it by more than 0.2, there is dependence that a straight line does not capture; if not, nearly all the dependence detected is also captured by a line. If the two ξ differ by more than 0.15, it says so and explains in which direction the dependence is stronger: in a parabola, Y is a function of X but not the reverse. It also gives the MIC nonlinearity index, MIC – r^2.

WITH FEW DATA, NOT EVERY MEASURE WORKS

With the 16 years of this example, MIC can only use a 2 × 2 grid, and the maximal correlation three classes per variable, which tend to give high values. The distance correlation, ξ and HSIC work well with few data and come with their own test; MIC and mutual information need at least about 50.

6.3 MIC and its characteristic matrix

MIC tries many grids over the cloud—from 2 × 2 up to those with as many as $n^{0.6}$ cells—and in each one it measures the mutual information between the column and the row each point falls in, normalized by the logarithm of the smaller side. The **characteristic matrix** shows that value for each combination of columns and rows; MIC is its maximum, framed in red. With very few data only the 2 × 2 grid fits, and the app says so instead of drawing a one-cell matrix.

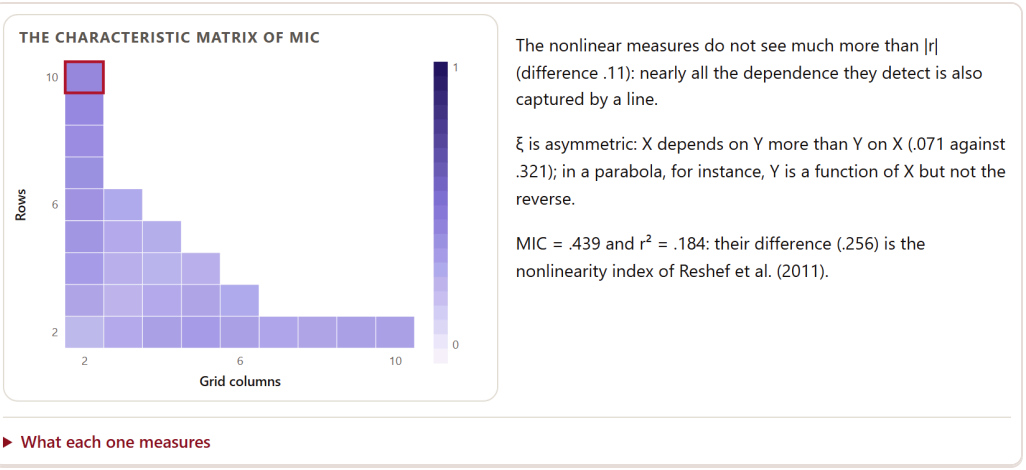


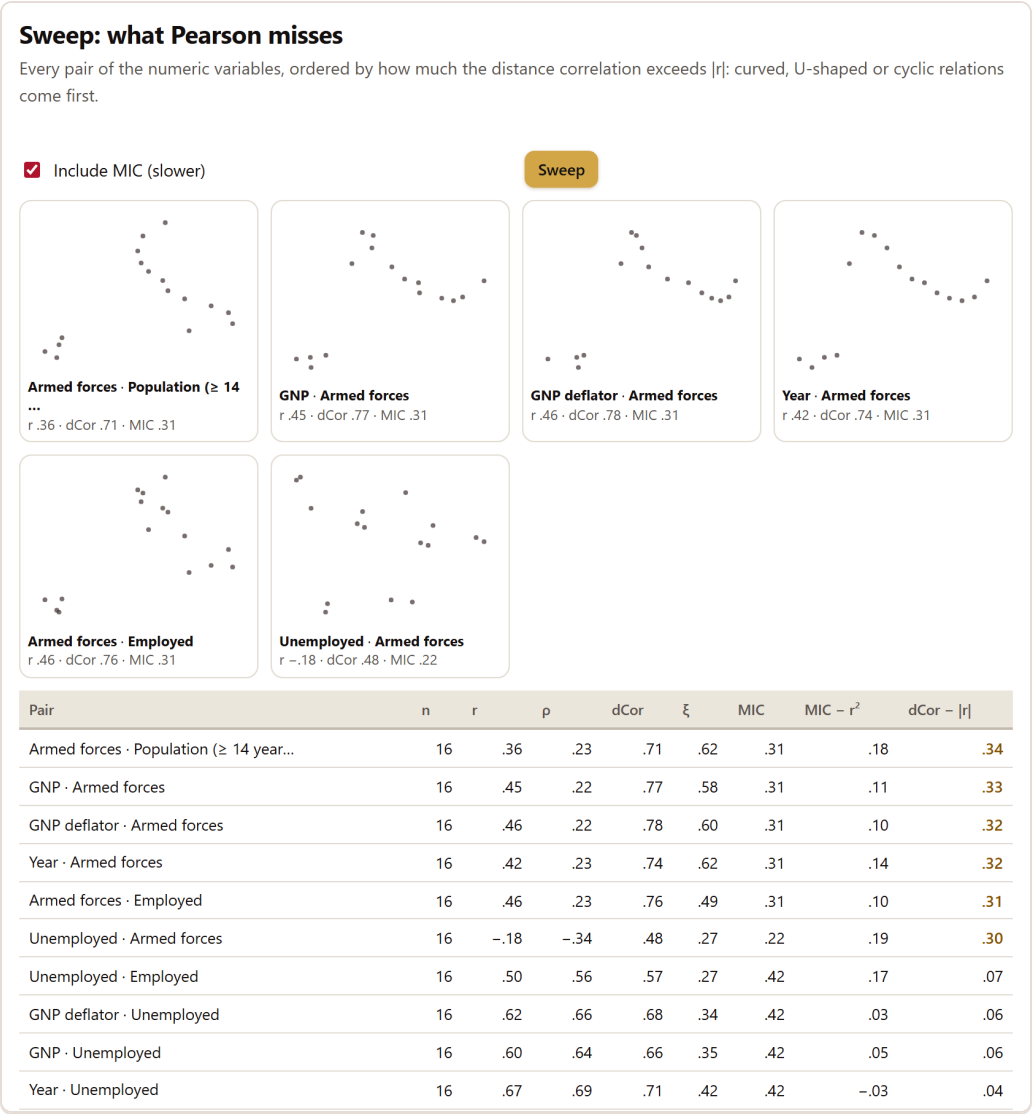
Figure 6.3. The sepal width and the petal length of the irises. The maximum is in the grid of 2 columns by 10 rows: MIC = .439 against r^2 = .184. ξ is asymmetric: the petal length says a lot about the sepal width (.321), and not the other way around (.071), because for the same sepal width there are both short and long petals.

THEORY IN BRIEF · WHAT EACH ONE SEES

The distance correlation compares the distances between all pairs of points in X with those in Y; HSIC does the same with the similarities given by a Gaussian kernel, whose width is set by the median of the distances. Both are zero only under independence. ξ sorts the data by X and looks at how much the ranks of Y jump from one neighbor to the next: if Y is a function of X, they hardly jump. MIC and mutual information come from information theory; the maximal correlation, from searching for the best transformations of the two variables.

6.4 The sweep: what Pearson misses

Sweep computes, for every pair of numeric variables (up to 30 variables), r, ρ, the distance correlation and ξ in its stronger direction, and sorts them by **dCor − |r|**: curved, U-shaped or cyclic relations rise to the top. Include MIC (slower) adds MIC and MIC − r^2. The first six pairs appear as cards with a miniature cloud; the table shows up to 40, with the difference in amber when it exceeds 0.2. A click on a card or a row takes that pair to the study of Section 6.1.



Pair	n	r	ρ	dCor	ξ	MIC	MIC − r^2	dCor − |r|
Armed forces · Population (≥ 14 year...	16	.36	.23	.71	.62	.31	.18	.34
GNP · Armed forces	16	.45	.22	.77	.58	.31	.11	.33
GNP deflator · Armed forces	16	.46	.22	.78	.60	.31	.10	.32
Year · Armed forces	16	.42	.23	.74	.62	.31	.14	.32
Armed forces · Employed	16	.46	.23	.76	.49	.31	.10	.31
Unemployed · Armed forces	16	−.18	−.34	.48	.27	.22	.19	.30
Unemployed · Employed	16	.50	.56	.57	.27	.42	.17	.07
GNP deflator · Unemployed	16	.62	.66	.68	.34	.42	.03	.06
GNP · Unemployed	16	.60	.64	.66	.35	.42	.05	.06
Year · Unemployed	16	.67	.69	.71	.42	.42	−.03	.04

Figure 6.4. The 21 pairs of the United States economy. The first six involve the armed forces, which rise and then fall against the other series; the series that grow together end up at the bottom, with dCor − $|r|$ close to zero.

EXAMPLE · WHAT THE SWEEP IS FOR

A matrix of 20 variables has 190 pairs, and nobody looks at 190 clouds. The sweep brings forward the few pairs on which a Pearson matrix—the one in Block 5—is misleading, because it says "little relation" where there is a lot. Those pairs deserve a cloud in the report, and a dependence measure instead of *r*.

6.5 Copulas: the shape of the dependence

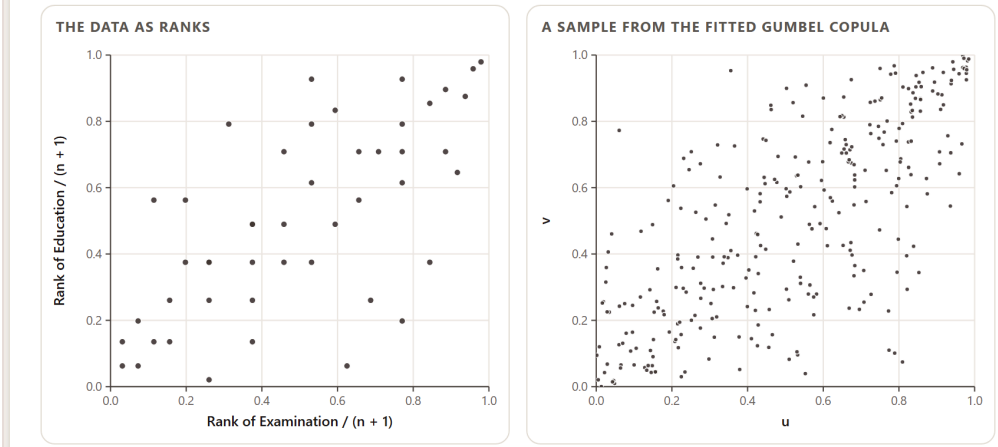
Sklar's theorem (1959) separates each variable from its dependence: when the data are turned into ranks divided by $n + 1$—the pseudo-observations—only the **copula** remains, the dependence without the marginals. Two pairs with the same Kendall's τ can have very different copulas, and the difference lies above all in the tails: do extreme values happen together?

The block fits five families to the chosen pair, each with two estimates of its parameter θ: the one given by inverting Kendall's τ and the maximum pseudo-likelihood one (Genest, Ghoudi and Rivest 1995). It gives their log-likelihood, their AIC—the one with the lowest AIC carries the label lowest AIC—and their lower and upper tail dependence. Clayton and Gumbel are fitted only with positive dependence.

Copulas: the shape of the dependence

Sklar's theorem (1959) separates each variable from its dependence: turning the data into ranks (pseudo-observations) leaves only the copula. Five families are fitted; they differ above all in the tails: do extreme values happen together?

Family	θ from τ	θ (MPL)	log-likelihood	AIC	λ_{low}	λ_{up}
Gaussian	0.739	0.735	15.87	−29.75	0.000	0.000
Clayton	2.246	1.426	11.60	−21.19	0.615	0.000
Gumbel lowest AIC	2.123	2.128	17.28	−32.57	0.000	0.615
Frank	6.289	5.934	14.51	−27.02	0.000	0.000
Student t	0.739	0.719 ν = 2.5	16.84	−29.68	0.499	0.499



EMPIRICAL LOWER TAIL
.50
P(V ≤ 0.1 | U ≤ 0.1)

EMPIRICAL UPPER TAIL
.50
P(V > 0.9 | U > 0.9)

IF THEY WERE INDEPENDENT
.10
both probabilities would be 0.1

KENDALL'S τ
.529
what fixes each family

▸ The families and their tails

Figure 6.5. Examination and education in the districts of Switzerland, 1888 (τ = .529). Gumbel wins, with the dependence concentrated at the top (λ_{up} = .615): the districts with more high-scoring conscripts are also those with more education. On the right, a sample from the fitted copula to compare with the data.

Table 6.2. The five families and their tails

Family	Tails
Gaussian	no tail dependence: the extremes become independent
Student t	the same dependence in both tails, larger the fewer the degrees of freedom (ν)
Clayton	dependence at the bottom: low values happen together
Gumbel	dependence at the top: high values happen together
Frank	no tail dependence, symmetric; allows negative dependence

Below, four figures give the tails of the data themselves: the probability that Y is in its lower 10% when X is, the same for the upper 10%, what both would be if the variables were independent (0.1), and the Kendall's τ that fixes each family.

THEORY IN BRIEF · TAIL DEPENDENCE

Upper tail dependence λ_{up} is the limit of the probability that V exceeds u when U exceeds u, as u approaches 1 (Joe 1997); the lower one is the same at the bottom. It matters for risk: if the droughts of two regions, the falls of two prices or the pests of two crops share a tail, they happen together more often than a Gaussian with the same correlation would suggest. Inverting τ gives a starting value that does not depend on the marginals; the pseudo-likelihood refines it.

6

READING BLOCK 6

If dCor is much larger than $\lvert r \rvert$…	look at the cloud and report a dependence measure, not r; describe the shape in words.
If the two ξ differ a lot…	one variable is nearly a function of the other, but not the reverse: say in which direction.
If there are fewer than about 50 data…	rely on dCor, ξ and HSIC; take MIC, mutual information and the maximal correlation as a guide.
If the best copula has a tail…	extremes happen together: keep it in mind in any simulation or risk analysis.
If two families are close in AIC…	the data are not enough to choose: with few data, the empirical tails rest on very few points.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

7

CHAPTER SEVEN · BLOCK 7

Partial, multiple and between sets

Two variables are almost never alone. This block removes the effect of third variables (partial and semipartial correlation), measures how much several variables together explain of one (multiple correlation, with the breakdown of its R^2) and correlates groups of variables with each other: canonical correlation, the RV coefficient, Procrustes, the Mantel test and co-inertia.

7.1 Partial and semipartial correlation

7.2 Multiple correlation

7.3 Canonical correlation

7.4 Between sets: RV, Procrustes, Mantel and co-inertia

7.1 Partial and semipartial correlation

How much of the relation between X and Y is left once you remove what both share with Z? The **partial** correlation removes Z from both; the **semipartial**, from Y only.

You choose X, Y and the Coefficient—Pearson, Spearman or Kendall—and one or more variables in Control for (Z). The figures are the uncontrolled correlation, the partial with its test, the semipartial, the **partial distance correlation** (Székely and Rizzo 2014), which removes Z without assuming linearity, with its permutation p-value (up to 800 rows), and the number of controls with the degrees of freedom, $n - 2 - k$.

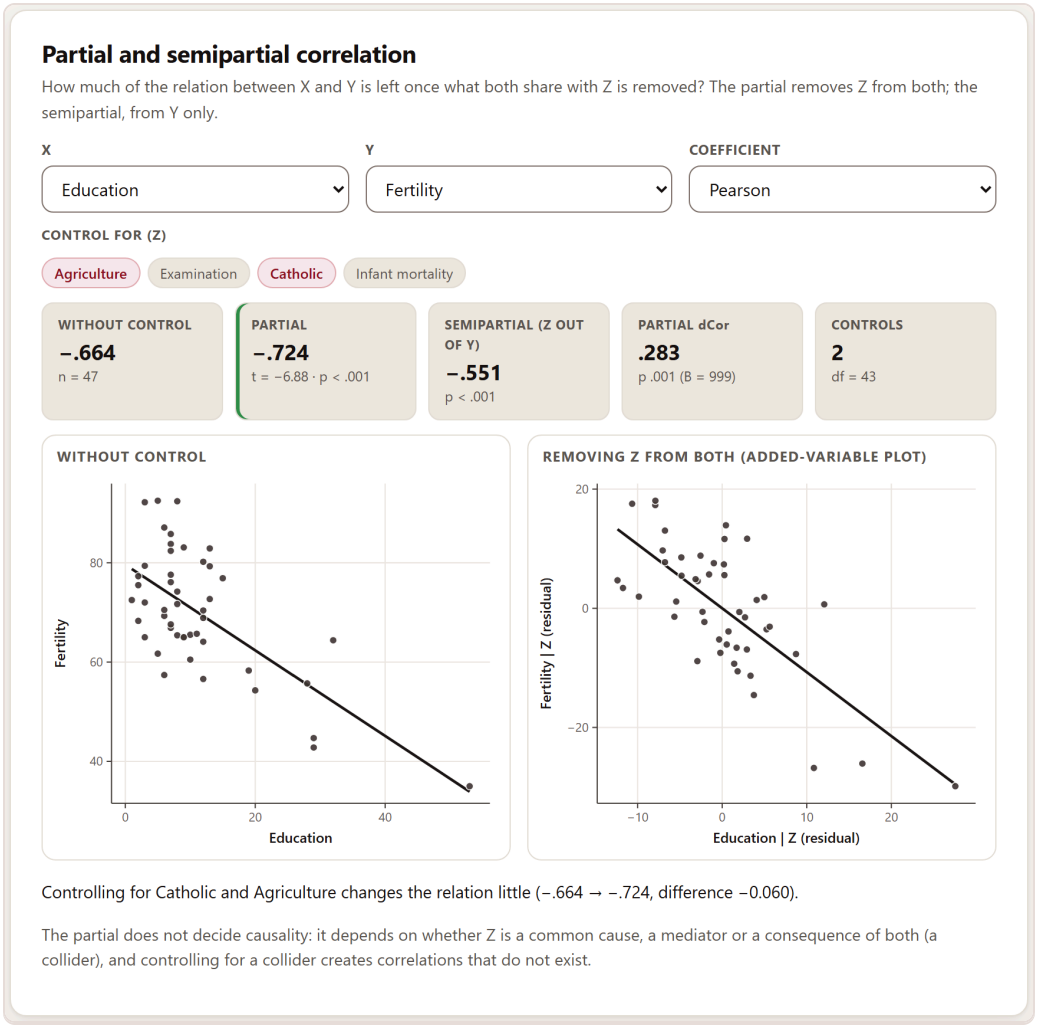


Figure 7.1. Education and fertility in the Swiss districts, controlling for Catholics and agriculture. The relation is not due to them: uncontrolled −.664, partial −.724 with 43 degrees of freedom. On the right, the added-variable plot: the residuals of each variable once *Z* is removed.

The two charts are the key to the partial correlation. On the left, the usual cloud; on the right, the **added-variable plot**: the residuals of *X* and of *Y* after removing what *Z* predicts. With Pearson, the partial is exactly the *r* of that second cloud. Below, a reading says what controlling did: whether the relation drops to less than half, whether it changes sign or grows (a suppression effect), or whether it changes little.

> **! THE PARTIAL DOES NOT DECIDE CAUSALITY**
>
> A relation that drops when you control for *Z* may mean two opposite things: that *Z* is a common cause of *X* and *Y* (and the original correlation was partly spurious) or that *Z* is the path through which *X* acts on *Y* (a mediator). And if *Z* is a consequence of both—a **collider**—controlling for it creates a correlation that does not exist. Choose the controls by what you know about the phenomenon, not by how they change the result.

7.2 Multiple correlation

The correlation between *Y* and the best linear combination of several *X*: the *R* of a multiple regression. You choose Y and, in Predictors (X), the others. Four figures summarize it: *R*, R^2 with its version adjusted for the number of predictors, the *F* test that all the coefficients are zero, and the **shared** part of R^2, the part that belongs to no predictor in particular.

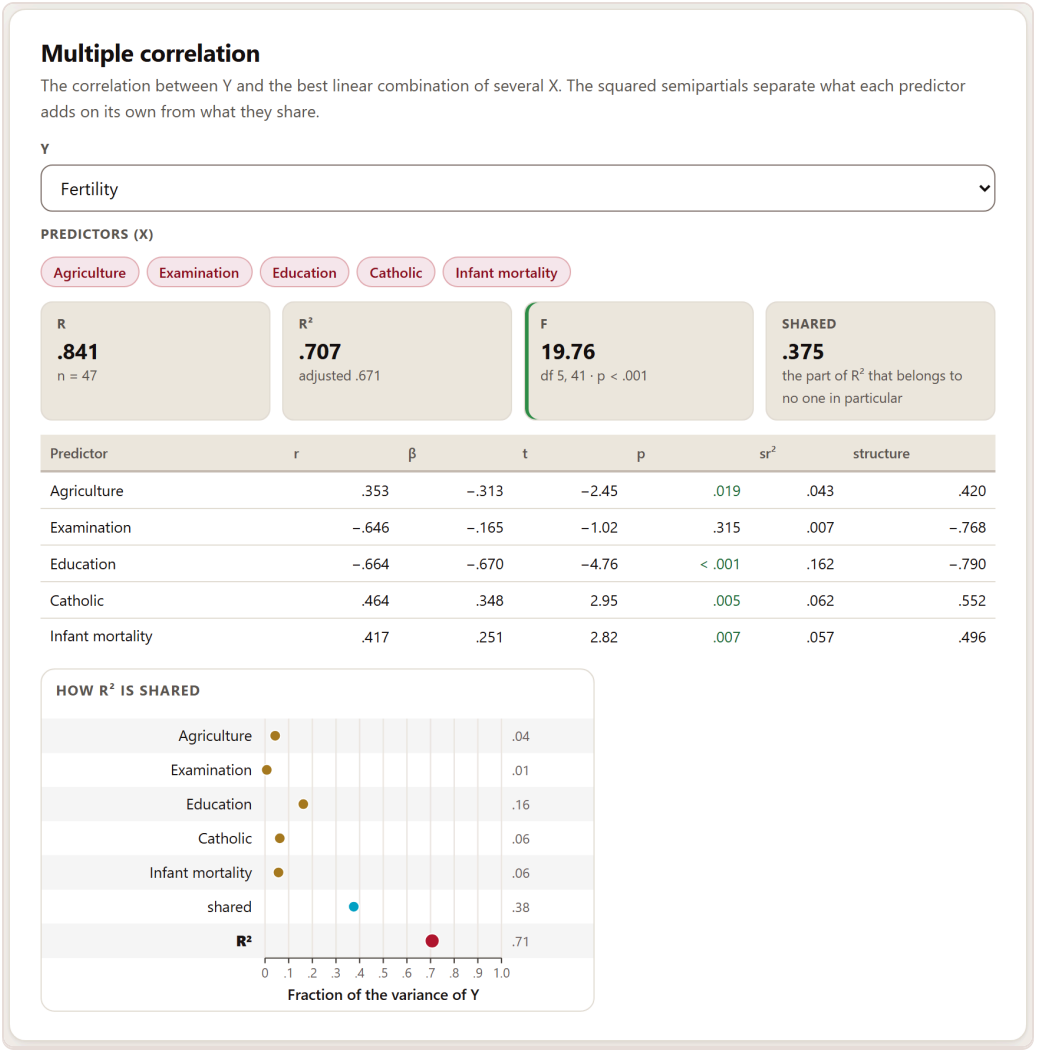


Figure 7.2. Fertility and the other five indicators: $R = .841$, $R^2 = .707$. What each predictor contributes on its own (sr^2) adds up to .331; the other .375 is shared. Education contributes the most by itself (.162).

The table gives, for each predictor, its simple correlation *r* with *Y*, its standardized coefficient β with its *t* test, its squared semipartial sr^2—what R^2 would lose if that predictor were removed—and its **structure coefficient**, r/R, the correlation of the predictor with the prediction. The chart splits R^2 into what is unique to each predictor and what is shared.

EXAMPLE · WHEN B AND R HAVE OPPOSITE SIGNS

In Figure 7.2, agriculture has $r = .353$ with fertility—the more agricultural districts have more children—but $\beta = -.313$. It is not a mistake: among districts with the same education, the same examination scores, the same proportion of Catholics and the same mortality, more agriculture goes with somewhat lower fertility. The simple correlation was loaded by education, which is lower in the agricultural districts. That is why a β is not interpreted without also looking at *r* and the structure.

7.3 Canonical correlation

With two **sets** of variables, canonical correlation (Hotelling 1936) looks for the linear combination of each set that correlates most with one of the other; then the next one, independent of the first, and so on up to the size of the smaller set. The sets are built with the buttons of Set A (in red) and Set B (in gold); a variable cannot be in both. There must be more rows than variables in total.

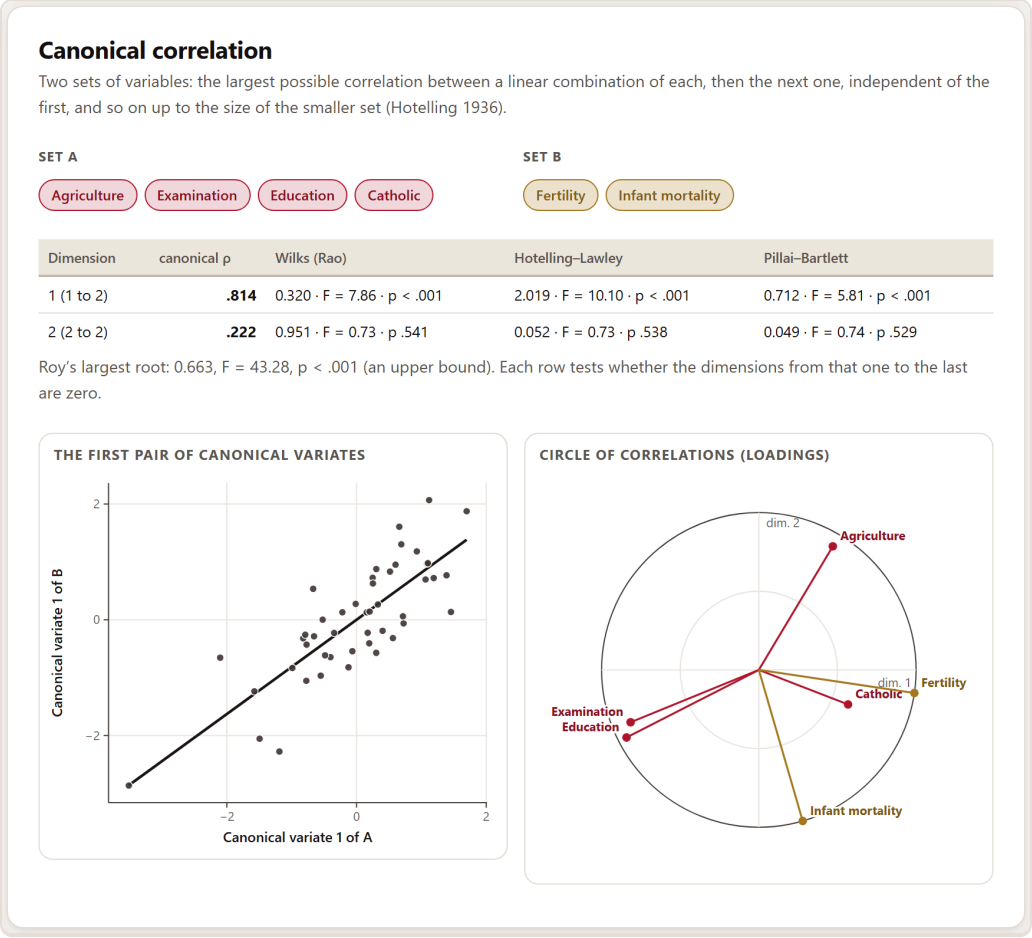


Figure 7.3. Four social indicators (A) against fertility and infant mortality (B). The first dimension, ρ = .814, is significant by all three tests; the second, .222, is not. In the circle, the first dimension sets fertility and Catholics against education and examination.

The table gives each canonical correlation with three tests: Wilks' lambda (with Rao's *F*), the Hotelling–Lawley trace and the Pillai–Bartlett trace. Each row tests whether the dimensions from that one to the last are zero: if the first row is significant and the second is not, only the first dimension is real. Below comes Roy's largest root, which is an upper bound and therefore tends to reject too often. The cloud shows the first pair of canonical variates; the **circle of correlations** places each variable according to its correlation with the first two canonical variates of its set (the structure loadings).

7.4 Between sets: RV, Procrustes, Mantel and co-inertia

With the same sets A and B, four measures that do not look for linear combinations but compare the two whole tables. All of them use the standardized variables, and the first three come with a permutation test: 999 shuffles of the rows (fewer with more than 300 rows).

- **RV** (Escoufier 1973): an *r* between two tables, from 0 to 1. The figure also gives the value it would have with no relation, which is not zero: it grows with the number of variables and falls with the number of rows.
- **Procrustes** (Gower 1975; Peres-Neto and Jackson 2001): rotates and scales one configuration to fit the other; m^2 is what remains unfitted and $r = \sqrt{(1 - m^2)}$.
- **Mantel** (1967): the correlation between the distances of all pairs of rows in one table and in the other (with up to 400 rows).
- **Co-inertia**: the analysis that looks for the axes along which the two tables vary together; the figure gives the share of the co-inertia taken by the first axis and the *r* between the scores of the two tables on it.

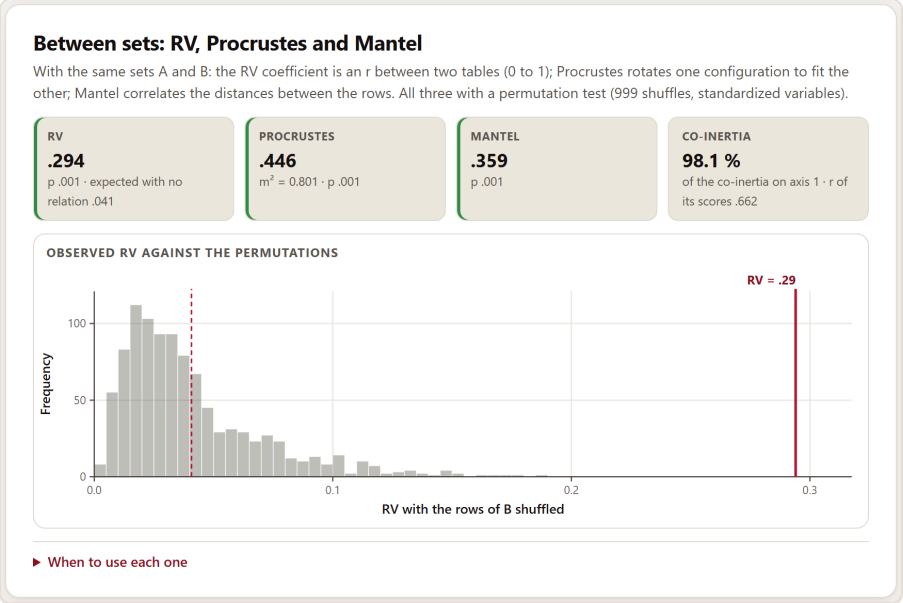


Figure 7.4. The same sets: RV = .294 against the .041 that chance would give, Procrustes .446 and Mantel .359, all three with p = .001, the smallest possible with 999 shuffles. The first axis takes 98.1% of the co-inertia. The chart sets the observed RV against those of the shuffles.

THEORY IN BRIEF · WHEN TO USE EACH ONE

Canonical correlation looks for linear combinations and breaks down with many variables and few rows; **RV** summarizes the whole table and is the basis of co-inertia; **Procrustes** compares configurations, for example two ordinations of the same sites; **Mantel** is useful when what you have are distances—genetic, geographic, ecological—although its power is low and it does not correct for spatial autocorrelation (Guillot and Rousset 2013; Legendre, Fortin and Borcard 2015).

READING BLOCK 7

If the relation drops a lot when controlling for *Z*...	decide, from what you know, whether *Z* is a common cause (the relation was spurious) or a mediator (the partial is the direct effect).
If the relation grows or changes sign when controlling...	there is suppression: report both, the simple and the partial.
If R^2 is high but no β is significant...	the predictors share almost everything: look at the shared part and the structure, not only at the β.
If only the first canonical dimension is significant...	interpret only that one, with the circle of correlations.
If there are almost as many variables as rows...	use RV or Procrustes instead of canonical correlation.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

8

CHAPTER EIGHT · BLOCK 8

Categorical, ordinal and mixed

Not everything is measured with numbers. This block analyzes the association in a contingency table—with its test, its mosaic and some twenty measures—, the relation between a continuous and a categorical variable, the latent correlations (tetrachoric, polychoric and polyserial) and the heterogeneous matrix, in which each pair gets the coefficient its types call for.

8.1 Two categorical variables

8.2 The tests and measures of a table

8.3 The 2 × 2 table

8.4 A continuous and a categorical variable

8.5 The heterogeneous matrix

8.1 Two categorical variables

You choose the Rows and the Columns among the binary, nominal or ordinal variables (from 2 to 30 categories; the number appears in parentheses), and the app builds the contingency table.

In each cell decides what is shown: the Count, the Row %, the Column % or the Standardized residual (Haberman's adjusted residual). Whatever the choice, the background of each cell tells you whether it has more cases than there would be if the two variables were independent (blue) or fewer (red), when the adjusted residual exceeds 1.96, and it is more intense when it exceeds 2.58. If any expected frequency is below 5, a note warns you and asks for the exact or Monte Carlo *p*-value.

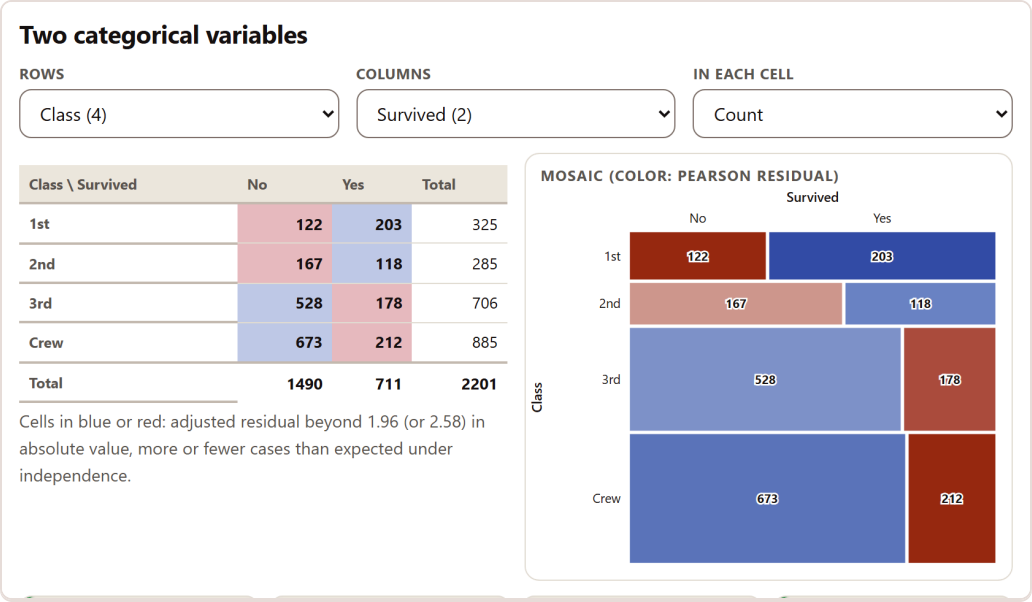


Figure 8.1. The Titanic, the example the block opens with: the class and the survival of 2,201 people. In first class more survived than expected (203, blue); in third class and in the crew, fewer (red). The mosaic draws it: the height of each band is the size of the class and the width of each rectangle, the proportion within it.

In the **mosaic** each row of the table is a horizontal band, with a height proportional to its total, and within it each column takes up the width of its proportion. The color is the Pearson residual of the cell, in the palette of Block 1: blue when there are more cases than expected, red when there are fewer.

8.2 The tests and measures of a table

Below come the main figures: Pearson's χ^2 with its degrees of freedom, the likelihood ratio G^2, Fisher's exact *p*-value (in 2 × 2 tables, with Yates' corrected χ^2) or the Monte Carlo one with 1,999 tables (in larger tables, with up to 5,000 cases), and the bias-corrected Cramér's V. The χ^2 says *whether* there is association; the measures say *how much*.

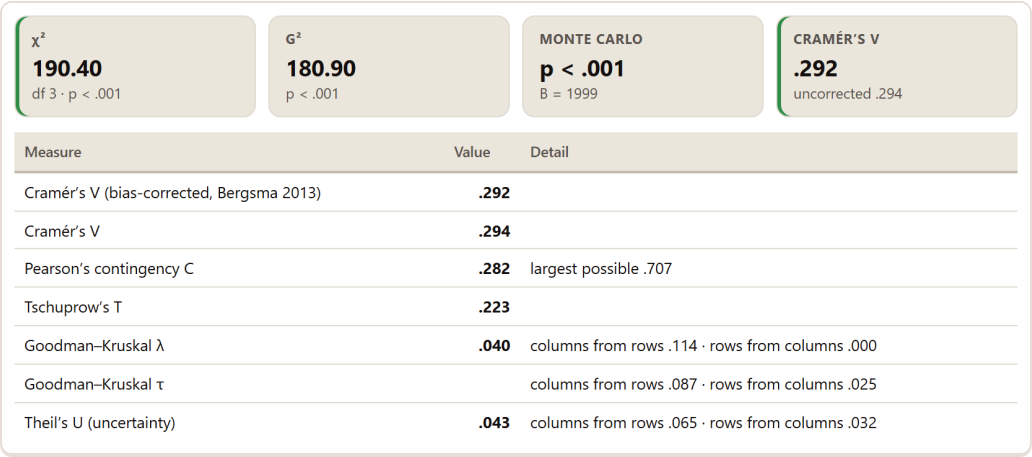


Figure 8.2. Class and survival: $\chi^2 = 190.40$ with 3 degrees of freedom and Cramér's V = .292. The asymmetric measures say that knowing the class somewhat improves the prediction of survival ($\lambda = .114$), and that knowing survival does not help at all to predict the class ($\lambda = .000$).

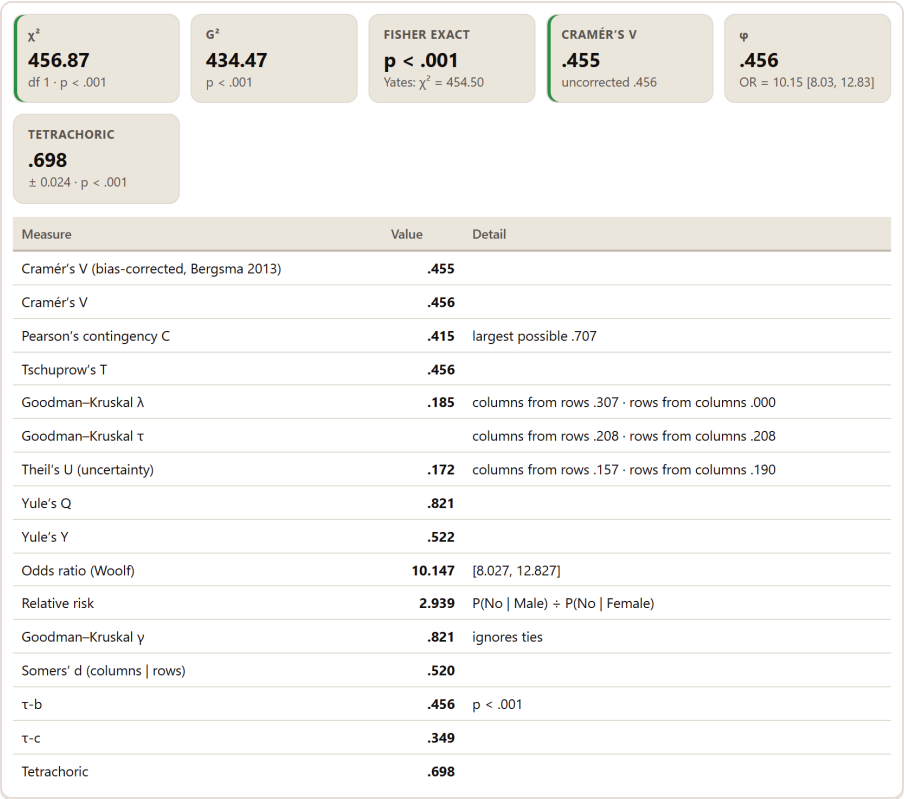
Table 8.1. The measures of a contingency table

Measure	What it says
Cramér's V	the strength of the association, from 0 to 1, for any table; the corrected one (Bergsma 2013) removes the upward bias of small samples
Contingency C	Pearson's measure; it never reaches 1, and the app gives its largest possible value for that table
Tschuprow's T	like V, but it reaches 1 only in square tables
Goodman–Kruskal λ	how much the error of guessing one variable by its most common category drops when you know the other; in both directions and symmetric
Goodman–Kruskal τ	the same, but guessing at random with the proportions; in both directions
Theil's U	the part of the uncertainty (the entropy) of one variable removed by the other; in both directions and symmetric
γ, Somers' d, τ-b, τ-c	with ordered categories (none nominal): the concordance in the order. γ ignores ties and therefore exaggerates; τ-c suits rectangular tables
Polychoric	if each variable is the cut of a latent normal, the correlation of those normals, with the test that the normal model fits

> **! WITH MANY CASES, EVERYTHING IS SIGNIFICANT**
>
> With 2,201 people, a tiny association already gives $p < .001$. The p-value of χ^2 does not measure strength: always report a measure—the corrected V, a λ, an odds ratio—and the percentages that explain it.

8.3 The 2 × 2 table

When both variables are binary, the measures specific to the 2 × 2 table appear: φ, which is Pearson's r of two variables coded 0 and 1; the **odds ratio** with its Woolf interval, which does not depend on the marginals; Yule's Q and Y, which bring the odds ratio to the scale from −1 to 1; the **relative risk**; and the **tetrachoric** correlation.



Measure	Value	Detail
Cramér's V (bias-corrected, Bergsma 2013)	**.455**	
Cramér's V	**.456**	
Pearson's contingency C	**.415**	largest possible .707
Tschuprow's T	**.456**	
Goodman–Kruskal λ	**.185**	columns from rows .307 · rows from columns .000
Goodman–Kruskal τ		columns from rows .208 · rows from columns .208
Theil's U (uncertainty)	**.172**	columns from rows .157 · rows from columns .190
Yule's Q	**.821**	
Yule's Y	**.522**	
Odds ratio (Woolf)	**10.147**	[8.027, 12.827]
Relative risk	**2.939**	P(No | Male) ÷ P(No | Female)
Goodman–Kruskal γ	**.821**	ignores ties
Somers' d (columns | rows)	**.520**	
τ-b	**.456**	p < .001
τ-c	**.349**	
Tetrachoric	**.698**	

Figure 8.3. Sex and survival. Of the women, 73% survived; of the men, 21%: φ = .456 and an odds ratio of 10.15 [8.03, 12.83]. The relative risk, 2.939, compares the first column (not surviving) between the first row and the second: among the men, the proportion of deaths was almost three times that of the women.

THE ORDER OF THE CATEGORIES DECIDES THE SIGN

The odds ratio, the relative risk, φ and the tetrachoric compare the first category with the second in the order of the table. Here sex goes "Male, Female" and survival "No, Yes": that is why φ is positive (being a woman goes with surviving). If you reverse the order of a variable in Block 2, the sign changes, not the strength.

THEORY IN BRIEF · Φ VERSUS THE TETRACHORIC

φ measures the association of the two binary variables as they are. The tetrachoric assumes that each binary variable is the cut of a continuous normal variable—the propensity to survive, for example—and estimates the correlation of those two normals; that is why it is larger (.698 against .456). Use it only if that latent variable makes sense; for two genuine categories, such as sex, φ or the odds ratio are more honest.

8.4 A continuous and a categorical variable

You choose a Continuous and a Categorical variable. The **correlation ratio η** is always computed, with its η^2 and its *F* test (that of the one-way analysis of variance): what part of the variation of the continuous variable the categories explain. Depending on the type of the categorical variable, others are added:

- **Binary**: the **point-biserial**, which is Pearson's *r* with the binary variable as 0 and 1 and is equivalent to the two-group *t* test; the **biserial**, if the binary variable is the cut of a normal; the **rank-biserial**, which does not assume normality (with the Mann–Whitney *U*); and the polyserial.
- **Ordinal**: the **polyserial** (Olsson, Drasgow and Dorans 1982), the correlation of the continuous variable with the latent normal that is cut into the categories, with its standard error and the ad hoc estimate of the same authors.
- **Nominal**: only η, which has no sign because the categories have no order.

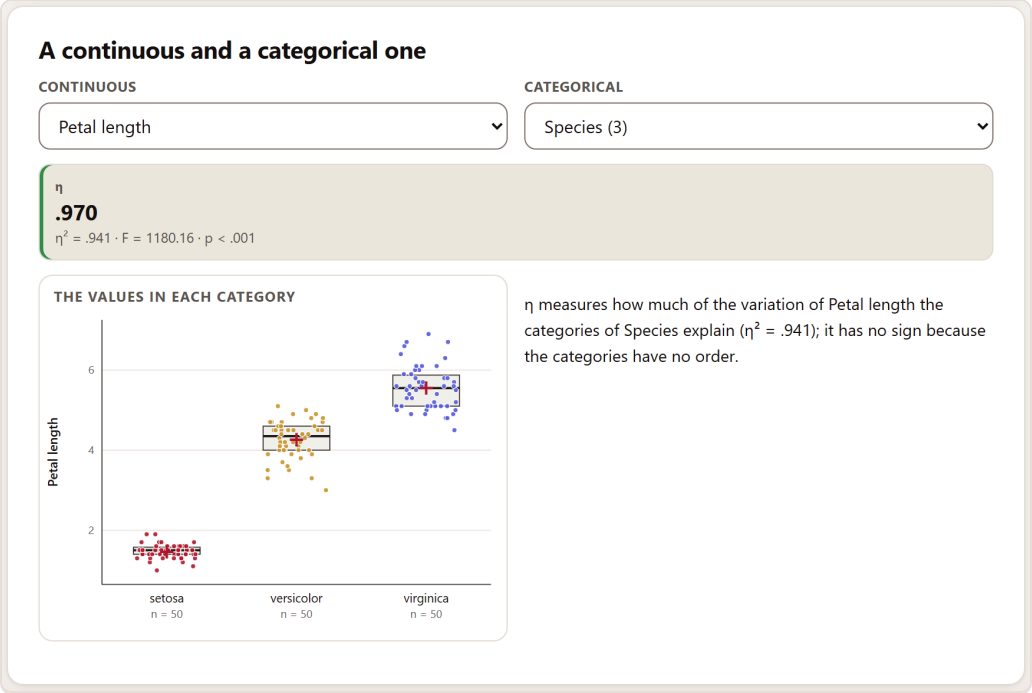


Figure 8.4. Petal length and species of the irises: η = .970, η² = .941. The species explain 94% of the variation in petal length. Each category shows its points, its box (median and quartiles) and its mean (the cross).

8.5 The heterogeneous matrix

When the data mix types, a Pearson matrix treats as numbers what are not numbers. The heterogeneous matrix gives each pair the coefficient its types call for, and a letter in the corner of each cell tells you which one it was:

Table 8.2. The coefficient for each pair of types

Pair of types	Letter	Coefficient
continuous and continuous	**P**	Pearson's *r*
continuous and ordinal or binary	**S**	polyserial
ordinal or binary with ordinal	**C**	polychoric
binary and binary	**T**	tetrachoric

Table 8.2. The coefficient for each pair of types *(continued)*

Pair of types	Letter	Coefficient
nominal and continuous	η	correlation ratio
nominal and categorical	V	bias-corrected Cramér's V

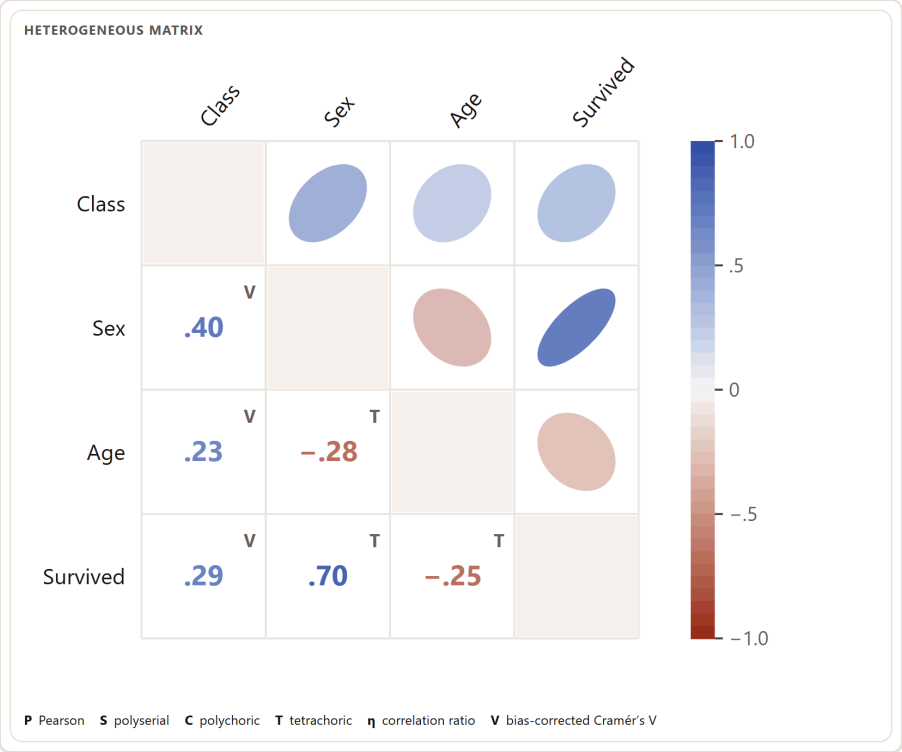


Figure 8.5. The four Titanic variables: Cramér's V for class, which is nominal, and tetrachorics for the pairs of binary variables. The strongest is that of sex with survival (.70).

The buttons at the top choose the variables, with their type beside them (at first, the first twelve). Since η and V have no sign, a note reminds you that, mixed with the others, they do not form a positive definite correlation matrix: the heterogeneous matrix is for description. For a factor analysis of ordinal variables, the right choice is a matrix of polychorics only.

READING BLOCK 8

If there are expected frequencies below 5…	report Fisher's exact *p*-value (2 × 2) or the Monte Carlo one.
If the table is 2 × 2…	report the odds ratio with its interval, and the percentages of each group.
If you want to predict one variable from the other…	use an asymmetric measure (λ, τ or *U*) in that direction.
If the categories are ordered…	use τ-b or the polychoric; V ignores the order.
If the tetrachoric or the polychoric turn out very different from φ or V…	remember that they assume a latent normal: say whether that assumption is reasonable.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

9

CHAPTER NINE · BLOCK 9

Agreement and reliability

Sometimes the question is not whether two measurements move together, but whether they say the same thing: two instruments, two laboratory methods, several raters scoring the same cases. Correlation is no use for that. This block computes Lin's concordance, the Bland–Altman plot, the six intraclass coefficients, κ with its variants and relatives, and Krippendorff's α.

9.1 What is compared

9.2 Two methods: concordance and Bland–Altman

9.3 Several raters: the intraclass coefficients

9.4 Categories: κ and its relatives

9.1 What is compared

The data come with one row per case and one column per rater, instrument or method. In What is compared you choose two or more columns that measure the same thing on the same cases.

If all the chosen columns are continuous, the block shows the concordance, the Bland–Altman plot of the first two and the intraclass coefficients of all of them. If all are categorical (binary, nominal or ordinal), it shows κ and its relatives; the columns must have the same categories, because they are compared by their position on the scale. If the two types are mixed, a message asks you to remove those of one type.

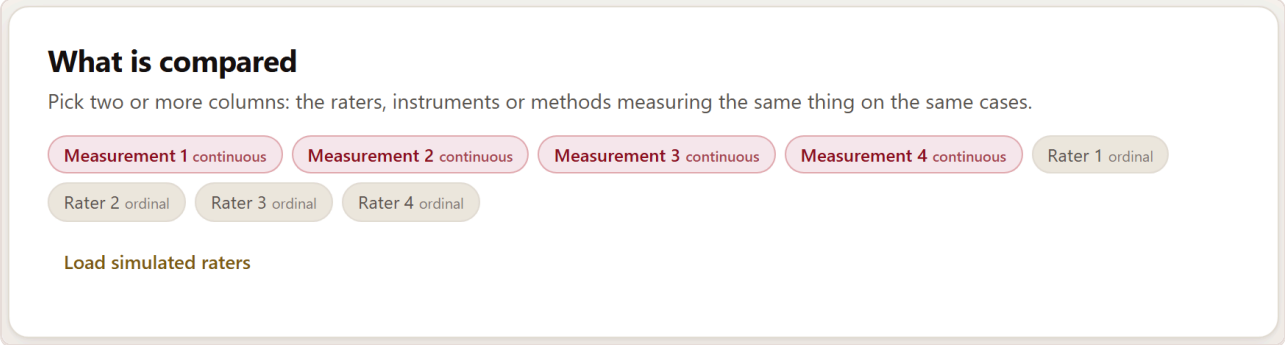


Figure 9.1. The simulated raters: 40 cases measured with four instruments, each with its own bias and noise, and rated by four raters in five categories. These are simulated data, not real ones; the app says so when it loads them.

THE TWO EXAMPLES OF THE BLOCK

Besides the Simulated raters, the block offers a real set: the extra sleep gained by ten patients with two drugs (Cushny and Peebles 1905), the data with which Student presented his *t* test. It serves to compare two measurements of the same patients, although with only ten cases.

9.2 Two methods: concordance and Bland–Altman

With the first two columns, five figures: **Lin's concordance correlation coefficient** (CCC) with its interval; Pearson's *r* and the accuracy factor C_b, because CCC = $r \times C_b$; the **mean bias** (the average difference, with its test); the 95% **limits of agreement**; and the **proportional bias**, the slope of the difference on the mean.

On the left, the cloud against the **line of equality** (dashed): if the two said the same thing, the points would fall on it, not just on some straight line. On the right, the **Bland–Altman** plot: the difference of each case against the mean of the two, with the mean bias (solid line), the two limits of agreement (dashed), the bands of their confidence intervals and, if there is one, the line of the proportional bias. Below, the reading says whether the concordance departs from *r*, whether there is mean or proportional bias, and how much a case can differ.

A HIGH R IS NOT AGREEMENT

Two instruments can correlate at .99 and always differ by ten units, or one can give twice the value of the other: *r* measures association, not agreement (Bland and Altman 1986). Lin's concordance penalizes both departures—in location and in scale—and the Bland–Altman plot shows how much they can differ on a case. Whether that difference is acceptable is not for statistics to decide: the person who uses the measurement decides, before seeing the data.

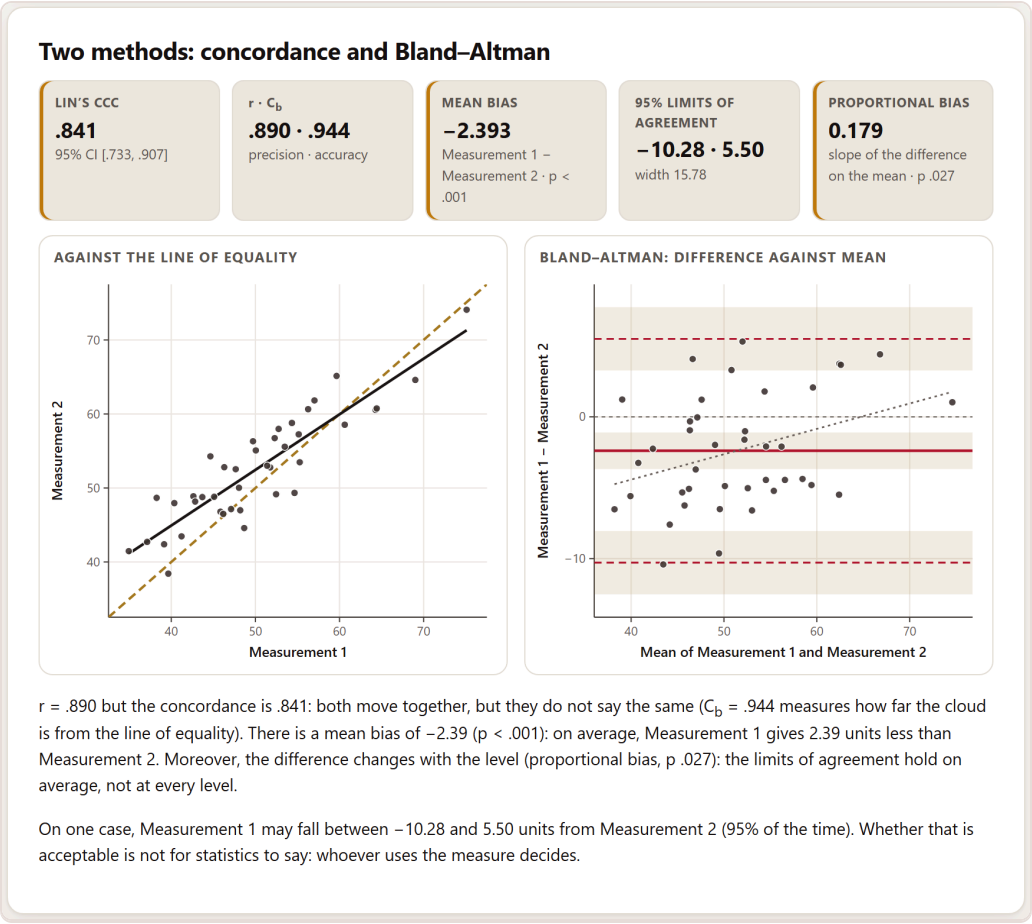


Figure 9.2. Two simulated instruments: r = .890 but CCC = .841. The first gives, on average, 2.39 units less than the second (p < .001), and the difference grows with the level (slope 0.179, p = .027). On any given case, the first may fall between −10.28 and 5.50 units from the second.

9

9.3 Several raters: the intraclass coefficients

The intraclass correlation coefficient (ICC) measures what part of the total variation is due to differences between cases rather than between raters. There are six forms (McGraw and Wong 1996), and the table gives all of them, with their interval, their F test and the reading of Koo and Li (2016). The right form is decided by the design, not by the result:

Table 9.1. How to choose the form of the ICC

If...	use
each case is rated by different raters, drawn at random	**one-way**: ICC(1) or, for the average of k raters, ICC(k)
the same raters rate all the cases, and it is enough that they rank them alike	**two-way, consistency**: ICC(C,1) or ICC(C,k)
the same raters rate everything, and they must give the same number	**two-way, absolute agreement**: ICC(A,1) or ICC(A,k)

Table 9.1. How to choose the form of the ICC *(continued)*

If…	use
in practice, the rating of a single rater will be used	the **single measure** form (…,1)
in practice, the average of the raters will be used	the **average** form (…,*k*)

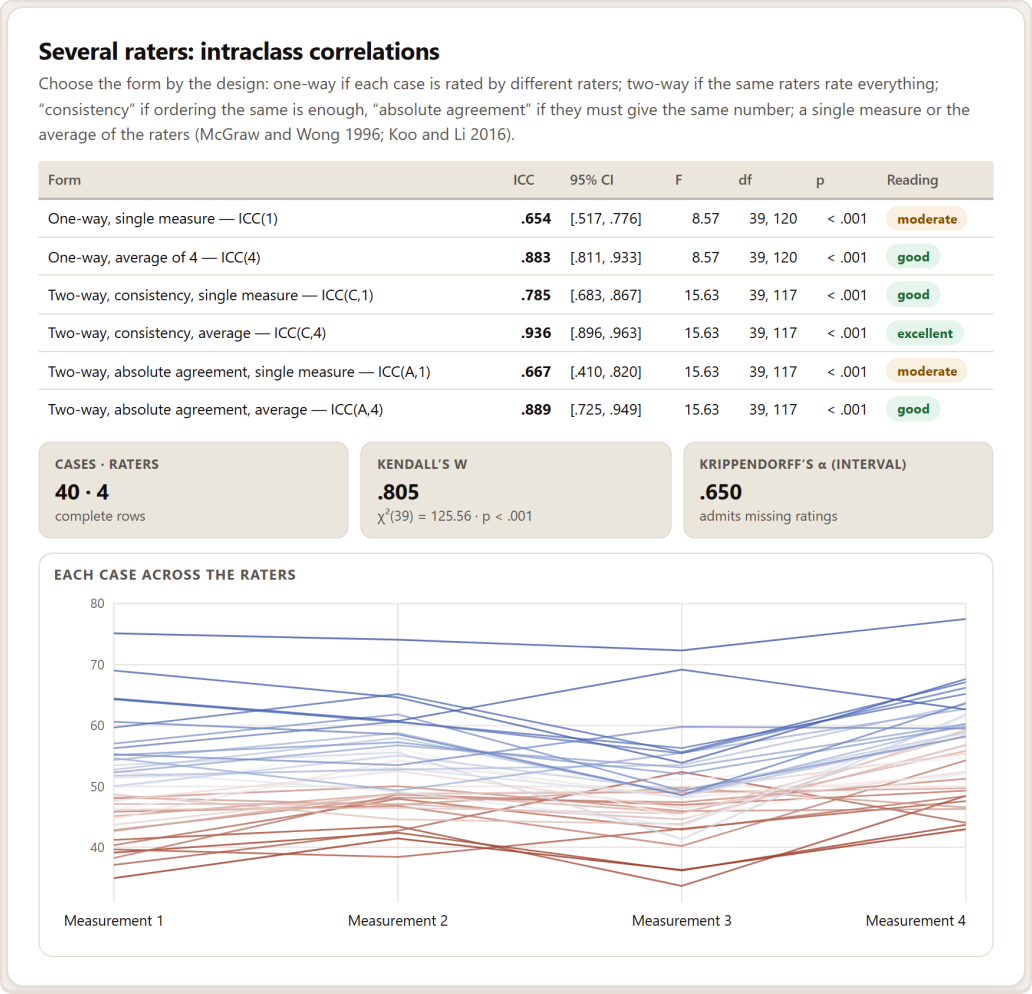


Figure 9.3. The four simulated instruments. The single-measure consistency is good (ICC(C,1) = .785), but absolute agreement drops to moderate (ICC(A,1) = .667), because the instruments have different biases. Averaging the four, both forms exceed .88. In the chart, each line is a case across the four instruments.

Below the table, three more figures: the cases and raters used (complete rows only), **Kendall's W**, the agreement in the ordering of the cases with its χ^2 test, and the interval **Krippendorff's α**, which admits missing ratings. The profile chart draws each case as a line across the raters, colored by its mean: if the raters agree, the lines are horizontal and do not cross.

Table 9.2. Reading the ICC (Koo and Li 2016)

ICC	Reliability
below .50	poor
.50 to .75	moderate
.75 to .90	good
above .90	excellent

9.4 Categories: κ and its relatives

With categorical columns, the main measure is **κ**: the observed agreement corrected for the agreement expected by chance. With two raters it is Cohen's κ (1960) or, if the categories are ordinal and there are more than two, the quadratic weighted κ; with three or more raters, Fleiss' κ. The figure comes with the reading of Landis and Koch (1977), the observed agreement and the agreement expected by chance, Gwet's AC1 and Krippendorff's α, ordinal or nominal according to the type.

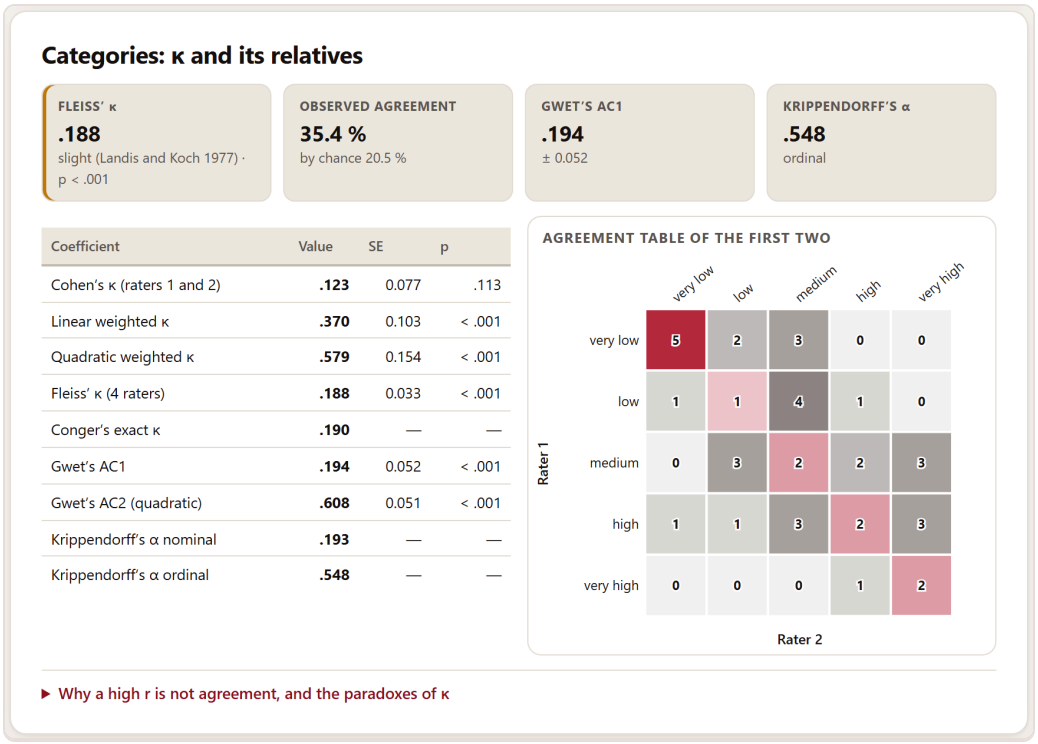


Figure 9.4. The four simulated raters, in five ordered categories. Fleiss' κ is "slight" (.188): they agree on the exact category only 35% of the time. But their disagreements are usually by one or two categories, rarely more, and the measures that use the order see it: quadratic κ of the first two .579, AC2 .608, ordinal α .548.

The table gives all the variants, with their standard error and p-value when they exist: Cohen's κ of the first two raters, unweighted, with linear and with quadratic weights; Fleiss' κ and Conger's exact κ (with three or more); Gwet's AC1 and its weighted version AC2; and Krippendorff's nominal and ordinal α. On the right, the agreement table of the first two raters: the diagonal is the exact agreement.

Table 9.3. Reading κ (Landis and Koch 1977)

κ	Agreement
below 0	poor
0 to .20	slight
.21 to .40	fair
.41 to .60	moderate
.61 to .80	substantial
.81 to 1	almost perfect

THEORY IN BRIEF · THE PARADOXES OF K

κ corrects agreement for chance, and chance depends on the marginals. With very unbalanced categories—almost all cases "healthy", for example—the agreement expected by chance is high, and κ can come out low even though the raters agree almost always (Feinstein and Cicchetti 1990). Gwet's AC1 (2008) computes chance in another way and is more stable in that case. Krippendorff's α admits any number of raters, missing ratings and nominal, ordinal, interval or ratio scales (Hayes and Krippendorff 2007).

READING BLOCK 9

If two instruments correlate highly…	do not say that they agree: report the concordance and the limits of agreement.
If there is mean or proportional bias…	correct it or describe it; the limits of agreement are not enough.
If you are going to report an ICC…	say which of the six forms, and why your design calls for it.
If the categories are ordinal…	report a weighted κ, the AC2 or the ordinal α, not the unweighted κ.
If κ is low but the observed agreement is high…	look at the marginals: it may be the κ paradox; report the AC1 as well.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

10

CHAPTER TEN · BLOCK 10

Structured data and special data

The test of Pearson's *r* assumes independent rows. Many data are not: grouped cases, subjects measured several times, consecutive years, neighboring plots, related species. And some numbers do not lie on a straight line: angles wrap around and percentages add up to one hundred. This block gathers seven tools, one for each of those cases.

10.1 Groups and Simpson's paradox

10.2 Repeated measures

10.3 Time series

10.4 Spatial data

10.5 Related species

10.6 Circular data

10.7 Compositions

Each card has its own menus, and almost all of them a **simulated example** button: there are not public-domain data sets for every case, so the app generates data with just the right structure to show the tool. When they load, the title and a message say that they are simulated, not real. Loading an example replaces the open data; to go back to yours, open them again in Block 2.

10.1 Groups and Simpson's paradox

When the cases come from different groups, there are three correlations, not one: the one within the groups, the one between their means, and the pooled one, which mixes the other two.

You choose X, Y and the Group. Four figures: the r pooling everything; the r **within the groups**, computed with the data centered on the means of their group, with $n - K - 1$ degrees of freedom; the r **between the means** of the K groups; and the Q test that all the groups have the same r (with Fisher's z).

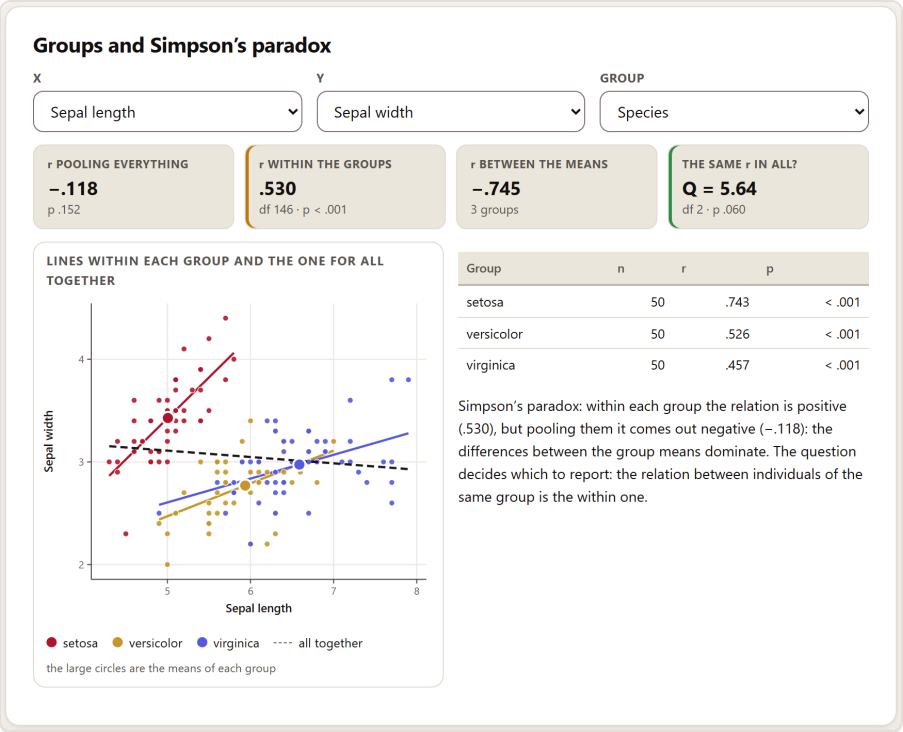


Figure 10.1. The iris sepals, the example the block opens with. Within each species the relation is positive (.530); between the species means, negative (−.745); pooled, −.118. The dashed line is for all together; the large circles are the means. The Q test (p = .060) does not reject a common r for the three species.

The table gives the r of each group. The reading warns of **Simpson's paradox** when the relation within the groups has the opposite sign to the pooled one; otherwise, it reminds you that the pooled r describes neither level when the two differ much.

> **THE QUESTION DECIDES WHICH TO REPORT**
>
> If the question is about individuals—do flowers with longer sepals have wider ones?—the answer is the within r; if it is about the groups—do species with longer sepals have wider ones?—the one between the means, with K cases, not n. Judging individuals by the correlation of the groups is the **ecological fallacy** (Robinson 1950).

10.2 Repeated measures

When each subject contributes several rows, the usual r mixes the relation within subjects with the differences between them, and also counts the degrees of freedom as if each row were a new subject. The **repeated measures correlation** (Bakdash and Marusich 2017) uses only the variation within each subject, with one common slope, and has $N - k - 1$ degrees of freedom, with N rows and k subjects.

You choose the Subject column—the one with the subject role in Block 2, or any column with repeated values—and the two variables. The figures are the correlation with its interval, the degrees of freedom and the r ignoring the subjects, in red if it has the opposite sign. The chart shows each subject in its own color, with lines of equal slope.

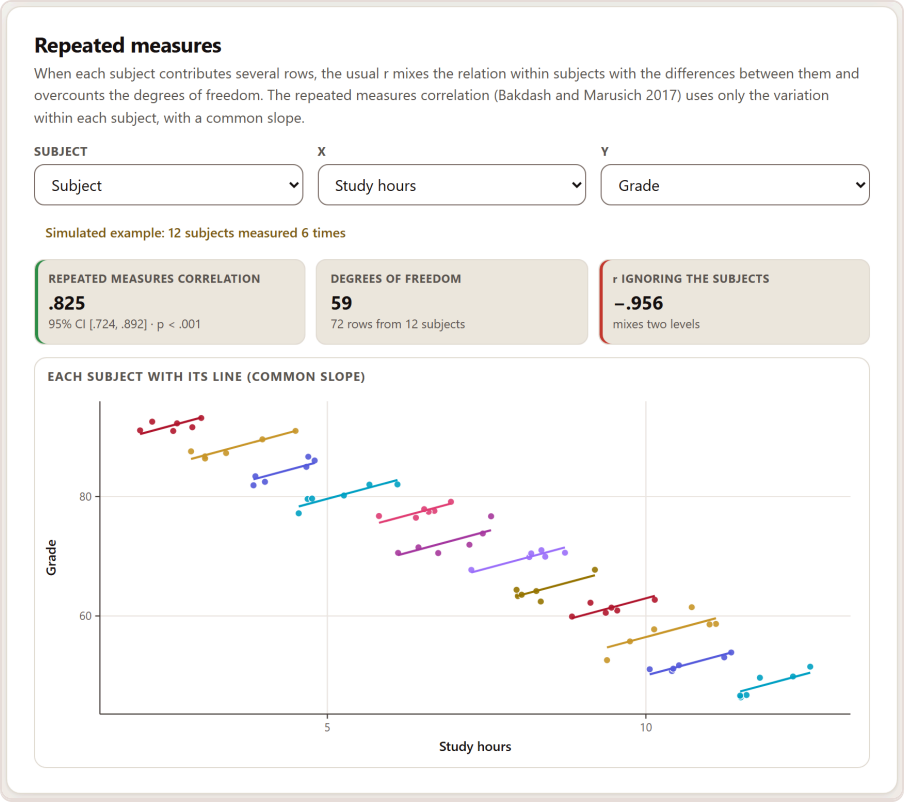


Figure 10.2. Simulated example: 12 students measured 6 times. Within each one, more study hours go with a better grade (.825, with 59 degrees of freedom); ignoring the subjects, the relation comes out the other way (−.956), because those who study more are those who start from lower grades.

10.3 Time series: cross-correlation and prewhitening

Two series with a trend or with memory correlate even if they have nothing to do with each other: this is Yule's (1926) "nonsense correlation". **Prewhitening** (Box and Jenkins 1976) fits an autoregressive model to X—the order is chosen by Akaike's criterion—, filters both series with it and only then correlates them, at each lag.

You choose X, Y and the column to Order by (if there is one with the time role, it is selected; if not, the order of the rows). At least 12 time points are needed. The figures are the r of the series, the model of X with its φ coefficients, the prewhitened r at lag zero with its limit, and the strongest lag. The two charts give the correlation at each lag k—with X ahead by k steps—before and after prewhitening; the band is the approximate limit of $\pm 2/\sqrt{n}$, and the bars that exceed it are in red.

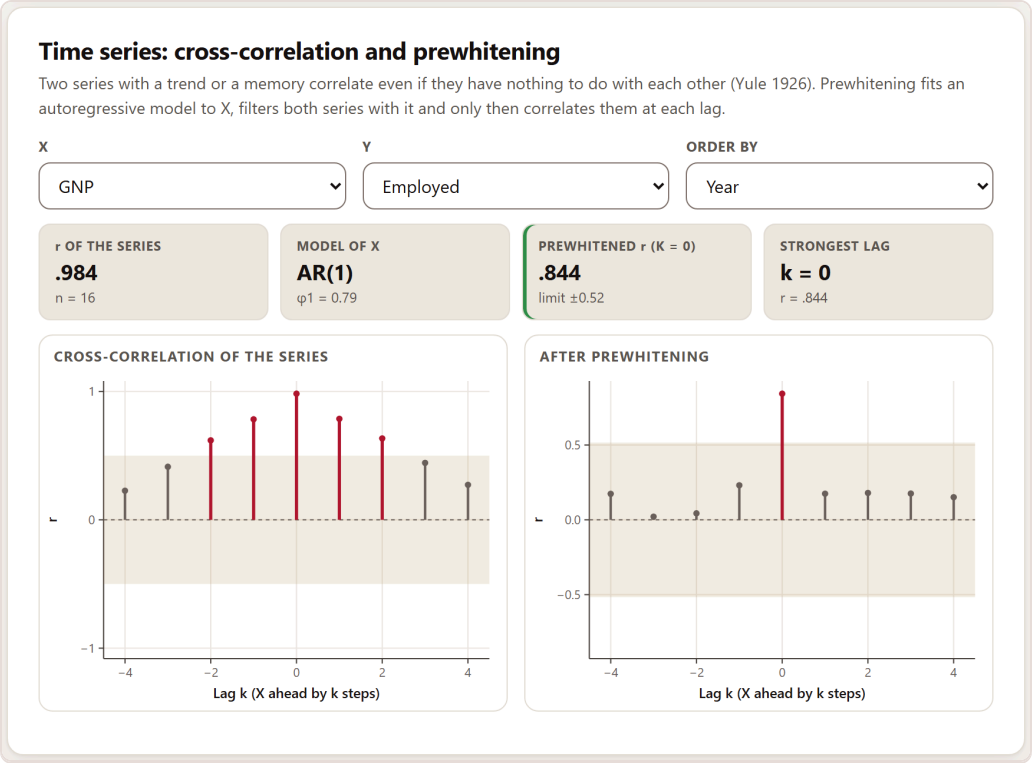


Figure 10.3. The gross national product and the employed in the United States, 1947 to 1962. Without prewhitening, the two series that grow together correlate at five lags; afterward, only at zero (.844): the relation is within the same year, and is not carried over from one year to the next.

10.4 Spatial data: Dutilleul's modified t

If nearby places resemble each other, there is less independent information than rows, and the usual p-value is too optimistic. **Dutilleul's** test (1993), which follows Clifford, Richardson and Hémon (1989), uses the autocorrelation of each variable—Moran's I by distance classes—to estimate the **effective sample size**: how many independent places the data are worth.

You choose the two variables and the two coordinates (if there are columns with the coordinate role, they are selected). The figures are r, the usual p-value, the effective size (in amber if it is less than half the places) and Dutilleul's p-value with its F. It is computed with up to 1,500 places. The map colors the places by X; the **correlogram** gives Moran's I of each variable by distance: if it starts high and falls, nearby places resemble each other.

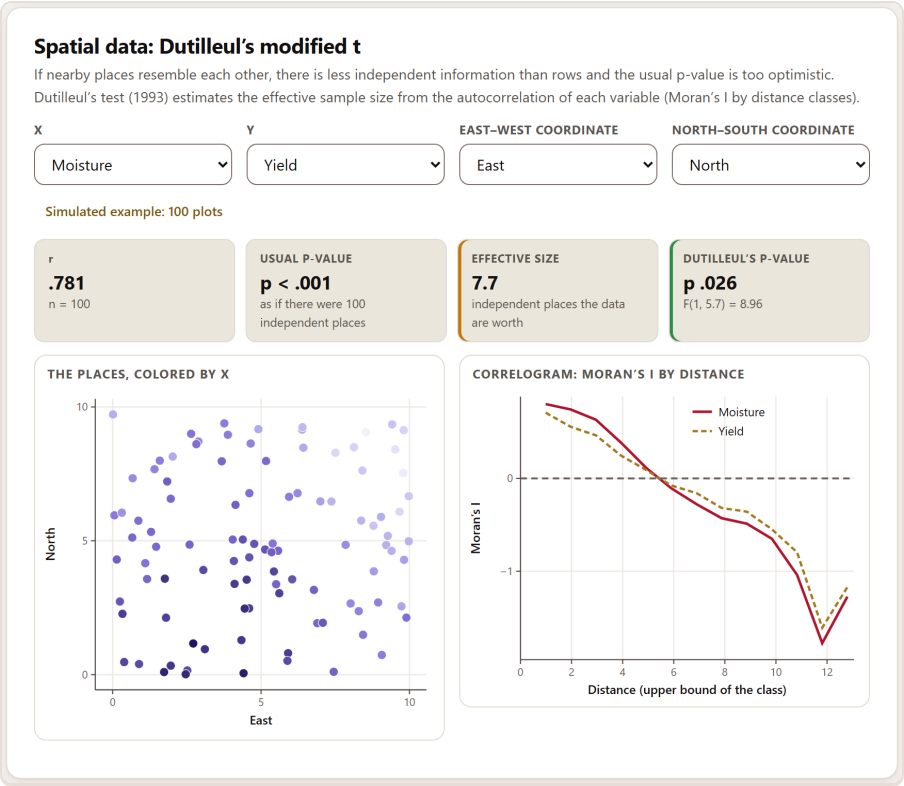


Figure 10.4. The simulated example: 100 plots with gradients of moisture and yield. The *r* of .781 looks compelling ($p < .001$), but the data are worth only 7.7 independent places, and the corrected *p*-value rises to .026.

10.5 Related species: independent contrasts

Species share evolutionary history: two sister species resemble each other because they come from the same ancestor, not because they are two independent observations (Felsenstein 1985). With the phylogenetic tree, each internal node gives an **independent contrast**: the difference between its two branches, standardized by the branch lengths. The correlation of the contrasts, which goes through the origin (Garland, Harvey and Ives 1992), is the one tested.

You choose the Column with the species names and the two variables, and paste the tree in Newick format, with branch lengths, into the box; Compute the contrasts does the calculation. The names in the tree must match those in the column exactly; if any is missing, a message says which.

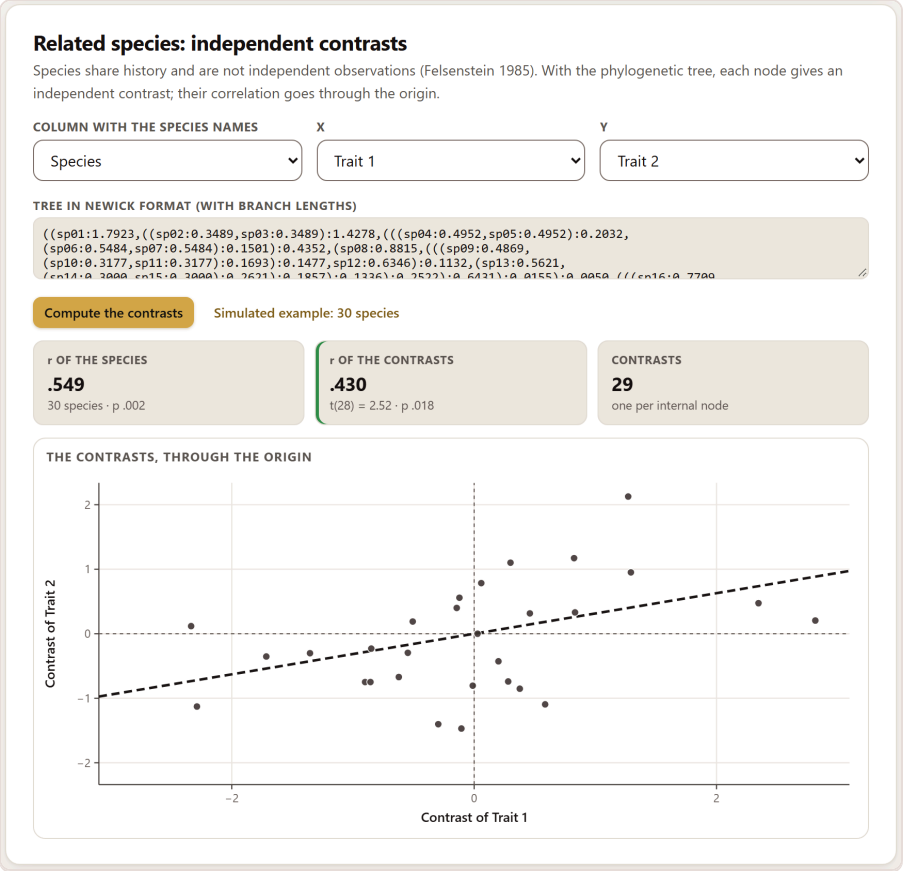


Figure 10.5. The simulated example: two traits that evolve together on a tree of 30 species. The *r* of the species (.549) exaggerates; that of the 29 contrasts is .430, with 28 degrees of freedom.

10.6 Circular data

Angles, hours, days of the year, months: 359° and 1° are next to each other, and Pearson does not know it. For two angles, the **circular–circular** correlation of Jammalamadaka and SenGupta (1988), with its *z* test; for an angle and a linear variable, Mardia's (1976) **circular–linear** correlation, which fits to the variable a sine and cosine wave of the angle. Alongside, the "naive" Pearson's *r*, to see how wrong it is.

The Angle menu offers the circular-type columns of Block 2, with their unit (degrees, radians, hours, day of the year or month); With, another circular column or a continuous one.

> **AN ANGLE AND A LINEAR VARIABLE**
>
> With the temperature in With, the same example gives the circular–linear correlation: the temperature follows the wind direction in a wave, higher when the wind comes from about 210°. The chart plots the linear variable against the angle, with the fitted wave.

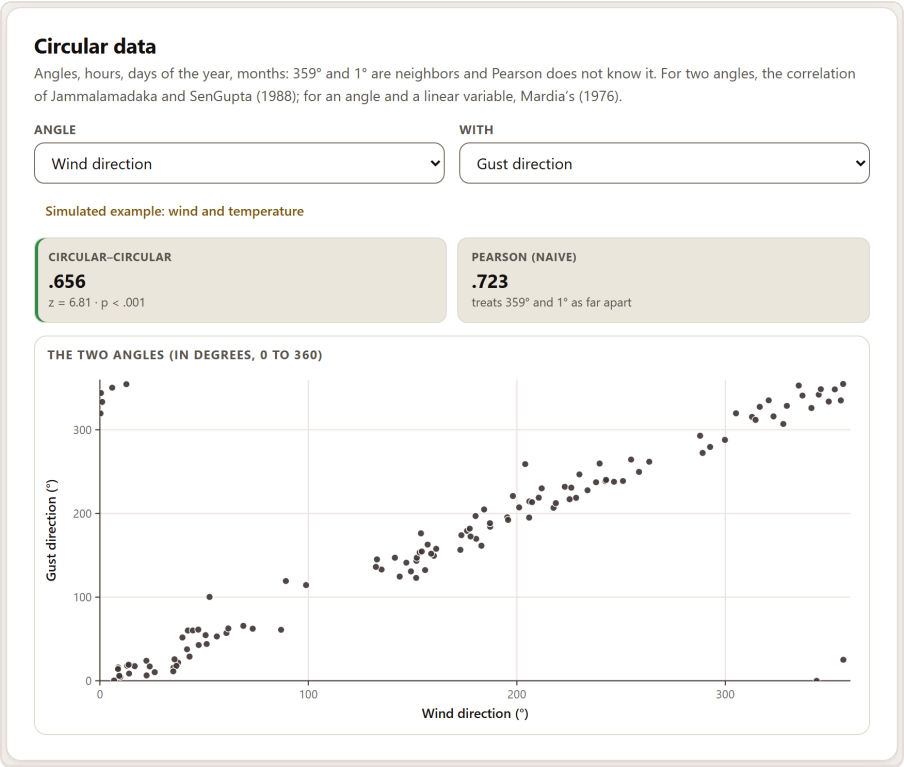


Figure 10.6. The simulated example: the wind direction and the gust direction. The pairs near 0° and 360° (top left and bottom right) are almost equal, but Pearson sees them as opposite. The circular correlation gives .656 ($p < .001$).

10.7 Compositions: parts of a whole

If the variables add up to a constant—percentages, proportions—their Pearson correlation is spurious by construction (Pearson 1897): if one rises, others must fall, and the mean of the correlations tends to be negative even if the parts are unrelated. You work with the **centered log-ratios** (Aitchison 1986) and with the **proportionality** ρ_p (Lovell et al. 2015; Erb and Notredame 2016).

Block 2 detects on its own the groups of adjacent columns that add up to a constant in every row; the Compositional group menu offers them. The three matrices compare the Pearson correlation of the parts, the Pearson correlation of the centered log-ratios and the proportionality.

THEORY IN BRIEF · PROPORTIONALITY

Two parts are proportional if their ratio is nearly constant from one sample to another. With a and b the centered log-ratios of two parts, $\rho_p = 1 - \mathrm{var}(a - b)/(\mathrm{var}(a) + \mathrm{var}(b))$; it equals 1 with perfect proportionality, and since $a - b$ is the logarithm of the ratio of the two parts, it does not depend on how the composition was closed. With only three parts, no measure fully escapes the closure: with more parts, the centered log-ratios come closer to what each part does on its own.

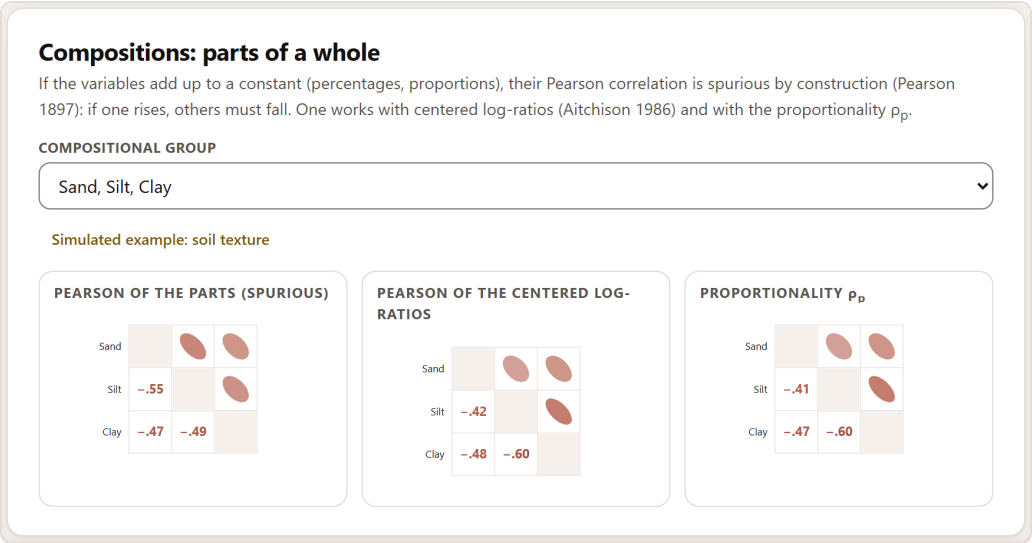


Figure 10.7. The simulated example: sand, silt and clay adding up to 100%, generated independently before being closed to 100. The three Pearson correlations of the parts are negative only because of the constant sum. Organic carbon, which is also in the data, is not part of the group.

READING BLOCK 10

If there are groups…	report the within or the between *r*, according to the question; never only the pooled one.
If each subject contributes several rows…	use the repeated measures correlation, with $N - k - 1$ degrees of freedom.
If the data are time series…	prewhiten before concluding; a high *r* between two trends says nothing.
If nearby places resemble each other…	report the effective size and Dutilleul's *p*-value.
If the cases are species…	use independent contrasts with a published tree.
If a variable is an angle or the parts add up to 100…	use the circular correlation or the log-ratios; Pearson misleads by construction.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

11

CHAPTER ELEVEN · BLOCK 11

Correlation networks

A large matrix reads better as a network: each variable is a node and each relationship, an edge. This block builds the network from correlations or from partial correlations—with or without regularization—, draws it in five ways that morph into one another, measures who sits at the center and finds the communities: the groups of variables that are more related to one another than to the rest.

11.1 Which network

11.2 The network and its five views

11.3 Centralities and communities

11.1 Which network

The numeric variables with the variable role go in; when you open a dataset, the first forty are selected. At least three are needed, and the network uses only the rows that are complete in all of them.

Edges decides which relationships are drawn; Coefficient, whether to start from Pearson or from Spearman.

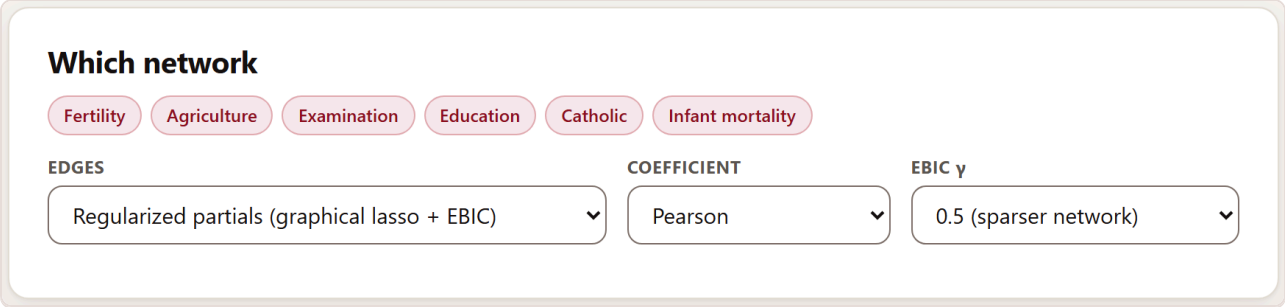


Figure 11.1. Switzerland, 1888, the example the block opens with: six variables, the regularized partial network, Pearson and γ = 0.5.

Table 11.1. The four ways of building the network

Edges	What is drawn
Regularized partials (the default)	the partial correlations of the **graphical lasso** (Friedman, Hastie and Tibshirani 2008), which sets the weak ones to zero; how many is chosen by the EBIC criterion (Foygel and Drton 2010) with its EBIC γ: 0.5 gives the sparsest network, 0 is the BIC
Significant partials	the partial correlations—each pair removing all the others—whose *p*-value, corrected by Holm, is below .05
Correlations with |r| ≥ threshold	the marginal correlations that pass the Threshold |r|, from .05 to .90
Significant correlations	the significant marginal correlations with Holm's correction

If the matrix is not positive definite, the two partial networks use the nearest one (Section 5.5) and a message says so. With fewer than three complete rows per variable, another message warns that the network is unstable. With more than 25 variables, the graphical lasso runs through its 100 penalties with a progress screen.

> **! MARGINAL OR PARTIAL: TWO DIFFERENT NETWORKS**
>
> In a network of marginal correlations almost everything is connected to everything, because many variables share a common cause. Partial correlations show only the relationships that remain after controlling for all the other variables: an edge means "they are related even knowing everything else." Both are useful, but they answer different questions, and the report must say which one was drawn.

11.2 The network and its five views

The title says how many nodes and edges the network has and how it was built (with the chosen lasso λ). The thickness of each edge is its |weight|, and its color, the sign: blue for positive and red for negative ones, in the Block 1 palette. The size of each node is its strength and its color, its community; nodes without edges are gray.

Hovering the pointer over a node highlights its edges and its neighbors; a click pins it and writes below it its community, its strength and its neighbors ordered by weight. The buttons at the top change the view, and the change is animated: each node and each edge travels to its new place.

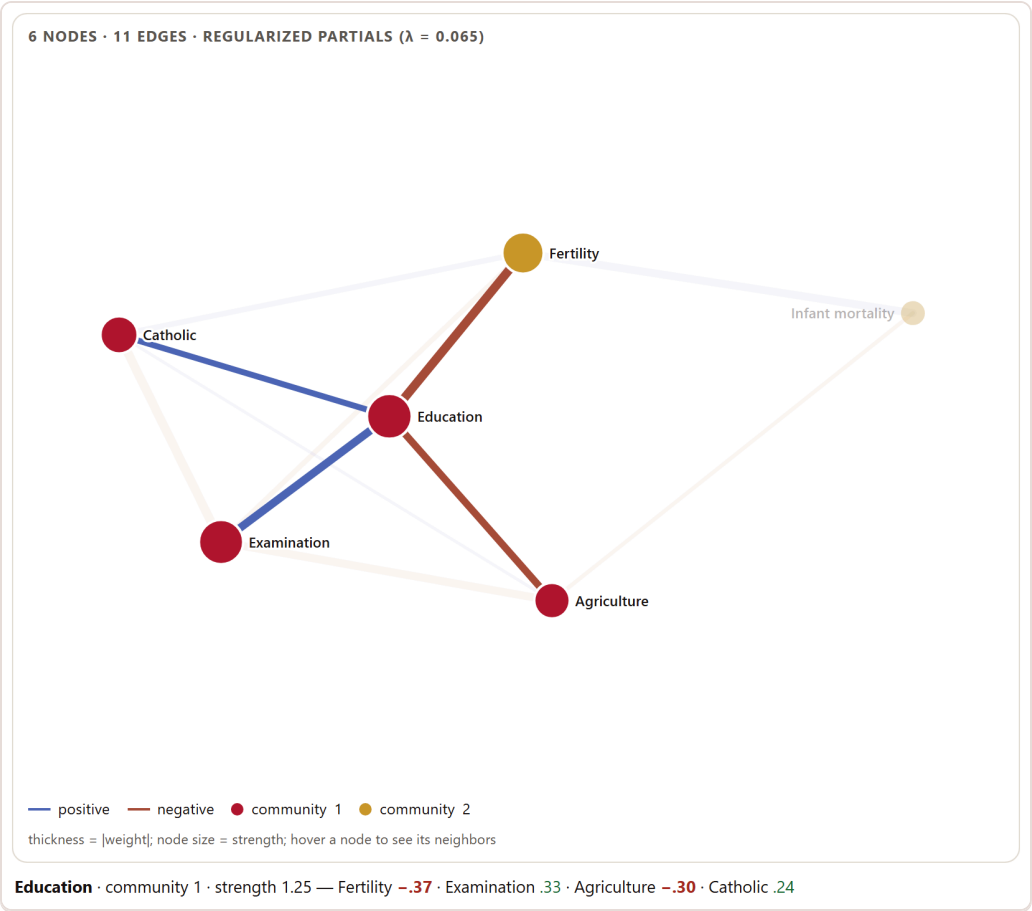


Figure 11.2. Switzerland, 1888: 11 edges out of 15 possible (λ = 0.065). With a click on Education, its edges are highlighted and the line below gives its neighbors with their weights: Fertility −.37, Examination .33, Agriculture −.30 and Catholic .24.

- Forces: the nodes arrange themselves as if the edges were springs (Fruchterman and Reingold 1991): strongly related ones end up close together.
- Circle: the nodes on a circumference, ordered by community.
- Bundled chords: the same circle, with the edges curved toward the center of their community: the relationships between communities show up as bundles.
- Arcs: the nodes on a line, with the edges as arcs above it.
- Matrix: the weights as colored cells, in the order of the communities.

▶ Tour: matrix → network → chords goes through the matrix, the forces, the circle and the chords, one after another, to show that they are the same information drawn in four ways.

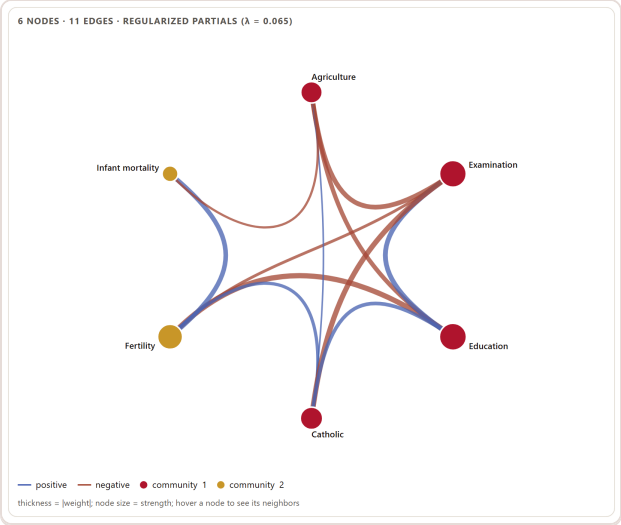


Figure 11.3. The same network, as bundled chords.

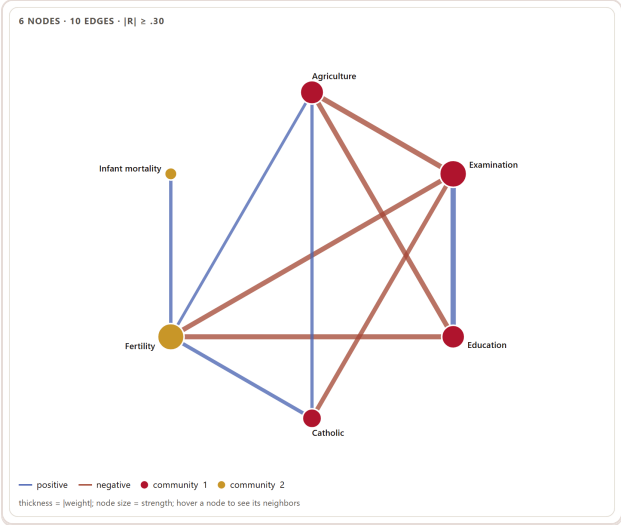


Figure 11.4. The marginal network with $|r| \geq .30$, as a circle.

The two figures above show the difference described in the previous section. In the marginal one, Agriculture and Fertility are joined ($r = .35$); in the regularized partial one that edge is missing, because the correlation runs through Education.

11.3 Centralities and communities

Four figures summarize the network: the edges with the density (the share of the possible edges that are present), the communities with their modularity Q, the strongest node and the rows used. The table gives, for each node, sorted by strength, five centralities, and the chart below, its strength relative to that of the strongest node, in the color of its community.

Table 11.2. The centralities

Centrality	What it measures
Degree	how many edges the node has
Strength	the sum of the |weights| of its edges: the most stable and the most widely used
Expected influence	the sum of the weights with their sign: if it is negative, the node holds its neighbors back more than it drives them
Closeness	how close it is to all the others, with path lengths of 1/|weight| (Opsahl, Agneessens and Skvoretz 2010)
Betweenness	how many shortest paths between other nodes pass through it (Brandes 2001)

The **communities** maximize modularity (Newman 2006) with the Louvain method (Blondel and colleagues 2008): the network is split into groups of nodes with more edges inside them than between them. A modularity near zero says that the partition can hardly be told apart from a random one.

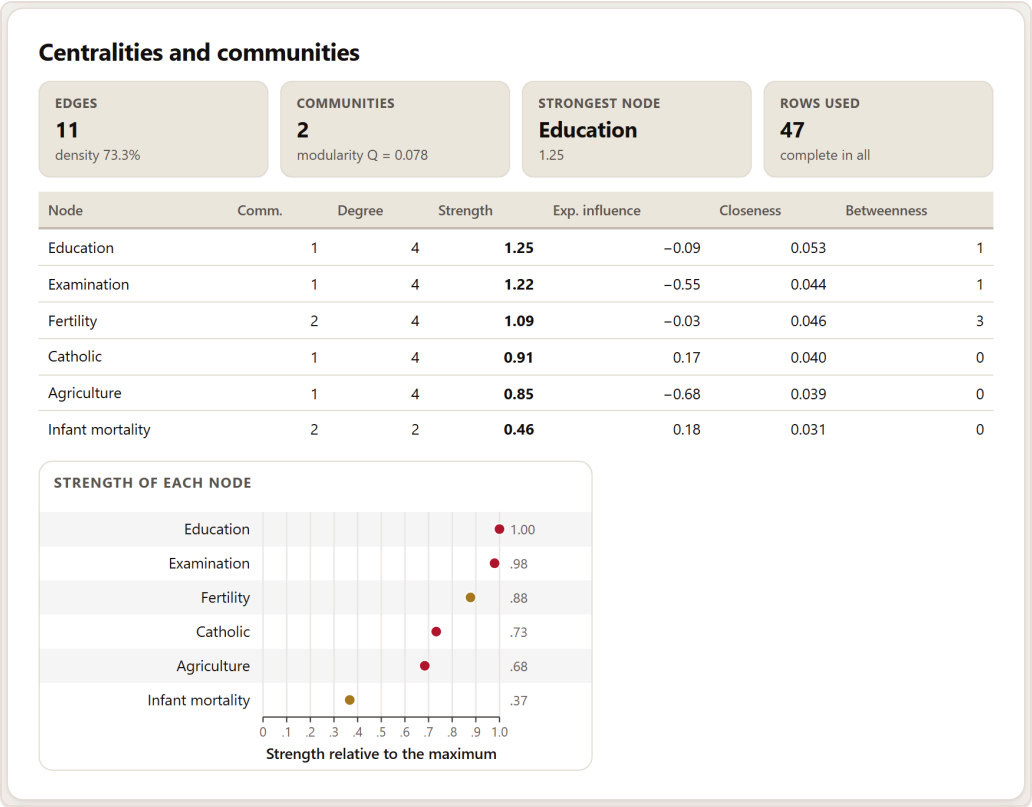


Figure 11.5. Education and Examination are the strongest nodes; Infant mortality, the weakest. Modularity is low ($Q = 0.078$): with six variables this closely related, the two communities barely separate.

THEORY IN BRIEF · HOW MUCH TO TRUST A NETWORK

A network estimated from few rows changes a lot from one sample to another: weak edges appear and disappear, and the order of the centralities shifts. Strength is the most stable centrality; closeness and betweenness, the least (Epskamp and Fried 2018). With many variables and few rows, the graphical lasso with $\gamma = 0.5$ is conservative on purpose: it would rather lose a weak edge than invent one.

READING BLOCK 11

If the question is what is directly related…	use a partial network, regularized if there are many variables.
If there are few rows per variable…	interpret only the strong edges and the strength; not betweenness.
If modularity is low…	do not talk about groups of variables: the communities cannot be told apart.
If you are going to publish the network…	state how the edges were built, the coefficient, the EBIC γ or the threshold, and how many rows were used.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

12

CHAPTER TWELVE · BLOCK 12

Compare, correct, plan and combine

The questions that come after computing a correlation: is it different from another one? Is the structure of two groups the same? What would it be worth without measurement error or without the selection of the cases? How many cases are needed? What do all the studies say together? Almost all of this block works without open data: typing in the published figures is enough.

12.1 Comparing two correlations

12.2 Comparing two matrices

12.3 Correcting for attenuation and range

12.4 Power and sample size

12.5 The Bayes factor of a published r

12.6 Meta-analysis

The block has five sections, chosen with the buttons at the top: Compare two correlations, Compare matrices, Correct, Power and sample size and Meta-analysis. Each one is recomputed as soon as a number changes.

12.1 Comparing two correlations

Is the correlation larger in one group than in another, or does one variable predict better than another? The right test depends on the **design**, and picking the wrong one is one of the most common mistakes when comparing correlations.

- Two independent groups: the same correlation in two different samples. You type the r and n of each group.
- Same cases, one variable in common: r_{jk} against r_{jh}, for example, whether the exam is more related to the final grade or to attendance. You also need r_{kh}, the correlation between the other two.
- Same cases, no variable in common: r_{jk} against r_{hm}, with the four cross correlations.

You also choose the Alternative hypothesis (different, or one greater than the other) and the Confidence of the interval. Take them from the open data computes the correlations of the variables and groups you choose; Example: iris by species compares the sepal and the petal of two species.

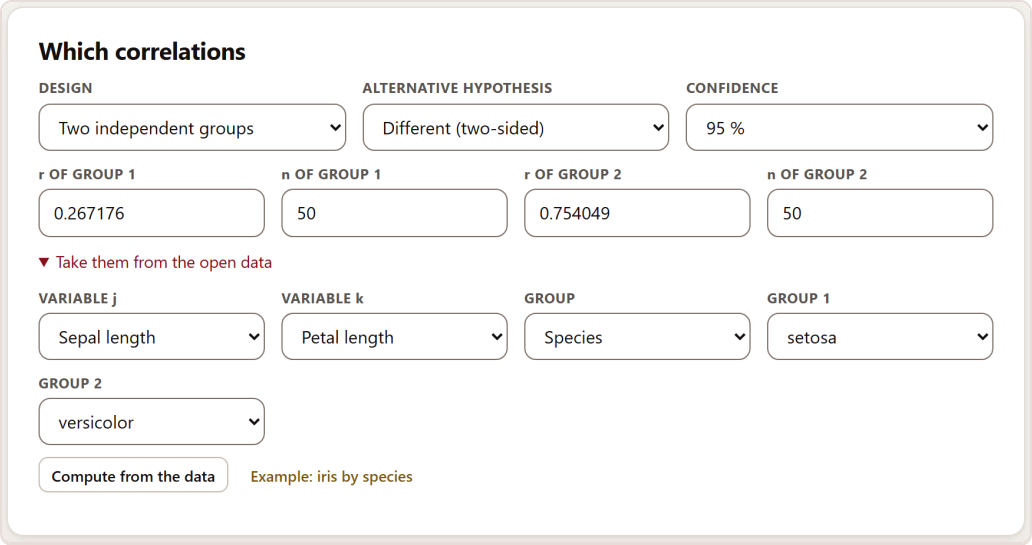


Figure 12.1. The iris example: the correlation of sepal length with petal length in setosa (.267) and in versicolor (.754), 50 flowers each.

The result gives the difference, **Zou's interval** (2007) for that difference—which says how much they differ, not only whether they differ—and the recommended test for the design. The table gives all the tests that apply, with the recommended one marked, and the chart shows the two correlations with their intervals and the difference with its own.

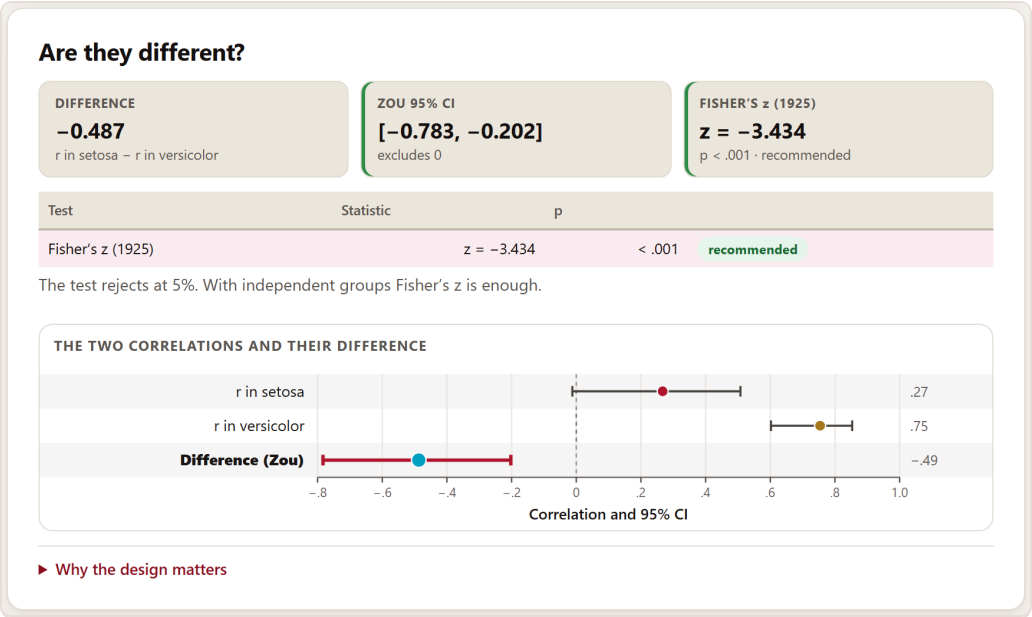


Figure 12.2. The difference is −.487, with interval [−.783, −.202]: in versicolor the relationship is clearly stronger (z = −3.43, p < .001).

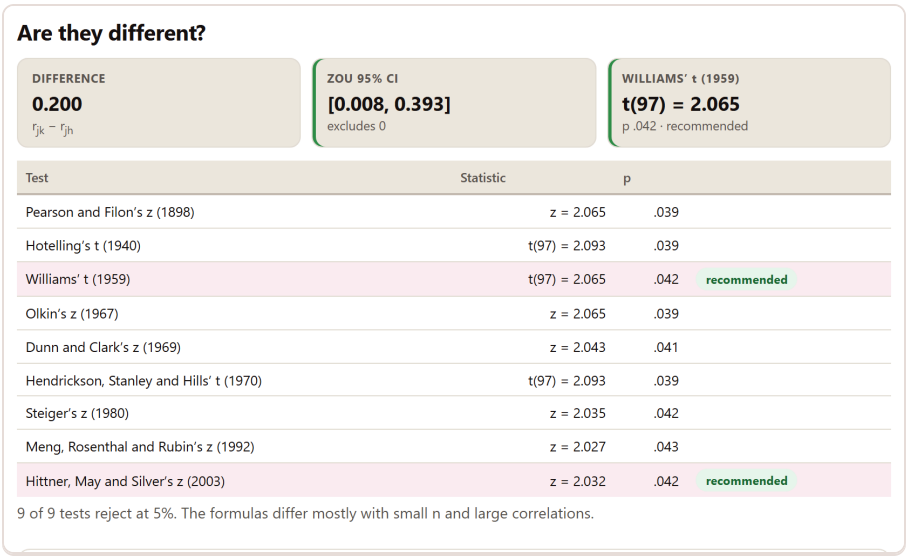


Figure 12.3. With one variable in common (the default values: r_{jk} = .50, r_{jh} = .30, r_{kh} = .40, n = 100) there are nine tests, from 1898 to 2003. They agree here, but they can differ with small n and large correlations.

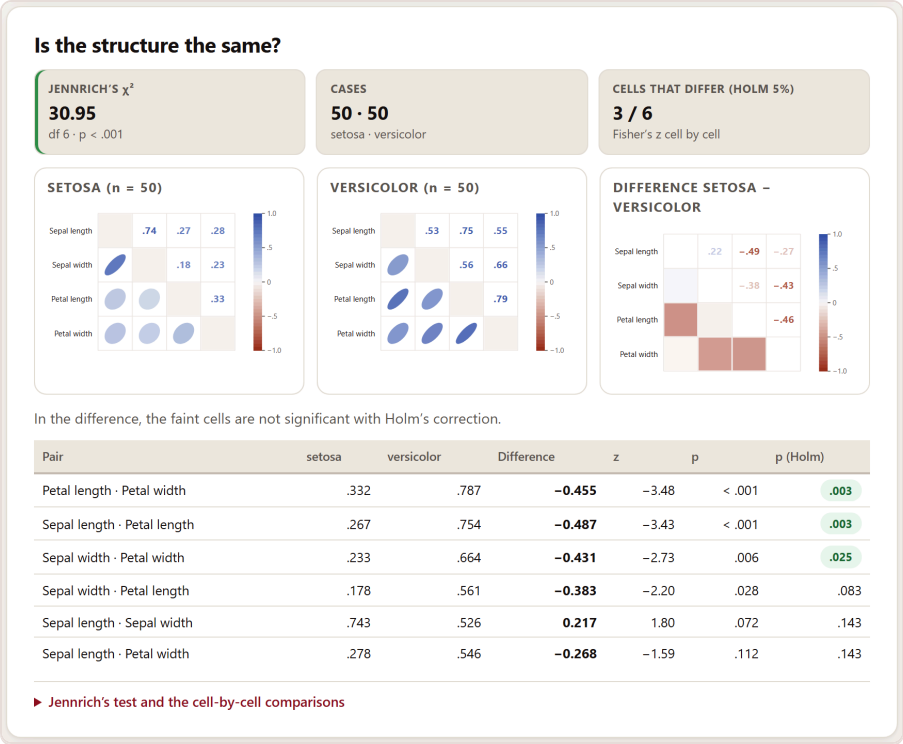
Table 12.1. The recommended tests for each design

Design	Recommended
two independent groups	the difference of Fisher's z (1925)
one variable in common	Williams' t (1959) and the z of Hittner, May and Silver (2003)
no variable in common	that of Silver, Hittner and May (2004) and that of Raghunathan, Rosenthal and Rubin (1996)

> **COMMON ERROR · TREATING TWO CORRELATIONS FROM THE SAME CASES AS INDEPENDENT**
>
> If the two correlations come from the same subjects, they are correlated with each other, and the independent-groups test gives a wrong *p*-value: almost always too large, when the variables are positively related. And one correlation being significant while the other is not does not prove that they are different: only the test of their difference says that.

12.2 Comparing two matrices

Are the variables related in the same way in two groups? You choose the Grouping column, the two groups, the coefficient and, with the buttons, the variables. **Jennrich's test** (1970) compares the two whole matrices with a χ^2 that takes into account that the correlations of a matrix depend on one another. Then each cell is compared separately with Fisher's *z* and Holm's correction.



Pair	setosa	versicolor	Difference	z	p	p (Holm)
Petal length · Petal width	.332	.787	**−0.455**	−3.48	< .001	.003
Sepal length · Petal length	.267	.754	**−0.487**	−3.43	< .001	.003
Sepal width · Petal width	.233	.664	**−0.431**	−2.73	.006	.025
Sepal width · Petal length	.178	.561	**−0.383**	−2.20	.028	.083
Sepal length · Sepal width	.743	.526	**0.217**	1.80	.072	.143
Sepal length · Petal width	.278	.546	**−0.268**	−1.59	.112	.143

Figure 12.4. Setosa against versicolor: the matrices are different (χ^2 = 30.95 with 6 degrees of freedom, $p < .001$), and three of the six correlations differ after correction. In the matrix of the difference, the faint cells are not significant.

WHY ADDING UP THE CELLS IS NOT ENOUGH

Adding up the z^2 of each cell only gives a χ^2 when all the correlations are close to zero. With correlated variables it rejects too often—in the simulation of this block's validation, 11% of the time instead of 5%—, and that is why here the cells serve to read where the difference lies, not to decide whether there is one. With Spearman, Jennrich's test is approximate, and a message says so.

12.3 Correcting for attenuation and range

Two reasons why an observed correlation can fall short, and their classical corrections.

Attenuation. If both measures carry error, the observed correlation underestimates that of the true scores. Spearman's correction (1904) divides by the square root of the product of the **reliabilities**—an α, a test–retest or an ICC from Block 9—. The figures are the corrected *r*, the **ceiling** of the observable *r* (no observed *r* can exceed it) and the part of the correlation taken away by measurement error.

$$r_{\text{corrected}} = r \,/\, \sqrt{(r_{xx} \cdot r_{yy})}$$

Restricted range. If only cases with values of *x* inside an interval were observed—the admitted, the survivors, the selected—, the correlation comes out lower than in the full population. Thorndike's **case II** (1949) corrects direct selection on *x*, with *U*, the ratio of the unrestricted standard deviation to the restricted one; **case III**, selection on a third variable *z*, with its two correlations.

The chart of the case II correction, below the figures, shows how the corrected *r* grows with *U* for several observed *r* values, with yours in red and its point: the correction is larger the stronger the selection was.

CORRECTING DOES NOT CREATE INFORMATION

Both corrections assume a linear, homoscedastic relationship, and that the selection was the one stated. A correction above 1 indicates that some reliability is underestimated or that the errors are not independent. And the corrected interval is the observed one transformed with the same formula: wider in the same proportion (Hunter and Schmidt 2004; Sackett and Yang 2000). Always report the observed *r* next to the corrected one.

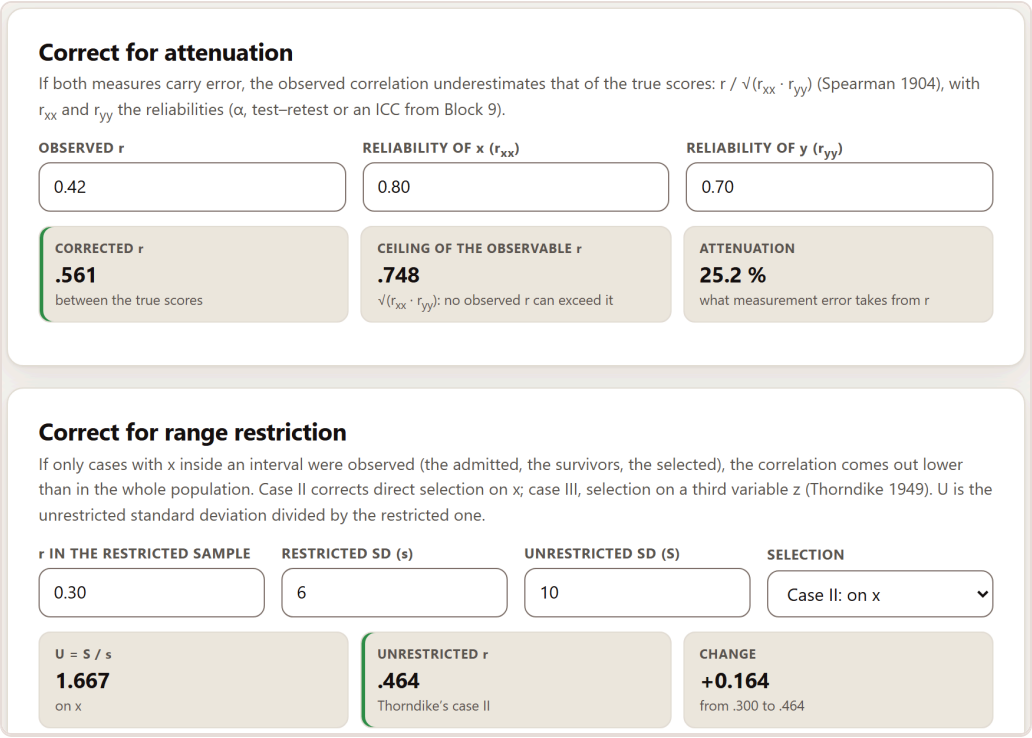


Figure 12.5. The default values. An *r* of .42 with reliabilities of .80 and .70 is worth .561 between the true scores: error takes away 25%. An *r* of .30 in a sample with a standard deviation of 6, when that of the population is 10, is worth .464 without the restriction.

12.4 Power and sample size

Before collecting the data: with the Expected ρ and the n of the two sliders, the α and the Test (two-sided or one-sided), the block gives four figures:

- the **exact power** with that *n*, and the approximate one;
- the ***n* for the desired power** (80, 90 or 95%), exact and approximate;
- the ***n* for an interval of the desired half-width** (from ±0.05 to ±0.20), with the expected interval;
- the **critical *r***: the smallest |*r*| that would be significant with that *n*.

> **THEORY IN BRIEF · EXACT AND APPROXIMATE**
>
> The exact power integrates the density of *r* when the true correlation is ρ (Fisher 1915; Hotelling 1953) over the region where the *t* test rejects; the approximate one uses Fisher's *z* with a bias correction and departs by a few thousandths with small *n*. With ρ = 0 the exact power is α, as it should be. Planning by the width of the interval, and not only by power, guarantees a useful estimate even if the effect turns out to be small (Bonett and Wright 2000).

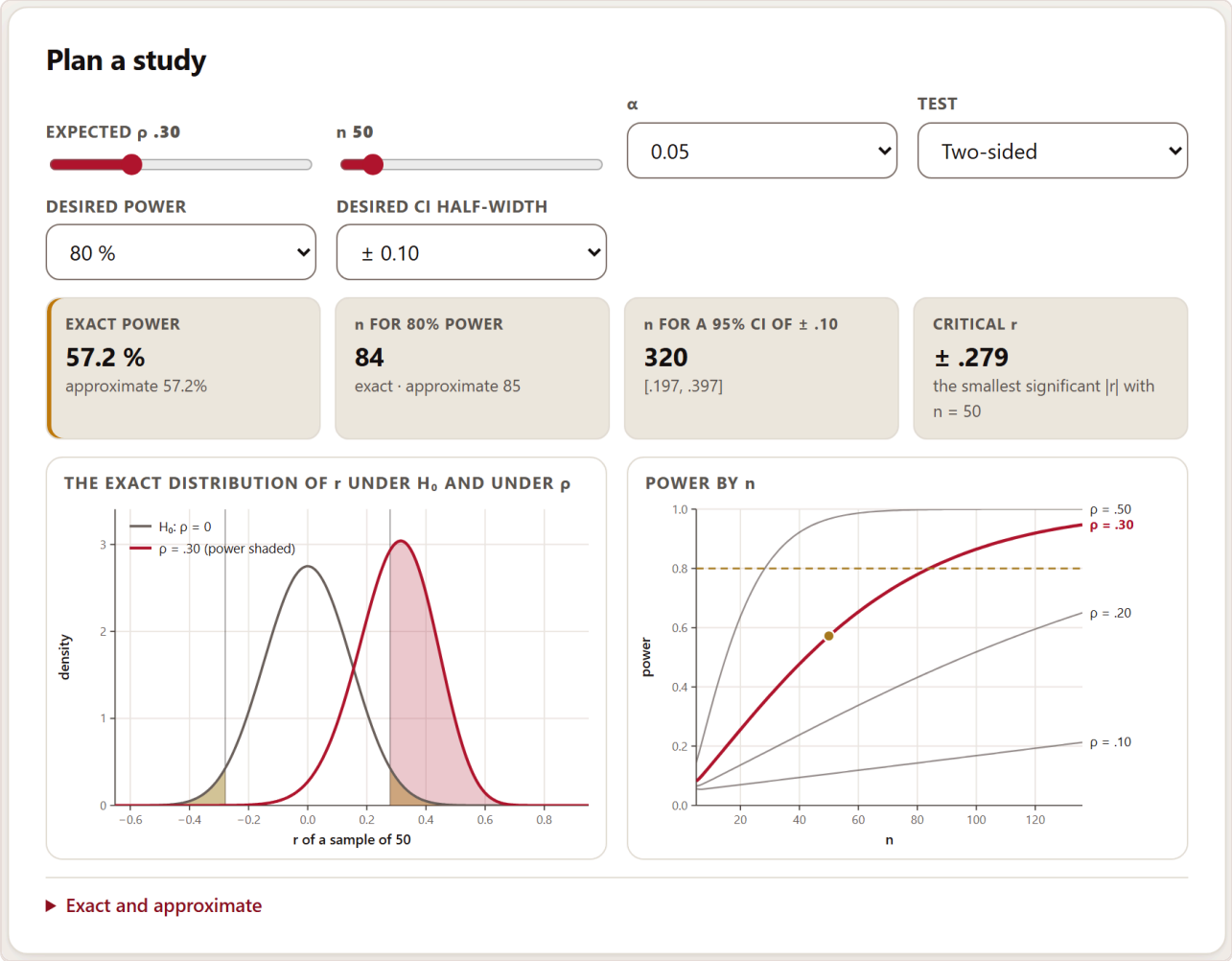


Figure 12.6. With ρ = .30 and 50 cases, the power is only 57.2%; for 80%, 84 are needed. But to estimate ρ with an interval of ±.10, 320 are needed. On the left, the exact distribution of *r* with no relationship and with ρ = .30; the shaded part is the power. On the right, power by *n* for four values of ρ.

12.5 The Bayes factor of a published r

With only the *r* and the *n* of a paper you get the evidence for there being a correlation against there being none, with the same prior as in Block 4 (Section 4.6) and its width κ. The figures are BF_{10} with its word, BF_{01}, the one-sided BF_{+0} ($\rho > 0$) and the posterior ρ with its credible interval.

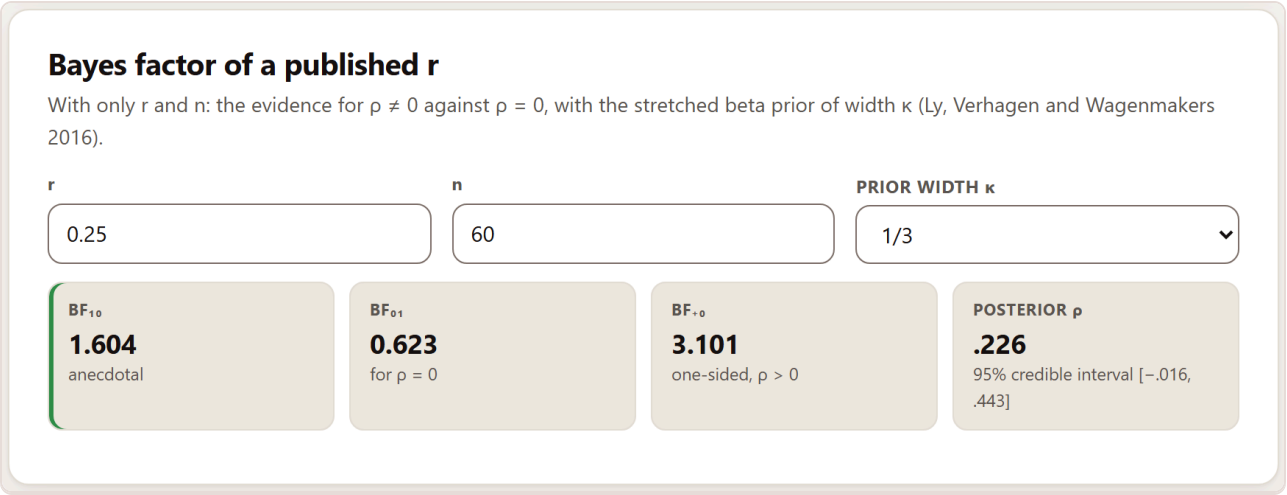


Figure 12.7. An *r* of .25 with 60 cases sits at the edge of significance: $p = .054$ two-sided and .027 one-sided. Its Bayes factor is 1.60, anecdotal evidence; the one-sided one, 3.10, barely moderate.

12.6 Meta-analysis

Combines the correlations of several studies. You type one study per line—name; *r*; *n*, separated by semicolons or tabs—or take them from columns of the open data with Use columns of the data. The correlations are combined on Fisher's *z* scale, with variance $1/(n - 3)$. The Model can be random effects—with REML (the default), DerSimonian–Laird or Hunter–Schmidt—or fixed effect.

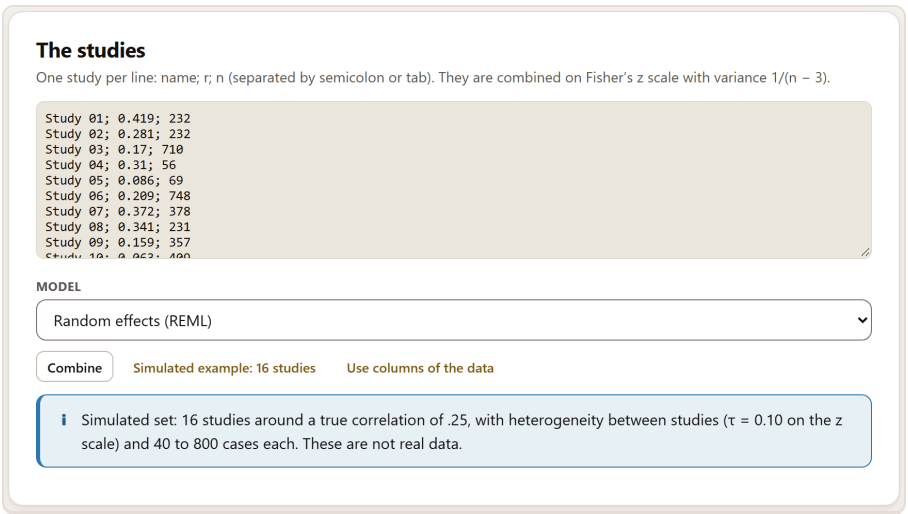


Figure 12.8. The example, loaded by the Simulated example: 16 studies button: studies generated around a true correlation of .25, with heterogeneity among them (τ = 0.10 on the *z* scale) and from 40 to 800 cases each. They are not real data, and the app says so when it loads them.

Five figures summarize the result: the combined *r* with its interval and its *p*-value; the **prediction interval**, where the ρ of a new study would fall; the heterogeneity τ and I^2 with the *Q* test; Egger's asymmetry test; and the Hunter–Schmidt *r* with the share of the variance explained by sampling error.

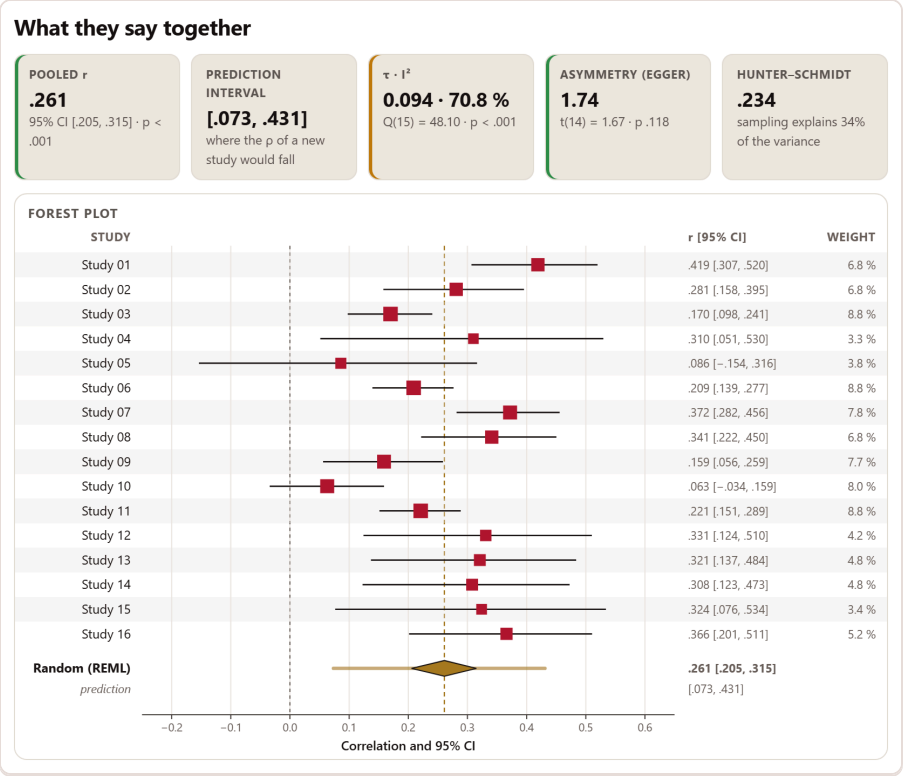
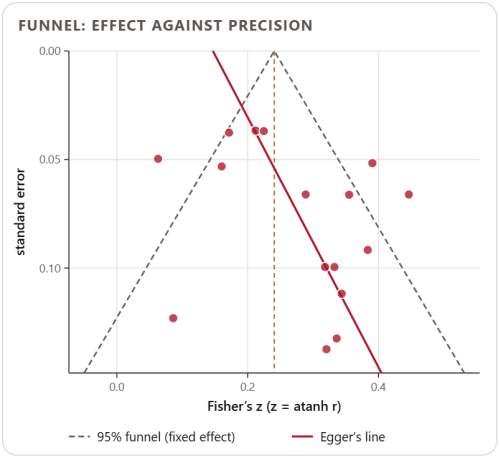


Figure 12.9. The 16 studies give r = .261 [.205, .315]—the interval contains the .25 they were simulated with—, but they are heterogeneous (I^2 = 70.8%), and a new study could find anything from .073 to .431. In the forest plot, the size of each square is its weight; the diamond is the combined estimate and the golden band, the prediction interval.

Below, the **funnel** places each study by its effect and its standard error: without bias, the studies form a symmetric triangle around the combined effect. Egger's line measures the asymmetry. Next to it, the reading in words, and below, the table with the four models side by side and the Hunter–Schmidt summary.

THEORY IN BRIEF · FIXED EFFECT, RANDOM EFFECTS AND HETEROGENEITY

The fixed effect assumes that all studies estimate the same ρ; random effects, that each one estimates its own, around an average, with variance τ^2 (DerSimonian and Laird 1986; Viechtbauer 2005). Q tests whether there is heterogeneity; I^2 says what part of the variation between studies is not chance (Higgins and Thompson 2002); the prediction interval (Higgins, Thompson and Spiegelhalter 2009) is usually much wider than the confidence interval. The test of Egger and colleagues (1997) looks for asymmetry, which may come from publication bias or from small studies being different; with few studies it has little power.



The 16 studies (N = 4633) give an average correlation of .261 (95% CI from .205 to .315).

Heterogeneity is substantial (I^2 = 70.8%, τ = 0.094 on the z scale): a new study could find a correlation between .073 and .431.

Egger's test finds no asymmetry in the funnel (p = .118); that does not prove there is no publication bias: with few studies it has little power.

Hunter and Schmidt: sampling error explains 34% of the observed variance between studies.

Model	r	95% CI	Prediction	τ^2	I^2
Fixed effect	**.236**	[.209, .263]	—	0.0000	68.8 %
Random (DerSimonian–Laird)	**.260**	[.206, .313]	[.081, .423]	0.0081	68.8 %
Random (REML)	**.261**	[.205, .315]	[.073, .431]	0.0089	70.8 %
Random (Hunter–Schmidt)	**.259**	[.208, .310]	[.092, .412]	0.0070	65.7 %

Hunter–Schmidt on r: r̄ = .234, SD of ρ = 0.078, 80% credibility interval [.135, .334]; total N = 4633.

Figure 12.10. The funnel shows no asymmetry (*p* = .118). The three random-effects models give almost the same *r* (from .259 to .261); the fixed-effect one gives .236 and a narrower interval, because it ignores the heterogeneity and gives more weight to the large studies.

READING BLOCK 12

If the two correlations come from the same cases…	use the test for dependent samples, never the independent-groups one.
If one correlation is significant and the other is not…	compare them: that does not prove they are different.
If you correct for attenuation or range…	report the observed *r*, the corrected one and the reliabilities or standard deviations used.
If you are planning a study…	plan by power and by the width of the interval, and take the expected ρ from previous studies, not from the one you would like.
If the studies are heterogeneous…	use random effects and report the prediction interval, not only the confidence interval.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

13

CHAPTER THIRTEEN · BLOCK 13

Figures, palettes and report

The end of the road: what goes into the paper or the thesis. This block gathers all the figures the other blocks drew and exports them at journal size, designs a palette of your own and puts it to the test, builds the correlation table for publication, writes the methods from what was actually computed, collects the references, prepares a report for printing and saves the complete project so you can pick it up another day.

13.1 The figures you have made

13.2 The figure studio

13.3 Designing a palette

13.4 The table for publication

13.5 The methods and the references

13.6 The report and the project

Like Block 12, this one has five sections chosen with the buttons at the top: Figures, Design a palette, Table to publish, Methods and references and Project and report. Everything they show comes from what was computed in this session: if a block was not opened, it does not appear.

13.1 The figures you have made

Each figure the blocks drew appears here as a card, with its title and the block it comes from.

Each card has two controls: In the report, which marks it for inclusion in the report of Section 13.6, and Export, which opens it in the figure studio. At the top, Typeface of the figures chooses the typeface family of all the figures in the app: Sans serif (the system's), Serif or Monospaced; the typeface is the computer's own, with nothing downloaded.

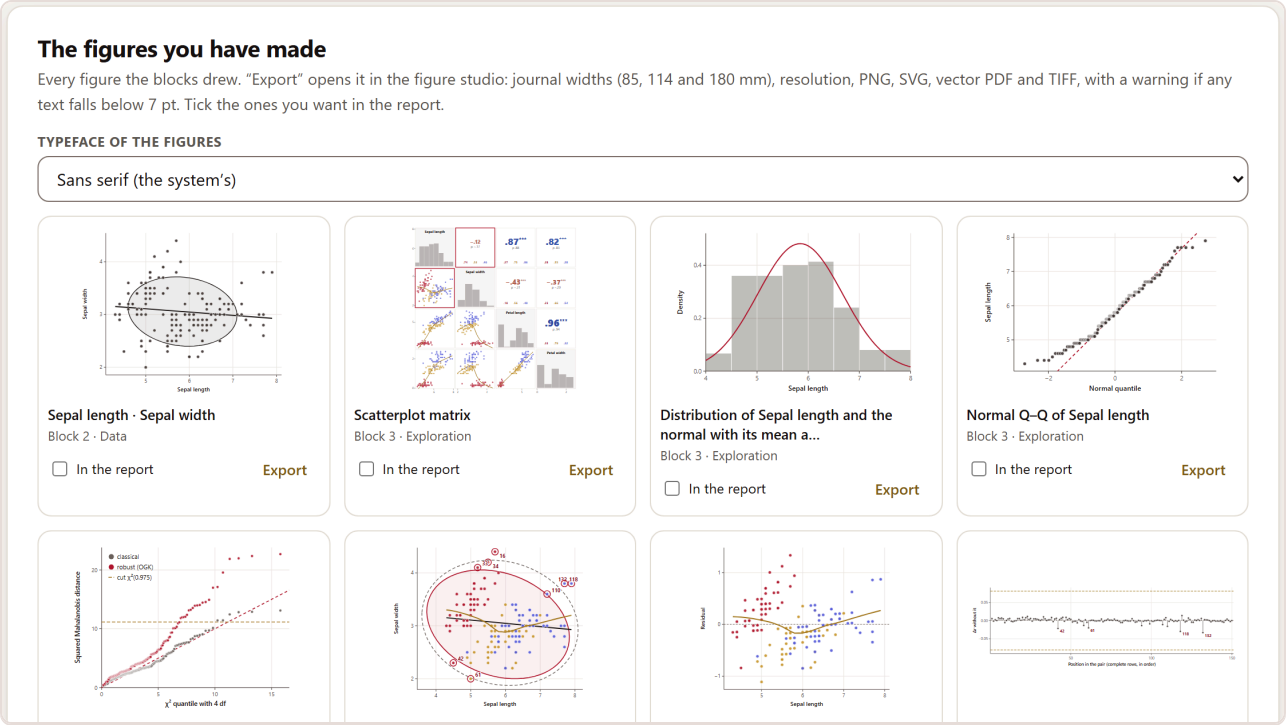


Figure 13.1. The first eight cards of a session with the iris data in Blocks 2 to 5. The gallery goes on with those of Block 4 (the scatter, the ranks, the intervals, the bootstrap, the permutation, the Bayes factor) and those of Block 5.

13.2 The figure studio

Export opens the figure in the **figure studio**, the same in every application of the suite: the figure in the middle, at its real size, and the controls on the right. It also opens from any block, with the pencil-and-ruler button that appears in the corner of each figure.

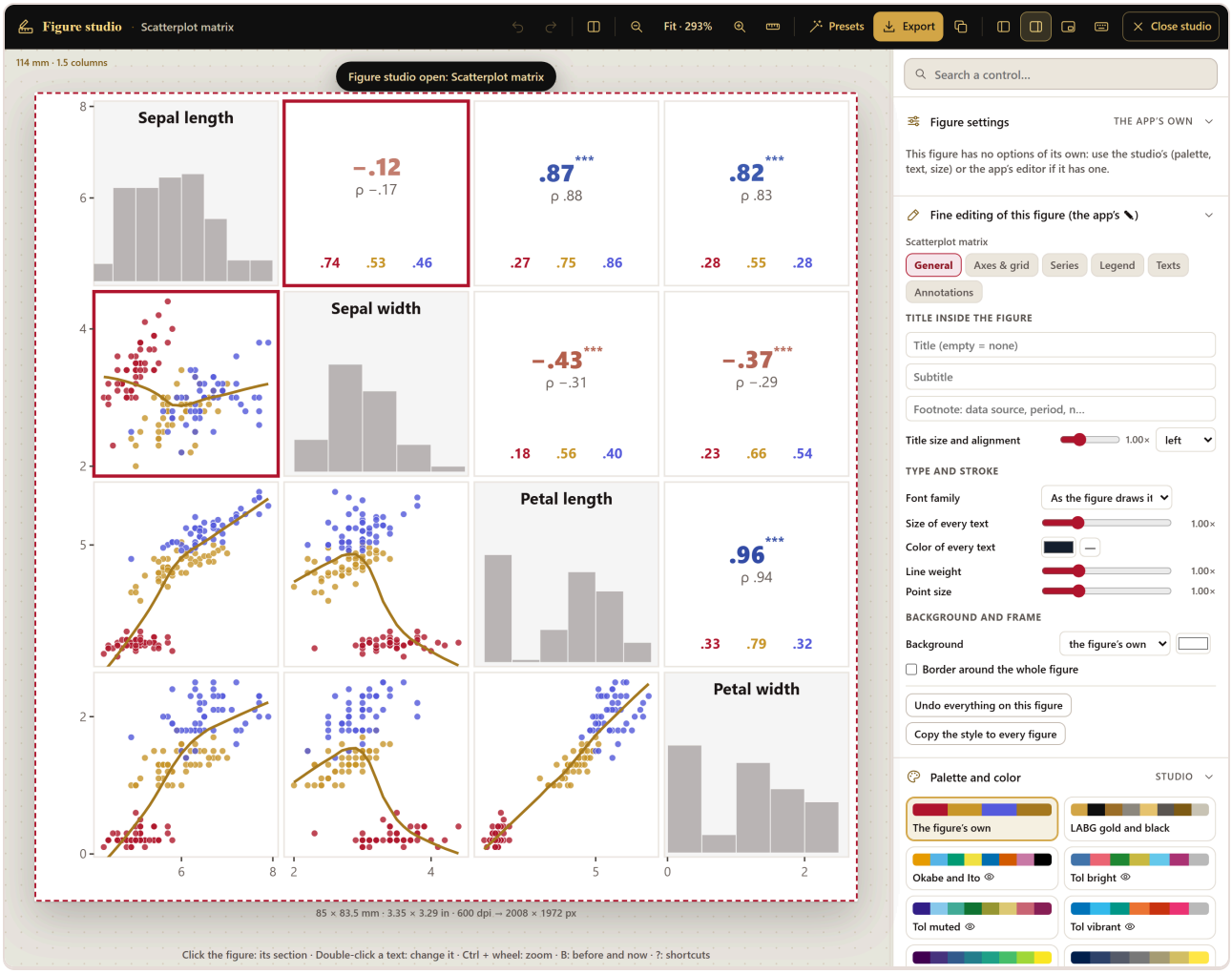


Figure 13.2. The iris scatterplot matrix in the studio. Below the figure, its final size (85 × 83.5 mm at 600 dots per inch, 2,008 × 1,972 pixels); on the right, the settings, the fine editing and the palettes.

- **Size**: the journal widths—one column (85 mm), one and a half (114 mm) and two (180 mm)—or a free one, and the resolution. The studio warns if any text falls below 7 points at the final size.
- **Format**: PNG, SVG, vector PDF with standard fonts, and TIFF for the journals that ask for it. Export downloads the file; nothing leaves the computer.
- **Fine editing**: title, subtitle and footnote inside the figure; the typeface, its size and color; the line width and point size; the background and the frame, and in its tabs, the axes and grid, the series, the legend—its place, inside or outside—, the texts and the annotations. Copy the style to every figure applies the same to the others.
- **Palette and color**: the figure's own or another of the app's palettes, including your own from Section 13.3. If the app is in dark mode, the figure can be exported with the light-mode colors.

On the figure, a click takes you to the section of controls for that part and a double click on a text lets you change it. Presets applies combinations of settings in one click and saves your own, and the arrows at the top undo and redo.

13.3 Designing a palette

A diverging palette for correlations: one hue for the positive ones and another for the negative ones, meeting at a light center. It is built in **OKLab**, a color space where equal distances look equal, so the color grows at the same pace as $|r|$. The controls are the Hue of positive r and the Hue of negative r (in degrees of the color wheel), the Lightness of the ends, the Chroma (intensity), the Name and, to start from one of the app's palettes, Start from.

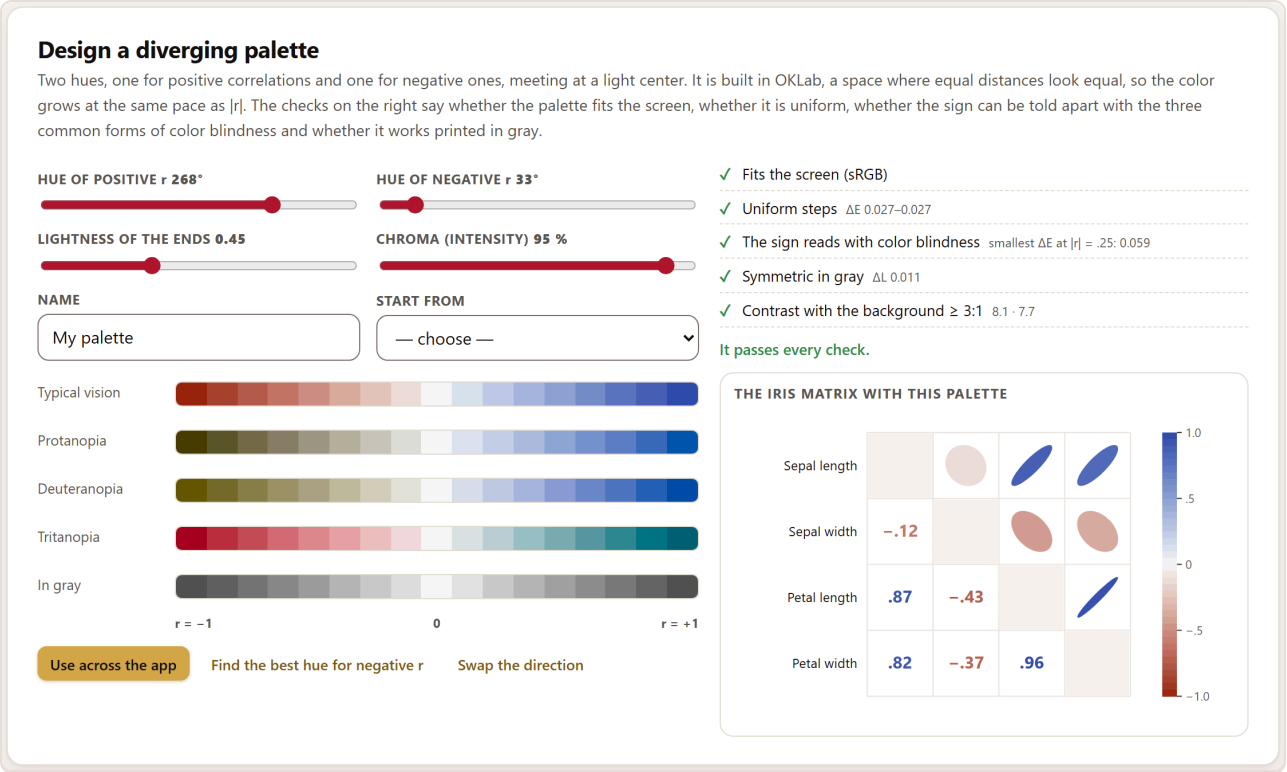


Figure 13.3. A blue-to-red palette that passes all five tests. The ramps show it as seen by people with the three common forms of color blindness and as it prints in gray; on the right, the iris matrix drawn with it.

Table 13.1. The five tests of a palette

Test	What it ensures
Fits the screen (sRGB)	that all its colors can be displayed without being clipped
Uniform steps	that each step of $\|r\|$ changes the color by the same amount (the ΔE between steps)
The sign reads with color blindness	that with protanopia, deuteranopia and tritanopia an r of .25 and one of −.25 can still be told apart
Symmetric in gray	that printed in black and white, r and $-r$ have the same lightness
Contrast with the background ≥ 3:1	that the ends can be read on both the light and the dark background

Find the best hue for negative r tries the possible hues and picks the one that best separates the sign under color blindness, keeping that of the positive ones fixed; Swap the direction exchanges the two. Use across the app makes it the palette of all the matrices, networks and figures; it is saved in the browser and travels in the project package.

13.4 The table for publication

The classic correlation table, with more options than the one in Block 5. You choose the variables with the buttons and, in the menus, the Coefficient (Pearson, Spearman or Kendall), the Significance (asterisks, the 95% interval or the *p*-value under each *r*, or nothing), the Multiple-testing correction (none, Holm, Benjamini–Hochberg or Bonferroni), the Decimals, the Style of the numbers —the same as in Section 1.7, which applies to the whole app—and the Descriptives (mean and SD, median and interquartile range, or none).

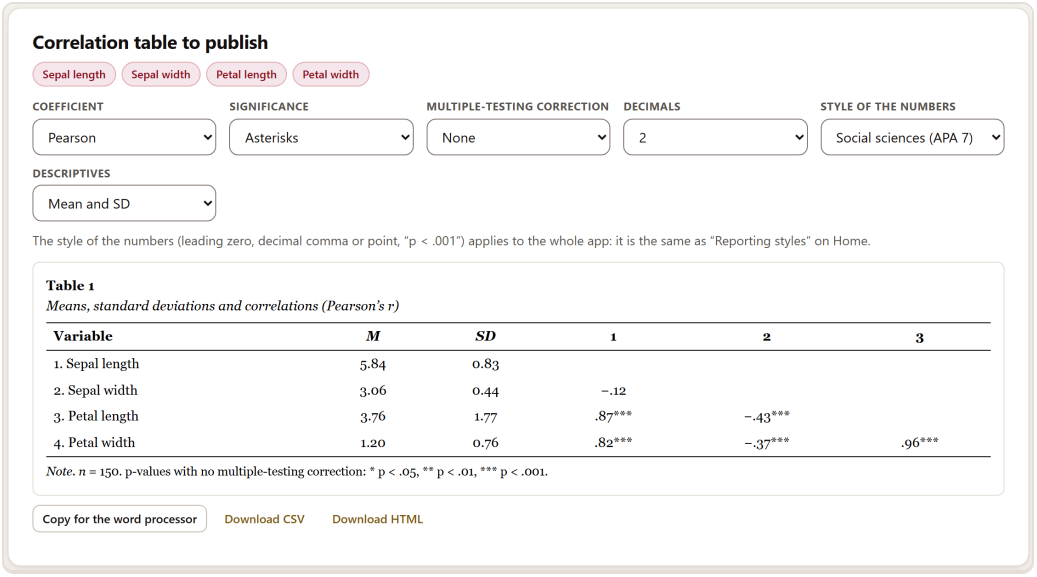


Figure 13.4. The iris data in the social-sciences style: title, means, standard deviations, the lower half of the matrix and the note with the *n* and the meaning of the asterisks.

Copy for the word processor puts it on the clipboard as a formatted table: when pasted into a word processor it keeps its columns, its rules and its italics. Download CSV and Download HTML save it as a file.

13.5 The methods and the references

The app writes the methods section from what was actually computed in this session: one paragraph per block used, with the coefficients, the tests, the corrections, the chosen settings—the number of resamples, the prior, the handling of missing values—and the citation of each method. At the end come the main results in a list and the version of the app.

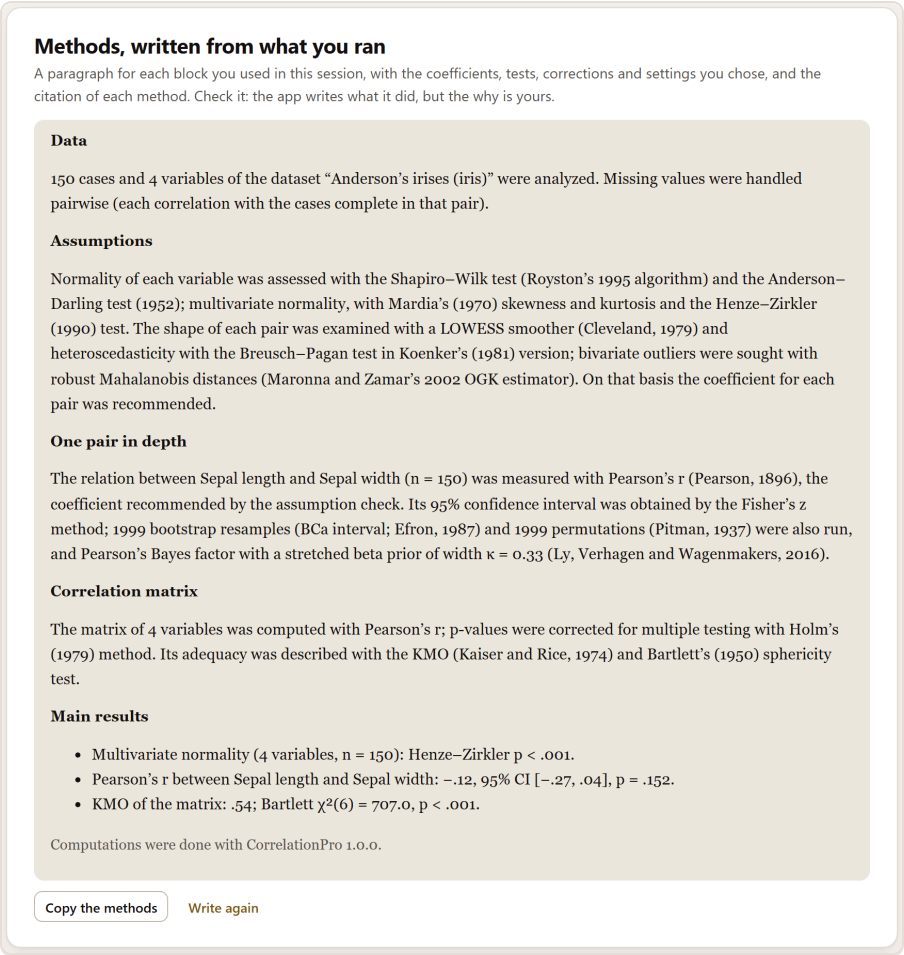


Figure 13.5. The methods of the iris session: the data, the assumptions of Block 3, the pair of Block 4 and the matrix of Block 5, with their main results.

Copy the methods puts them on the clipboard; Write again redoes them after you change something in another block. Below, References of the methods used gives, in alphabetical order and in social-sciences format, only the works cited in the methods, and Copy the references leaves them ready to paste.

> **ALWAYS REVIEW THE TEXT**
>
> The app writes *what* it did, not *why*. Why you chose that coefficient, why those control variables or that correction is something only you know, and a reviewer is going to ask. Use the text as a starting point: correct it, remove what you do not use and explain your decisions.

13.6 The report and the project

The report is a document with the data used, the results of each block, the table, the figures marked in Section 13.1, the methods and the references, always in the light-mode colors. You give it a Title and, if you wish, the Authorship (optional); a line says what it will include. Open the report to print opens it in a new tab, from which you print it or save it as PDF; Download the report (HTML) saves it as a web page.

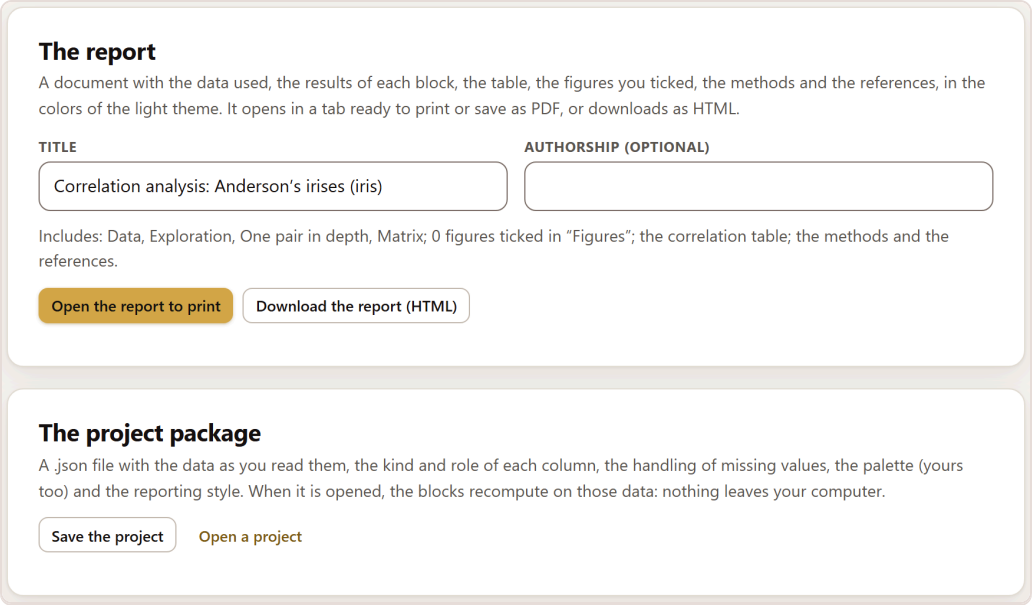


Figure 13.6. The report, with its title proposed from the data, and the project package.

The project package is a `.correlationpro.json` file with the data as they were read, the kind and role of each column, the transformed columns, the handling of missing values, the palette—your own as well—and the style of the numbers. Save the project downloads it; Open a project reads it and recomputes the blocks with those data. It is the way to resume an analysis another day, to send it to a colleague or to keep it alongside the paper.

> **NOTHING LEAVES YOUR COMPUTER**
>
> The app does not send the data anywhere: everything is computed in the browser, and the report, the figures, the table and the project are saved as files on your computer. That is why it works offline, and that is why, if you close the tab without saving the project, the analysis is not kept.

BEFORE YOU SUBMIT

If the journal asks for a width…	export the figures at that width and check the warning about small text.
If the figure will be printed in gray…	use a palette that passes the gray-symmetry test.
If you copy the table…	check that the style of the numbers is the journal's.
If you use the written methods…	add why you chose each method and remove what you do not report.
Always…	save the project package: it is the record of how each number was obtained.

These are guides for interpretation, not laws: what you know about the phenomenon, the design and how it was measured always comes first.

A

APPENDICES

Appendices

What is worth having at hand and did not fit in any chapter: which files the app reads and writes, all the reading rules of the thirteen blocks gathered in one place, the vocabulary, what to do when something does not work, the 229 references of the catalog, and the license of the tool with the way to cite it.

A The files the app reads and writes

B The reading rules, gathered

C Glossary

D When something goes wrong

E The tool: license and how to cite it

F References

A The files the app reads and writes

CorrelationPro runs in the browser and **sends nothing to any server**: everything that goes in and out is a file on your computer or the clipboard. This appendix says which ones.

A.1 · What it reads

Format	Where and what for
Delimited text `.csv .tsv .txt .dat`	Block 2. It detects the separator (comma, semicolon, tab or bar), the decimal mark and the encoding, and says what it chose.
Spreadsheet workbook `.xlsx .xlsm .ods`	Block 2, with its own reader. If the workbook has several sheets with data, it asks which one to open.
Copied cells	Block 2, with Paste: what you copy from a spreadsheet.
Tree in Newick format	Block 10, pasted into the box of the independent contrasts, with branch lengths.
List of studies	Block 12, typed or pasted: one study per line with its name, its *r* and its *n*.
Project `.correlationpro.json`	Block 13, with Open a project: the data and settings of a previous session.

A.2 · What it writes

File	What comes out and from where
The data `.csv`	Block 2, with Download CSV: the table as it stands, with the transformed columns.
The matrix `.csv`	Block 5: one line per pair with *r*, the uncorrected *p*, the corrected one and the *n*.
The table for publication	Block 5, copied with tabs; Block 13, copied with formatting for the word processor or downloaded as `.csv` and `.html`.
The figures	The figure studio: `.png`, `.svg`, vector `.pdf` and `.tiff`.
The methods and the references	Block 13, to the clipboard, ready to paste.
The report `.html`	Block 13: it opens for printing or saving as PDF, or it is downloaded.
The project `.correlationpro.json`	Block 13: the data, the kinds and roles, the missing values, the palette and the style of the numbers.

WHAT THE BROWSER REMEMBERS

The preferences—the language, light or dark mode, the palette, the style of the numbers, the typeface of the figures, your own palette—are kept in the browser's storage, on your computer. The data are not: they are lost when you close the tab, unless you save the project.

B The reading rules, gathered

The scales and cutoffs the app uses to put a word or a color next to a number, in one place, so you can cite and discuss them.

ALMOST ALL OF THESE CUTOFFS ARE CUSTOMS, NOT LAWS

The .05 of significance, the .30 of a "medium" correlation, the .60 of the KMO or the 3 of the Bayes factor are reading conventions, not thresholds that come out of a theory. Use them to find your bearings and always say which one you used; what you know about the phenomenon outweighs any table.

B.1 · The size of a correlation

Scale	Cutoffs in $\lvert r \rvert$ and words
Cohen (1988)	negligible < .10 ≤ small < .30 ≤ medium < .50 ≤ large
Evans (1996)	very weak < .20 ≤ weak < .40 ≤ moderate < .60 ≤ strong < .80 ≤ very strong
Gignac and Szodorai (2016)	very small < .10 ≤ small < .20 ≤ typical < .30 ≤ large

B.2 · Exploration and assumptions (Block 3)

Quantity	Cutoff and reading
Shapiro–Wilk *p*-value	≥ .05 "looks normal"; lower, "not normal" (in red if < .01)
Skewness g_1 · kurtosis g_2	$\lvert g_1 \rvert > 1$ or $\lvert g_2 \rvert > 2$ matter for the test of r
With more than 300 values	$p < .05$ without large skewness or kurtosis: "departs little"
Curvature · fan	flag if the quadratic term or Breusch–Pagan give $p < .01$
Influence of a point	flag if $\lvert \Delta r \rvert$ exceeds $\max(0.02, (1 - r^2)/\sqrt{(n - 1)})$
Few values	ten distinct values or fewer: τ-b instead of ρ
r versus ρ	if they differ by 0.1 or more, ρ
Not monotonic	dCor > |ρ| + 0.25
Groups	Simpson if in half the groups r has the opposite sign with $\lvert r \rvert \geq .3$; group effect if the mean within-group r differs by 0.3 or more
Multivariate outlier	squared Mahalanobis distance above the 0.975 quantile of χ^2 with p degrees of freedom

B.3 · Inference (Blocks 4 and 12)

Quantity	Cutoff and reading
Bayes factor BF_{10}	1–3 anecdotal · 3–10 moderate · 10–30 strong · 30–100 very strong · more than 100 extreme; the reciprocals, in favor of ρ = 0 (Jeffreys 1961; Lee and Wagenmakers 2013)
Heterogeneity I^2	< 25% low · < 50% moderate · < 75% substantial · more, high
Power	the desired one, 80, 90 or 95%

B.4 · Matrices and networks (Blocks 5 and 11)

Quantity	Cutoff and reading
KMO and MSA (Kaiser and Rice 1974)	≥ .90 marvelous · .80 meritorious · .70 middling · .60 mediocre · .50 miserable · below, unacceptable
Condition number	more than 1,000: nearly collinear
Rows per variable in a network	fewer than three: the network is unstable
EBIC γ	0.5, sparse network (the default); 0, the BIC

B.5 · Nonlinear, categorical and agreement (Blocks 6, 8 and 9)

Quantity	Cutoff and reading
dCor − |*r*|	more than 0.2: dependence that a straight line does not capture
Asymmetry of ξ	the two directions differ by more than 0.15
Adjusted residual of a cell	|·| > 1.96 (and > 2.58): more or fewer cases than expected
Expected frequencies	below 5: use the exact or Monte Carlo *p*-value
κ (Landis and Koch 1977)	< 0 poor · up to .20 slight · .40 fair · .60 moderate · .80 substantial · more, almost perfect
ICC (Koo and Li 2016)	< .50 poor · .75 moderate · .90 good · more, excellent

B.6 · Data with structure (Block 10)

Quantity	Cutoff and reading
Cross-correlation	outside ±2/√*n*: a lag that stands out
Effective size (Dutilleul)	less than half the sites: autocorrelation weighs in
Simpson's paradox	all the group *r* values with the sign opposite to that of pooling everything

C Glossary

Each English term is followed by its Spanish equivalent, the one the app shows when its interface is in Spanish.

Attenuation (*atenuación*)

The reduction of a correlation caused by the measurement error of its variables. It is corrected with their reliabilities (Section 12.3).

Autocorrelation (*autocorrelación*)

The resemblance of a variable to itself at nearby times or places. It reduces the independent information in the data.

BCa (*BCa*)

Bootstrap interval corrected for bias and acceleration (Efron 1987).

Bootstrap (*bootstrap*)

Resampling the cases with replacement and recomputing the statistic many times to learn its variation.

Centered log-ratio (*log-cociente centrado*)

The logarithm of each part of a composition divided by the geometric mean of all of them (Aitchison 1986).

Collider (*colisionador*)

A variable that is a consequence of two others. Controlling for it creates a correlation that does not exist.

Confidence · credible · prediction interval (*intervalo de confianza · creíble · de predicción*)

The range of values compatible with the data (frequentist), the one that contains a percentage of the posterior (Bayesian), and the place where the value of a new study would fall (meta-analysis).

Copula (*cópula*)

The shape of the dependence between two variables, separated from their marginal distributions.

Distance correlation (*correlación de distancias*)

A measure of dependence that is zero, in the population, only when the variables are independent.

Ecological fallacy (*falacia ecológica*)

Drawing conclusions about individuals from the correlation between groups (Robinson 1950).

Effective sample size (*tamaño de muestra efectivo*)

The number of independent cases that data with autocorrelation are worth.

Eigenvalue (*valor propio*)

The variance carried by each component of a matrix. Their sum is the number of variables.

Fixed effect · random effects (*efecto fijo · efectos aleatorios*)

In a meta-analysis, assuming that all studies estimate the same thing, or that each one estimates its own value around an average.

Graphical lasso (*lazo gráfico*)

An estimate of the partial correlations that sets the weak ones to zero (Friedman, Hastie and Tibshirani 2008).

Heteroscedasticity (*heterocedasticidad*)

A spread that changes along the line; in the scatterplot it looks like a fan.

ICC (*ICC*)

The intraclass correlation coefficient: the part of the variation due to the cases and not to the raters.

Lin's concordance (*concordancia de Lin*)

A measure of agreement that penalizes both the lack of correlation and the shifts in location and scale.

LOWESS (*LOWESS*)
A smoother that fits weighted local lines, to see the shape of a scatterplot without assuming it (Cleveland 1979).

Masking (*enmascaramiento*)
When several outliers inflate the mean and the covariance and the classical distance stops seeing them.

Modularity (*modularidad*)
How much more the nodes connect within their communities than would be expected by chance.

Monotonic (*monótona*)
A relationship that always goes up or always goes down, even if it is not straight.

Multiple testing (*pruebas múltiples*)
Running many tests at once. Without correction, some come out significant by chance alone.

Outlier (*atípico*)
A case far away from the others. It may be an error or a real value; the robust distance finds it even when there are several (Section 3.4).

***p*-value (*valor p*)**
The probability of obtaining a result as extreme as the one observed if there were no relationship. It is not the probability that there is none.

Partial · semipartial correlation (*correlación parcial · semiparcial*)
The correlation that remains after removing the linear effect of other variables from both variables (partial) or from only one (semipartial).

Permutation (*permutación*)
Shuffling one variable against the other to learn the distribution of a statistic when there is no relationship.

Polychoric · tetrachoric · polyserial (*policórica · tetracórica · poliserial*)
The correlation of the latent continuous variables assumed to lie behind two ordinal variables, two binary ones, or an ordinal and a continuous one.

Positive definite (*definida positiva*)
A matrix whose eigenvalues are all greater than zero. A valid correlation matrix is one.

Prewhitening (*preblanqueo*)
Filtering two series with the autoregressive model of one of them before correlating them, to remove the memory that creates spurious correlations.

Pseudo-observations (*pseudo-observaciones*)
The ranks of each variable divided by $n + 1$: what is left of the data when the marginals are removed.

Range restriction (*rango restringido*)
Having observed only part of the span of a variable, which lowers the correlation.

Reliability (*fiabilidad*)

The part of the variation of a measure that is not error. It is estimated with an α, a test–retest or an ICC.

Simpson's paradox (*paradoja de Simpson*)

A relationship that changes sign when different groups are pooled (Simpson 1951).

Suppression (*supresión*)

When controlling for a third variable makes a correlation grow or changes its sign.

D When something goes wrong

What you see	What it usually is and what to do
The columns come out stuck together in a single one, or the numbers as text	The separator or the decimal mark were not detected correctly. Choose them in Reading options in Block 2.
A value very far from the others, such as −999	It is a missing-value code. Add it in Other missing codes and read the file again.
A variable does not appear in a block	Its kind or its role are not the ones that block uses. Check them in Block 2: almost all blocks take only the columns with the variable role.
"Too few complete pairs."	The two variables have missing values in different rows. Look at the missing-value pattern in Block 2.
The matrix is not positive definite	Pairwise missing values, a robust coefficient or more variables than rows. Section 5.5.
The distance correlation or τ* are not in the table	They have size limits: distance correlation, 3,000 pairs; τ*, 80.
A computation takes a long time	Many rows, many resamples or a costly measure. Lower the resamples while you explore and raise them for the final result.
MIC says that only a 2 × 2 grid fits	There are too few data for MIC. Use distance correlation or ξ.
The network changes a lot when one variable is removed	There are few rows per variable. Interpret only the strong edges.
The report does not open	The browser blocked the new tab. Use Download the report (HTML) or allow pop-up windows for the app.
The table does not paste with formatting	The browser did not grant clipboard permission. Select the table with the mouse and copy it, or download it as HTML.
"It could not be opened: it is not a project of this app."	The file is not a `.correlationpro.json`, or it was edited by hand.
I closed the tab and lost the analysis	The data are not saved automatically. Save the project in Block 13 before closing.
The text looks different on another computer	The app uses the system fonts, which change from one computer to another. Figures exported as PDF or PNG no longer change.

E The tool: license and how to cite it

E.1 · What it is and what it is not

CorrelationPro is a program written from scratch: the computing engines—the coefficients, the tests, the matrices, the copulas, the networks, the meta-analysis—, the spreadsheet reader, the figures and the palettes are in its own code, which can be read. It uses no third-party computing libraries and downloads nothing when it opens. Its results were compared with those of reference programs—19,334 comparisons with no disagreement, Section I.9—, which served only for validation and are not part of the program.

Nor does it bring third-party fonts or icons: the typefaces are those of the computer, and the icons of the navigator and of the figure studio were drawn for the author's suite. The only things that are not the author's are the nine example datasets, which are in the public domain, and each one says why (Section I.7); the meta-analysis example is simulated and says so when it loads. The file `PROCEDENCIA.md`, in the program folder, says where each part comes from.

E.2 · License

It is distributed under the GNU General Public License, version 3 or later (GPL-3.0-or-later). The full text is in the `LICENSE` file, in the program folder.

> **I LICENSING IS NOT GIVING UP AUTHORSHIP**
>
> Choosing a free license is a decision about what whoever receives the program may do. Authorship is neither transferred nor waived. You may use the program, study it, modify it and redistribute it; what the license asks in return is that, if you distribute a modified version, you do so with the same freedoms.

E.3 · How to cite it

The How to cite section of the home page gives the citation ready to copy, in social-sciences format and as a bibliography entry. The short form, for the program and for this manual:

Σ

SUGGESTED CITATION

Barrera-Guzmán, L. Á. (2026). *CorrelationPro: a browser-based platform for correlation analysis of every kind, with its theory* (Version 1.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23867439

Barrera-Guzmán, L. Á. (2026). *CorrelationPro: User's Manual* (1st English ed.). Distributed with CorrelationPro, version 1.0.0. https://doi.org/10.5281/zenodo.23867439

Concept DOI, which always leads to the most recent version: 10.5281/zenodo.23867439; each version also has its own on Zenodo (1.0.0: 10.5281/zenodo.23867440). Code: https://github.com/luisangelbg/CorrelationPro · online app: https://luisangelbg.github.io/CorrelationPro/.

In the methods section, also cite the works behind the methods you used: the coefficient, the test, the correction, the model. Block 13 gathers them on its own, and only those your analysis actually used, which is exactly the list that has to go there.

F References

The 229 works of the app's catalog, grouped by topic and, within each one, in alphabetical order: those that define each measure, its tests and its intervals, those on the assumptions and the figures, and those of the example datasets. The same ones appear in the References section of the home page.

F.1 · History

Galton, F. (1886). Regression towards mediocrity in hereditary stature. *Journal of the Anthropological Institute of Great Britain and Ireland, 15*, 246–263.

Galton, F. (1888). Co-relations and their measurement, chiefly from anthropometric data. *Proceedings of the Royal Society of London, 45*, 135–145.

Rodgers, J. L., & Nicewander, W. A. (1988). Thirteen ways to look at the correlation coefficient. *The American Statistician, 42*(1), 59–66.

F.2 · Assumptions and exploration

Anderson, T. W., & Darling, D. A. (1952). Asymptotic theory of certain "goodness of fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics, 23*(2), 193–212.

Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica, 47*(5), 1287–1294.

Cleveland, W. S. (1979). Robust locally weighted regression and smoothing scatterplots. *Journal of the American Statistical Association, 74*(368), 829–836.

Cleveland, W. S. (1981). LOWESS: A program for smoothing scatterplots by robust locally weighted regression. *The American Statistician, 35*(1), 54.

D'Agostino, R. B., & Stephens, M. A. (Eds.). (1986). *Goodness-of-fit techniques*. Marcel Dekker.

Gnanadesikan, R., & Kettenring, J. R. (1972). Robust estimates, residuals, and outlier detection with multiresponse data. *Biometrics, 28*(1), 81–124.

Henze, N., & Zirkler, B. (1990). A class of invariant consistent tests for multivariate normality. *Communications in Statistics – Theory and Methods, 19*(10), 3595–3617.

Koenker, R. (1981). A note on studentizing a test for heteroscedasticity. *Journal of Econometrics, 17*(1), 107–112.

Mardia, K. V. (1970). Measures of multivariate skewness and kurtosis with applications. *Biometrika, 57*(3), 519–530.

Royston, P. (1992). Approximating the Shapiro–Wilk W-test for non-normality. *Statistics and Computing, 2*(3), 117–119.

Royston, P. (1995). Remark AS R94: A remark on algorithm AS 181: The W-test for normality. *Journal of the Royal Statistical Society, Series C (Applied Statistics), 44*(4), 547–551.

Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika, 52*(3–4), 591–611.

F.3 · Classical

Best, D. J., & Roberts, D. E. (1975). Algorithm AS 89: The upper tail probabilities of Spearman's rho. *Applied Statistics, 24*(3), 377–379.

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